01. A NEW VISION

DURATION : 16:19

STUDY SHEET

COURSE SUMMARY

In this captivating video, a new vision of our relationship with the Earth is discussed, and how it should influence our practice as therapists. Drawing on concepts such as the illusion of perceptions and interconnectedness, the speaker proposes a renewal of consciousness that not only alleviates patients' suffering but also contributes to their spiritual awakening. The three essential elements to integrate are mobilisation, creation, and spiritual work, each serving to bring forth new perspectives and encourage a simpler, more conscious life. An invitation to abandon sensory pleasures in favour of authentic joy emerges enduringly from this approach, thus bringing a new dimension to therapy.

A NEW VISION

Our perception of reality can often be a **huge illusion**. For example, you have the impression that I exist here in front of you, but in reality, I am a **mirage**. Yona Messi wrote a book that invites us to consider the **Earth** as an extension of ourselves, in a spirit of **respect** and love.

Ultimately, we are a sum of **aggregates**, a **momentary**, **impermanent**, and **interdependent** composition. These aggregates include our form and our relationship with the Earth, as well as the great play of **perceptions**. Our perceptions, if we observe them closely, are merely mirages. They are often **blurred**, and we tend to fix things when they actually remain quite uncertain.

This notion of **interpretation** is crucial. How do we perceive what surrounds us? The game that is built around us is also an interpretation of phenomena. We are a **phenomenology**, and perhaps it is time to change our mode of perception, to aspire to a vaster mirage, as grand as our planet. If we become aware that we are connected to the Earth, we might take better care of it.

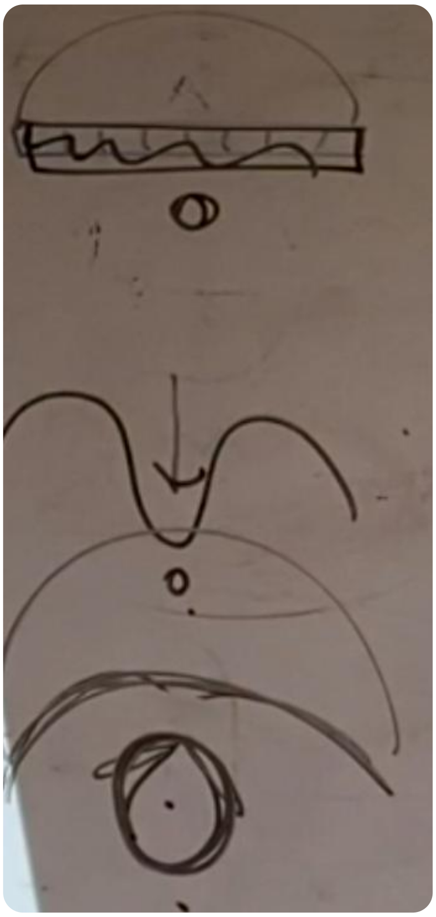


FIG. — The image is a general concept, and this paragraph, dealing with the awareness of our connection to the Earth, foreshadows the introduction and necessity of a 'new vision' that will follow

We live in a society focused on **growth**, often perceived as infinite, whereas our world is actually **finite**. It is essential to recognise this imminent change. I propose a **new vision**: either we make this turning point, or we head towards an inevitable **collapse**.

Theories of emergence teach us that everything emerges and then collapses. Current society is at a tipping point, and as **therapists**, we have a role to play. We can alleviate our patients' suffering while helping them to open up to new perspectives.

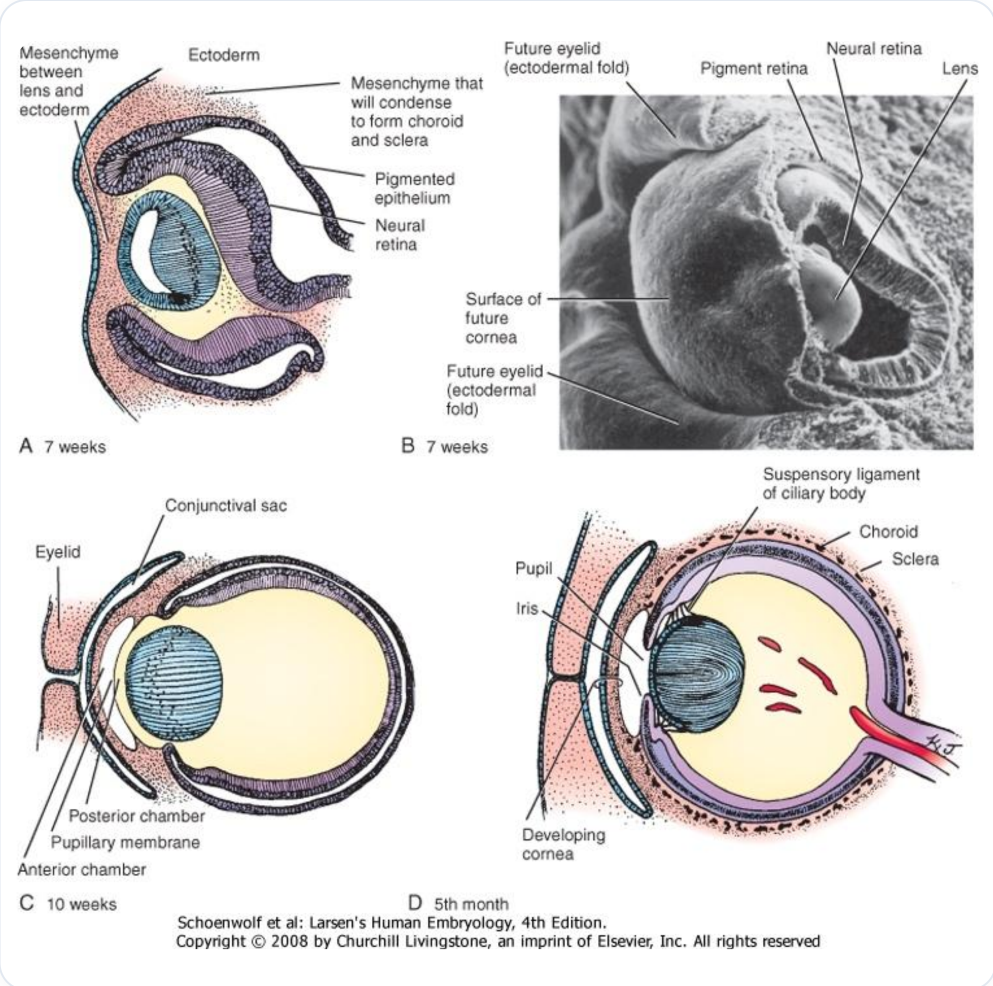


FIG. — Probably an aspect of 'emergence theories' or a visual concept related to transition and the role of therapists. The previous sentence concludes this general idea

Andreas Messie has identified three essential elements for making this turning point:

1. **Mobilisation**: It is crucial to engage and say no to what is not working, not only for oneself but also for others.

2. **Creation**: It is important to develop new ideas and forms, which pushes us to be **creators** and **thinkers**.

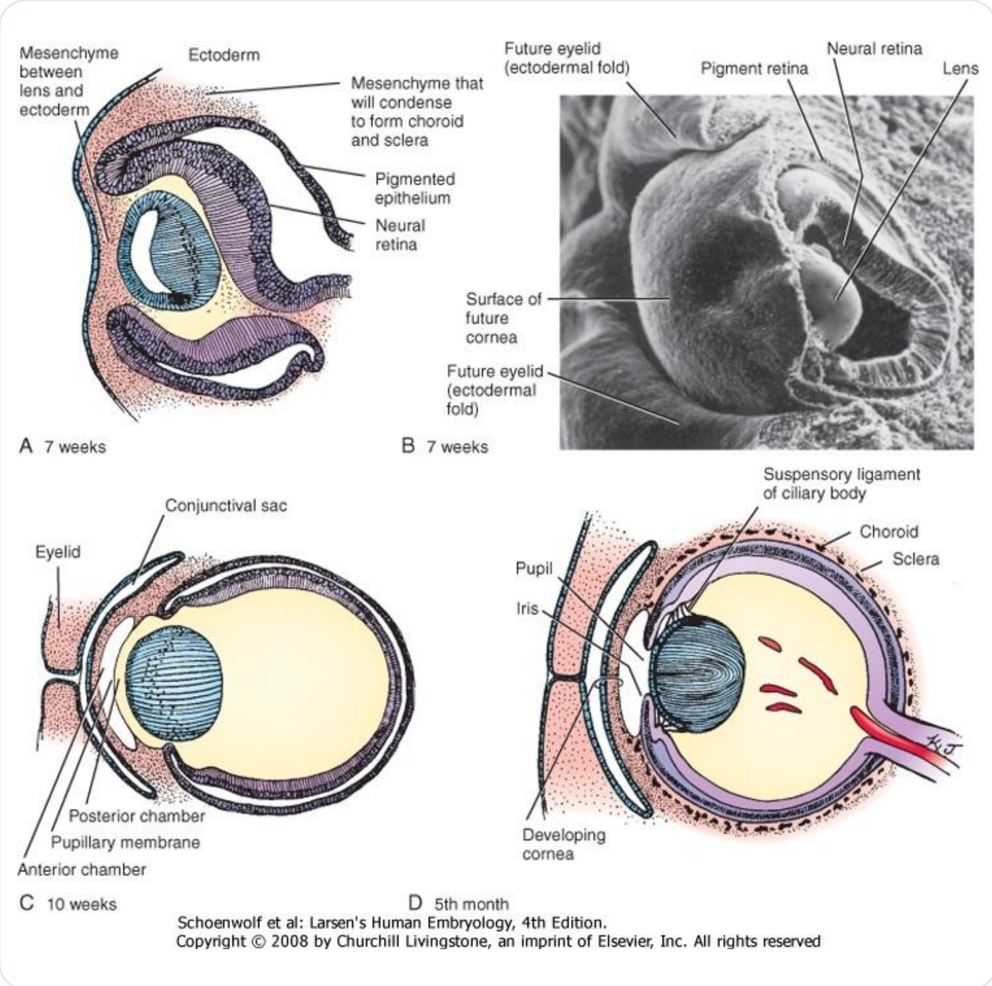


FIG. — This image could visually summarize or introduce aspects related to 'spiritual work' and 'change' mentioned in the previous paragraph by Iona Messie, before addressing more concrete concepts like benevolent consciousness

3. **Spiritual work**: According to Iona Messie, without this work, we will not be able to effect this change.

This work consists of developing a benevolent **consciousness** and a **good heart**. We must ask ourselves how to reduce suffering, not only our own but also that of others. Too often, we are absorbed by the pleasure of our senses.

I encourage you to learn to observe yourselves and examine your behaviours. We have a duty to teach others a form of pleasure that goes beyond simple sensory satisfaction. This can include the pleasure of **simplicity**, of **authentic presence**, or of a relationship with things rather than their possession.

Contentment can emerge from a certain **neutrality**. The joy of renunciation is a concept found in Buddhist texts. The Buddha himself expressed that, although he initially felt anger at the idea of renouncing, he ultimately found immense joy in this **letting go**.

I invite you to reflect on what you could give up to reduce suffering in your life and that of others. As therapists, we have the responsibility to open this consciousness and move towards a form of **simplicity**. This may even mean finding joy in giving up some of our **privileges**.

02. INTRODUCTION TO THE EYE

DURATION : 06:56 STUDY SHEET

COURSE SUMMARY

This video offers a fascinating dive into the anatomy and physiology of the eye, revealing its vital role not only in vision but also in balance and emotional perception. Key concepts include the close relationship between the eye and the heart, the influences of ocular tension on the liver, and the necessity of a multidisciplinary approach that combines embryology, anatomy, and spiritual dimensions. By learning to address the eye in a global context, therapists will be better able to understand the impacts of emotions on perception and, by extension, improve their practice by considering the links between physical structure and the environmental information that influences our vision.

INTRODUCTION TO THE EYE

The **eye** is a fascinating structure that plays a crucial role in our perception of the world. A part of our **brain**, the retina, is capable of detecting the smallest particle of light, the **photon**. The eye is actually an **extension** of our brain, with significant brain activity.

The importance of the eye is not limited to vision. It directly influences our **centre of gravity**, and the latter, in turn, affects the eye. In this sense, the eye acts as a **lighthouse** for all the body's **fascia**. The ocular tensions we observe in consultations are often linked to the **liver**. For example, a red or yellow eye can be an expression of liver problems.

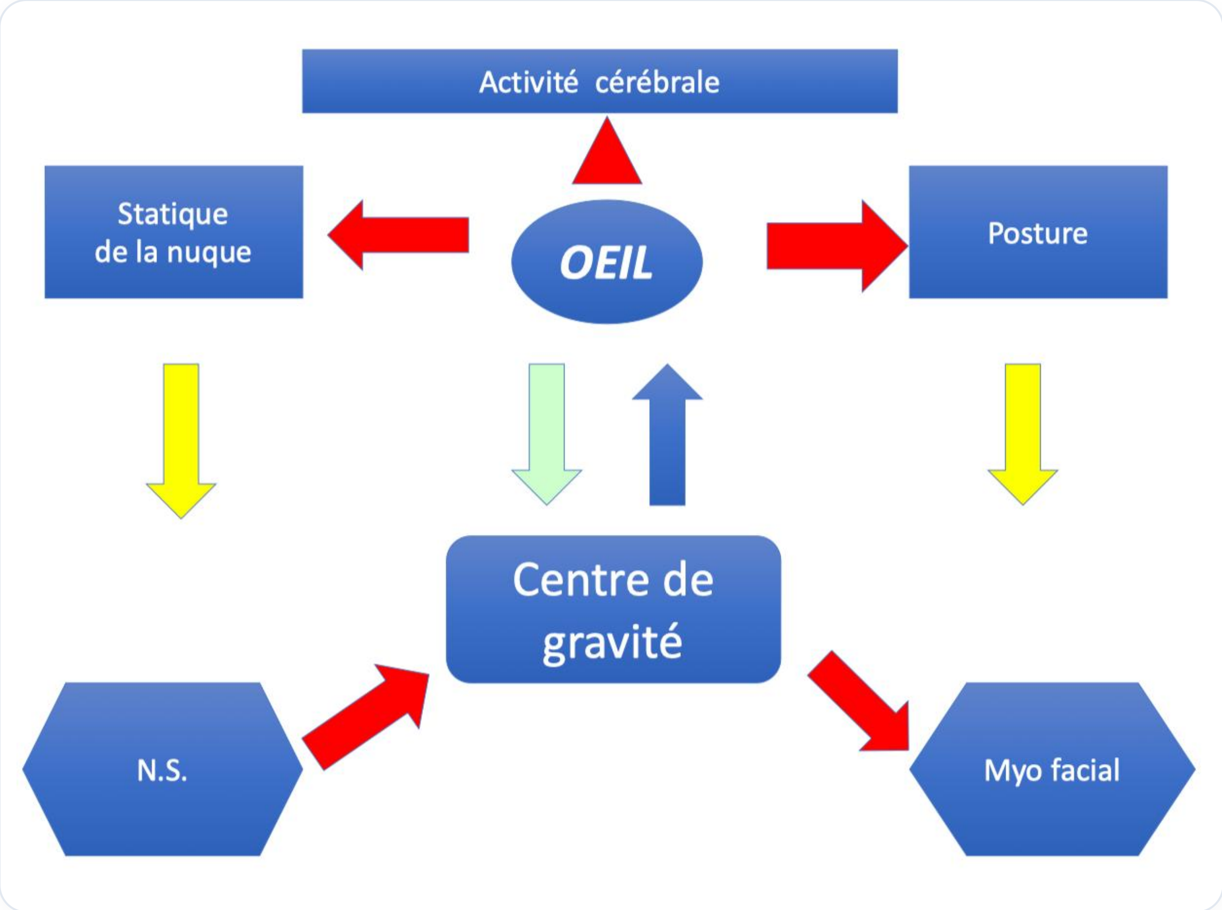


FIG. — Concept of the eye as the 'lighthouse of the fascias' and its influence on the center of gravity, as described in the text

It is also interesting to note that, from embryology, the eye develops in relation to the **heart**. It receives rhythmic information, just like the throat and hands. These different parts of our body transmit **electrical** and **electromagnetic charges**. For example, we can influence someone through our voice or through our gaze.

The eye receives information not only from the heart but also from our environment. **Brain frequencies**, particularly alpha frequencies, play a role in stabilising our perception. Buddhists speak of an **invisible thread** connecting the heart and the eyes, highlighting the importance of this connection.

It is essential to learn to **see with the heart**. This approach can change our perception and create subtle shifts in our vision. A simple word, gesture, or glance can have a profound impact on a person, either to block them or to free them.

Attention is a crucial tool to sharpen for improving vision. Emotions, such as fear or anger, can alter our perception of reality. For example, anger can offer temporary clarity, but it can also distort our vision.

Working on the eye requires a multidisciplinary approach, including **embryology**, **anatomy**, and a more **spiritual** dimension. We will explore the different **skandhas**, which are essential for understanding the world of perception, sensation, and consciousness. It is important to distinguish between illusion and reality, as our prejudices can influence our vision.

The eye is a **conformator** that is influenced by its environment. It is present even before the formation of bones and obeys laws of position, mobility, and information integration. We must also understand the **neurosensory** level of the eye and its **neuromotor** involvement.

When light touches the eye, it undergoes **transduction**, transforming light into electricity. This information is then transmitted to the **brain**, particularly to the lateral geniculate body, influencing our perception. It is important to note that we lose approximately **80%** of reality in our way of seeing, as our perception is strongly influenced by our experiences and emotions.

As an osteopath, it is crucial to receive this environmental and facial information. Treating the eye involves treating the entire body, as the eye functions as a **balance** between the body's structure and the problems that may manifest.

03. ORIGIN OF THE EYE

DURATION : 07:12 STUDY SHEET

COURSE SUMMARY

This video explores in depth the origin of the eye, which develops from the diencephalon, from the eighth day of embryonic development. The emphasis is placed on the importance of the amniotic cavity and cerebrospinal fluid (CSF), as well as how these elements influence the formation of the eye and its dynamic balance with the environment. The relationship between the amniotic cavity, the ventricular system, and the eye is highlighted, while also underlining the clinical and genetic implications that can affect the individual's health and manifest through the eye. This understanding can help therapists assess and restore balance in their patients, making the teaching essential for those who practice osteopathy and biodynamic embryology.

ORIGIN OF THE EYE

The eye shows a fascinating **resemblance** to its formation on the eighth day of embryonic development, when a small cavity called the **amniotic cavity** appears. This cavity is fundamental for the development of the eye, which originates from the **brain**, more precisely from the **diencephalon**, an expansion of the third ventricle.

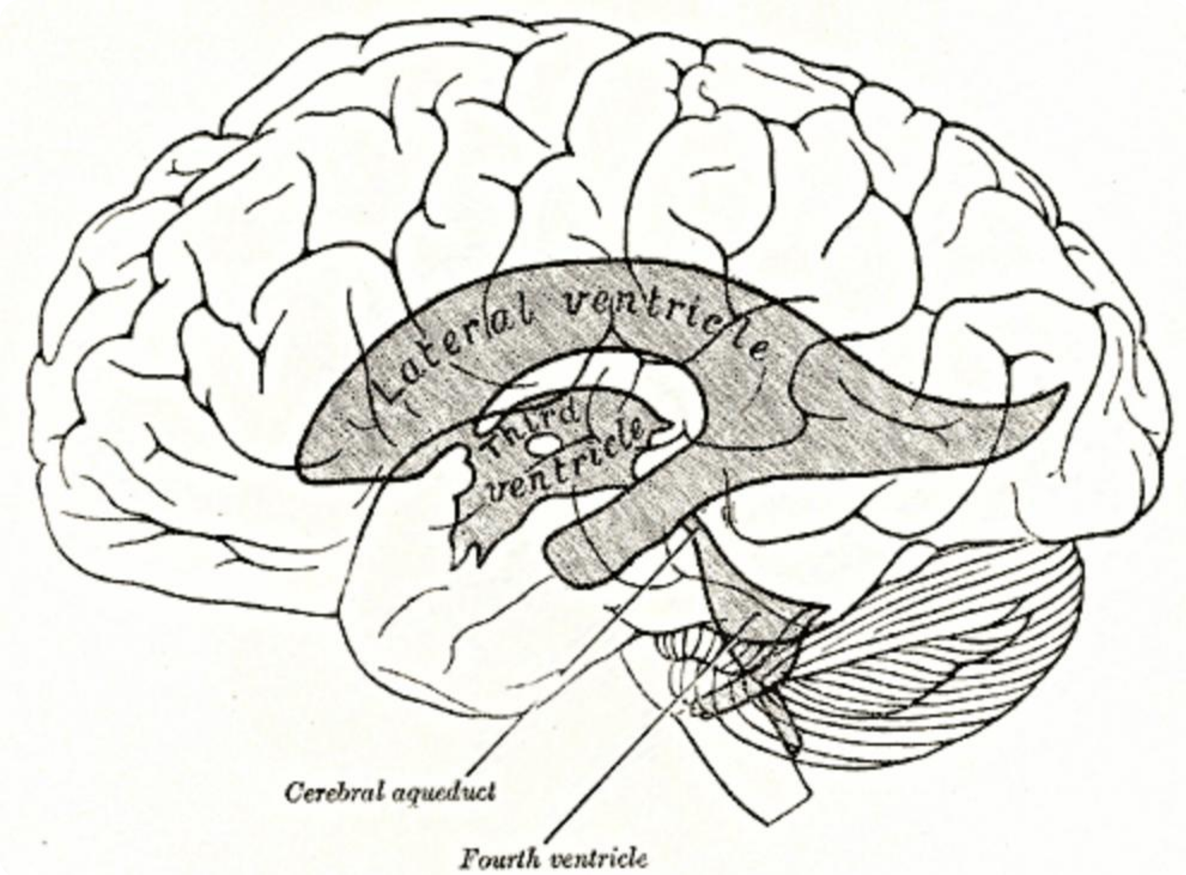


FIG. — Diagram related to the origin of the eye from the diencephalon, in connection with the third ventricle

During this development, the neural system receives **primitive amniotic fluid**, which will form the **cerebrospinal fluid (CSF)**. This initial CSF, derived from amniotic fluid, will be enclosed within the neural tube. Subsequently, this tube opens anteriorly, forming **primitive vesicles**.

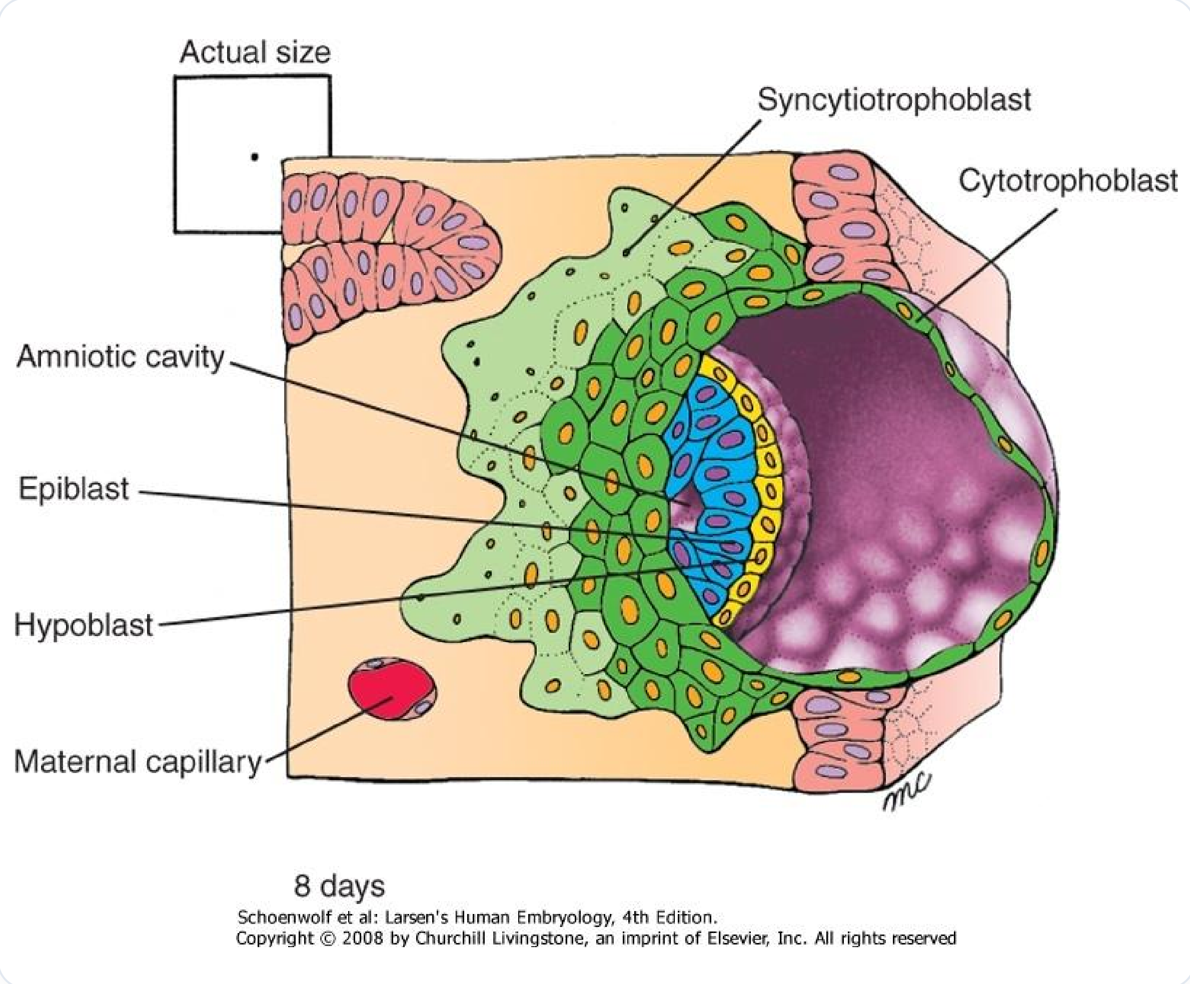


FIG. — Diagram of the appearance of the amniotic cavity, which is mentioned just after this sentence in the summary

The remnant of this amniotic cavity is found in what is called the **bone pocket** or **zone B**, where cerebrospinal fluid is located. It is interesting to note that the fluid inside and outside the embryo is the same. The eye, as a membrane, constantly seeks to balance this zone B.

Zone A is represented by the central axis of the physical body, while the amniotic cavity begins to form on the eighth day, marking the entry into the **mucosa**. At this stage, a central fluid loss creates a small cavity, and the **ectoderm**, called the **epiblast** at this stage, begins to establish itself. This already constitutes the subtle structure that will form the eye and the brain.

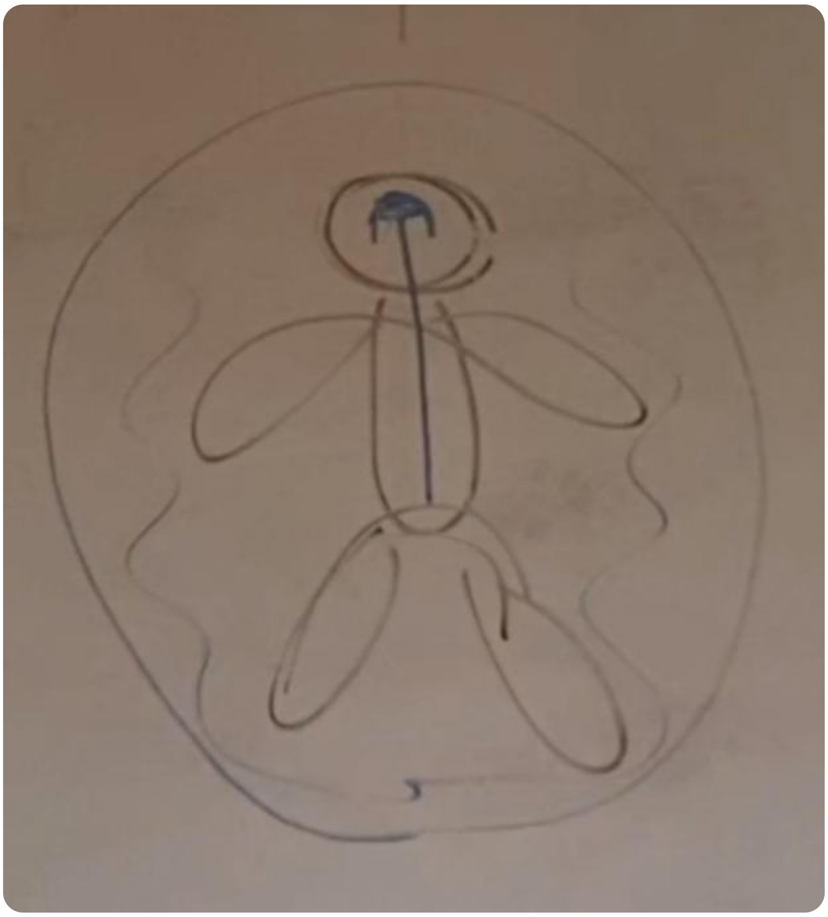


FIG. — Vestige of the amniotic cavity (zone B)

The amniotic cavity, which surrounds the embryo for nine months, then becomes a vaporous space. This space must be in balance with the internal fluid. Imbalances can occur, particularly in the event of accidents, affecting the position of the eye. The eye thus represents a balance between the depth of the fluid of the nervous system (CSF) and the surrounding space.

Observing the position of the eyes can reveal information about a person's balance. Each individual has a slightly different eye, often related to the architecture of their brain. The search for **horizontality** in relation to **verticality** is essential, as is the balance between the digestive system and the peritoneal system.

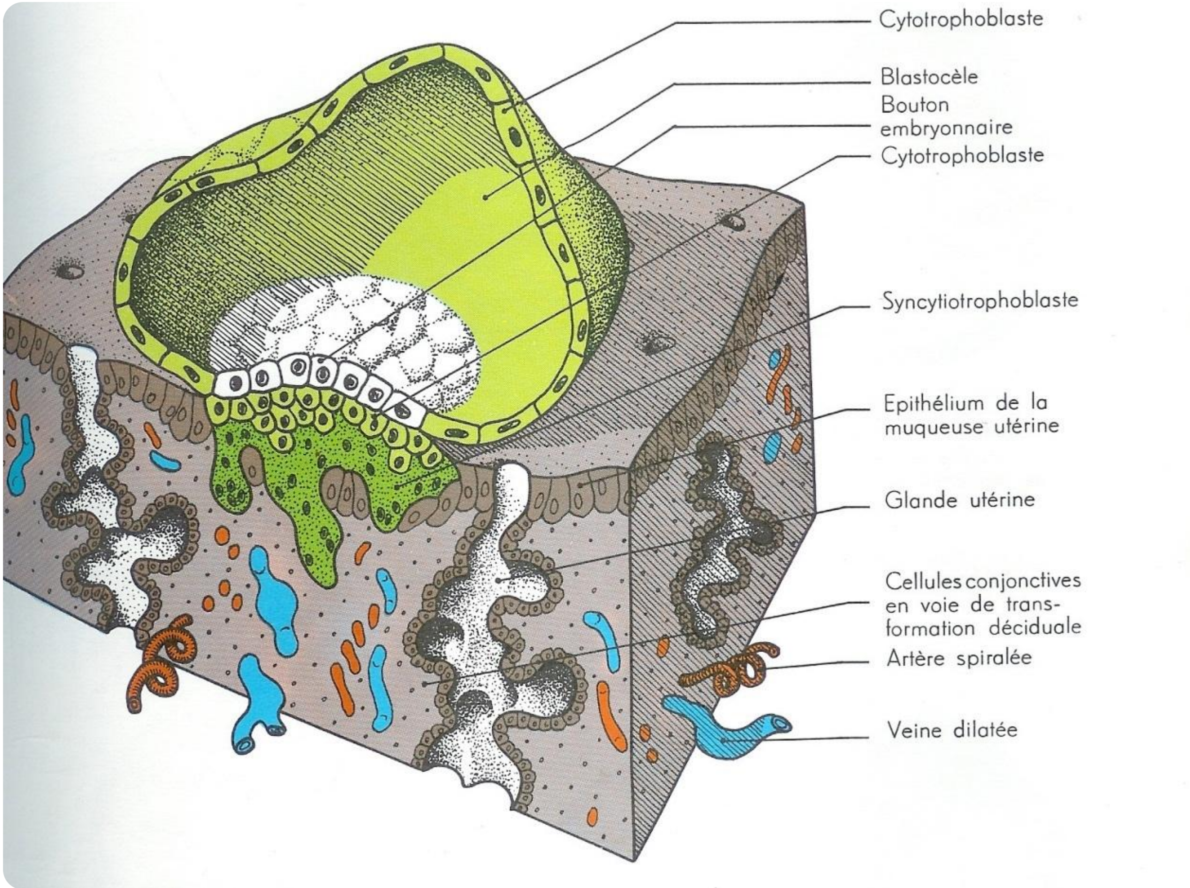


FIG. — Implantation process: cavity formation

It is crucial to observe the eyes after treatment. If the position of the eyes is not stable, it indicates that the balance is not yet restored. The amniotic cavity, although no longer in liquid form, remains present in other states, such as vaporous.

This relationship between the amniotic cavity, the eye, and the ventricular system is fundamental. The eye, as an expansion of the third ventricle, begins to develop before the closure of the **anterior neuropore**. Furthermore, there are interesting genetic implications, with co-localizations of information on impact zones, such as the hepatopancreatic, cardiac, and thyroid systems.

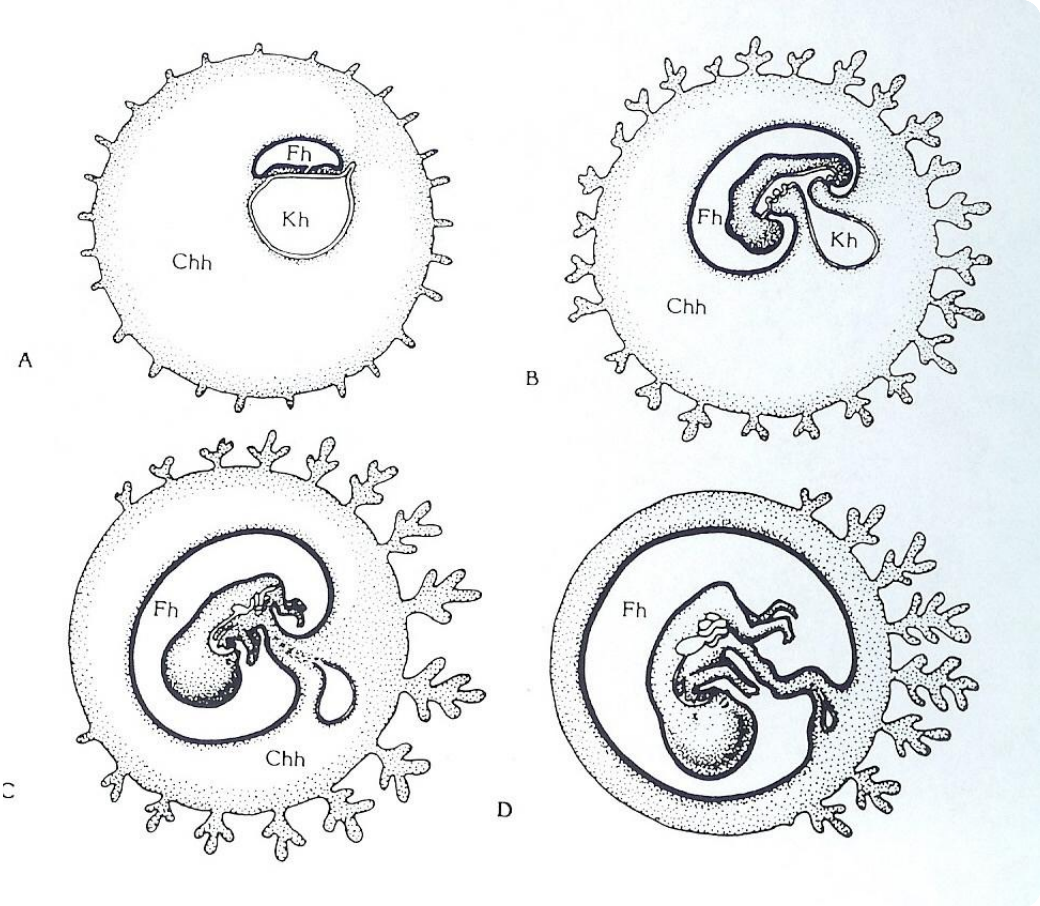


FIG. — Amniotic cavity

Pathologies such as **diabetes**, heart problems, and thyroid issues can manifest through the eye, thus revealing the importance of the **neural crest** in the organization of the midline and laterality.

04. EYE PLACEMENT

DURATION : 08:39

STUDY SHEET

COURSE SUMMARY

This video explores the development of the eye within the framework of biodynamic embryology and osteopathy. You will learn how the interaction between the endoderm and mesoderm, as well as the primary role of the notochord and neural tube, contribute to the formation of essential structures such as the eye. The movements of permeation and infusion, along with the concept of polarity, are key concepts underlying embryonic development. Furthermore, it will be discussed how rebalancing techniques, in response to emotional or physical whiplash, can restore the movements necessary for optimal eye development, thus linking it to events such as the first heartbeat.

EYE DEVELOPMENT

The **endoderm** is an epithelium, while the **mesoderm** is a connective tissue. It is essential to understand the balance between these two major tissue types: an internal tissue and an external tissue. For example, the digestive epithelium, although internal, acts as a boundary tissue.

The precursor to the **neural groove** and the **neural plate** is the **notochord**. This process of evagination along a longitudinal axis is related to the **primitive streak**, which forms the primitive axis of the body. This axis is crucial because it organises cells according to their space, time, and destiny.

By day 18, the neural plate develops, and the third cavity, that of the **external coelom**, appears. This change in polarity and the reorganisation of the axis create the primitive streak through a suction field from the caudal region, which "aspirates" information. This phenomenon is accompanied by movements called **permeation** and **infusion**.

The permeation movement, a major metabolic movement, occurs around the embryo, advancing the **amniotic cavity**. Thus, we distinguish the amniotic cavity, the vitelline cavity, and a tissue called the **epiblast**, which will give rise to the future nervous system, while the tissue beneath it will form the future digestive system.

This movement causes a change in the shape of the notochord, which takes on an S-shape. This differential growth is linked to polarity. The primitive axis organises the epithelial tissue, which reacts through phenomena of induction and inhibition in relation to the notochord.

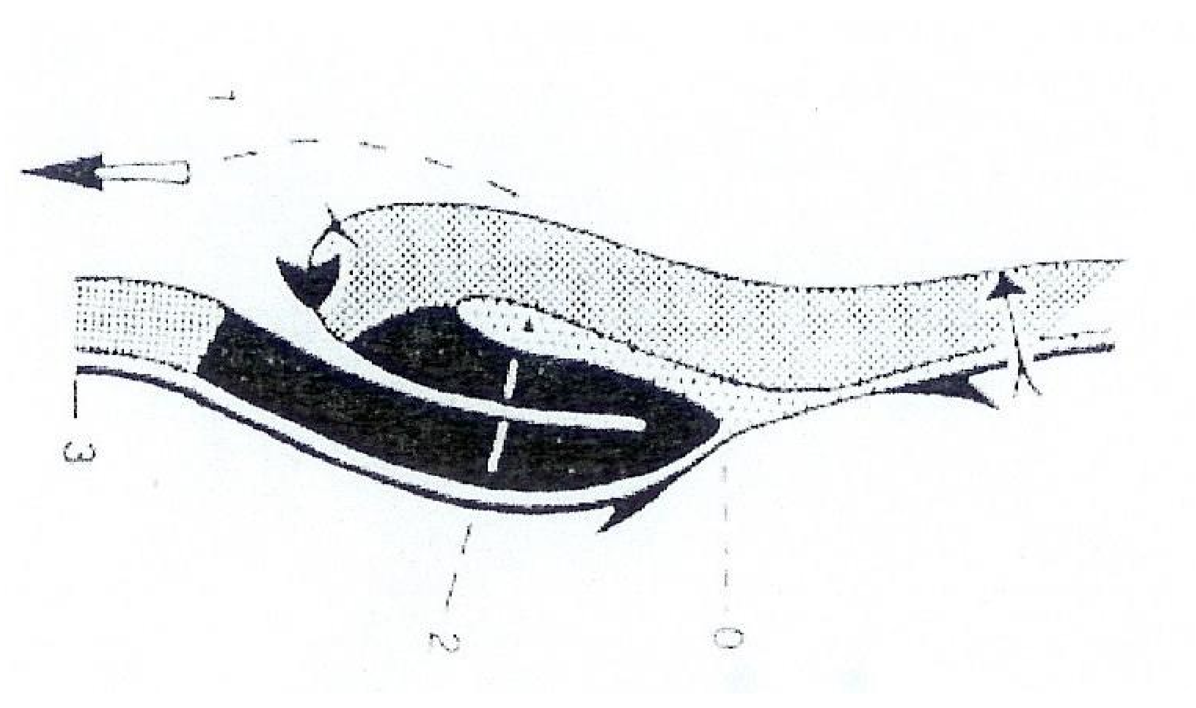


FIG. — Differential growth

The neural plate evolves to become a **neural groove**, which encloses cerebrospinal fluid, or rather primitive amniotic fluid. The closure of this groove forms a **neural tube**. The **neural crests**, considered a fourth embryonic tissue, play a crucial role in eye development.

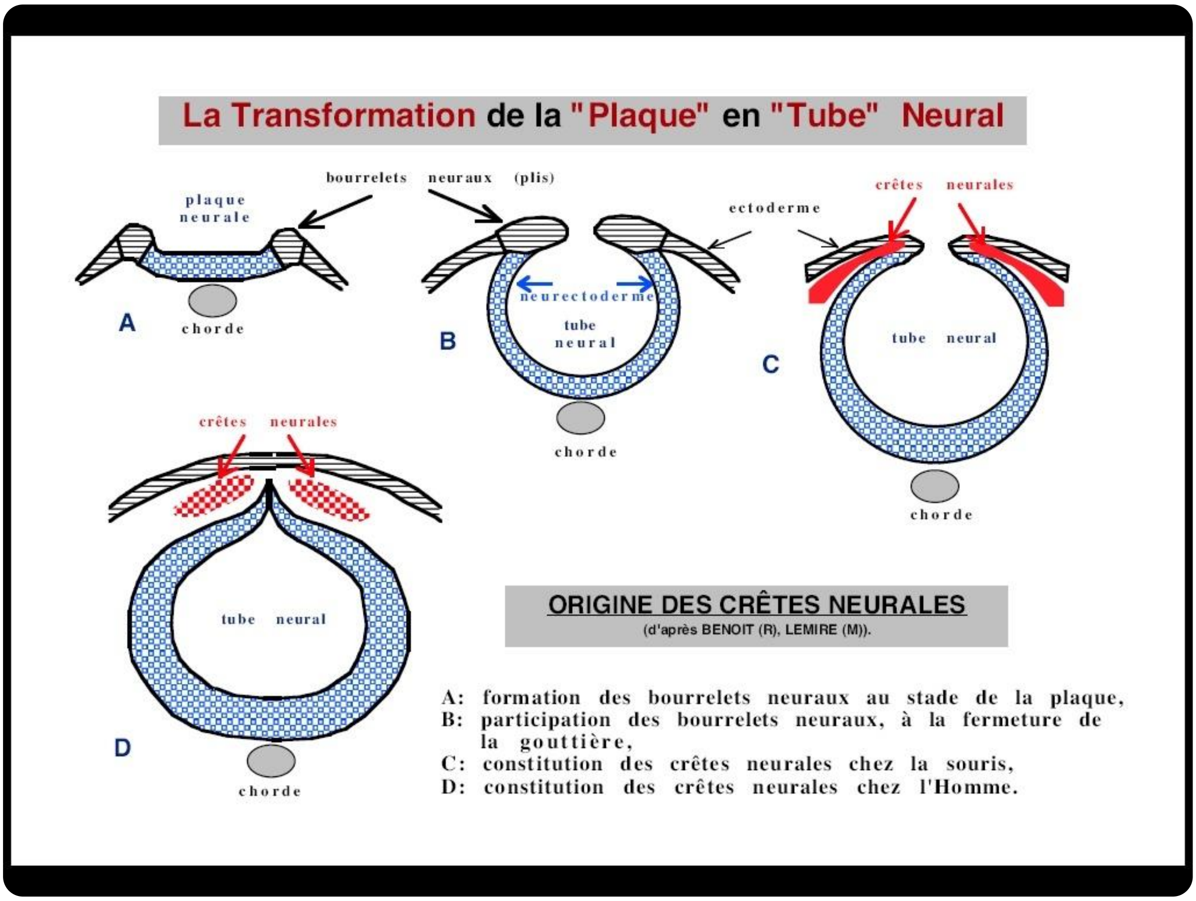


FIG. — Primitive amniotic fluid

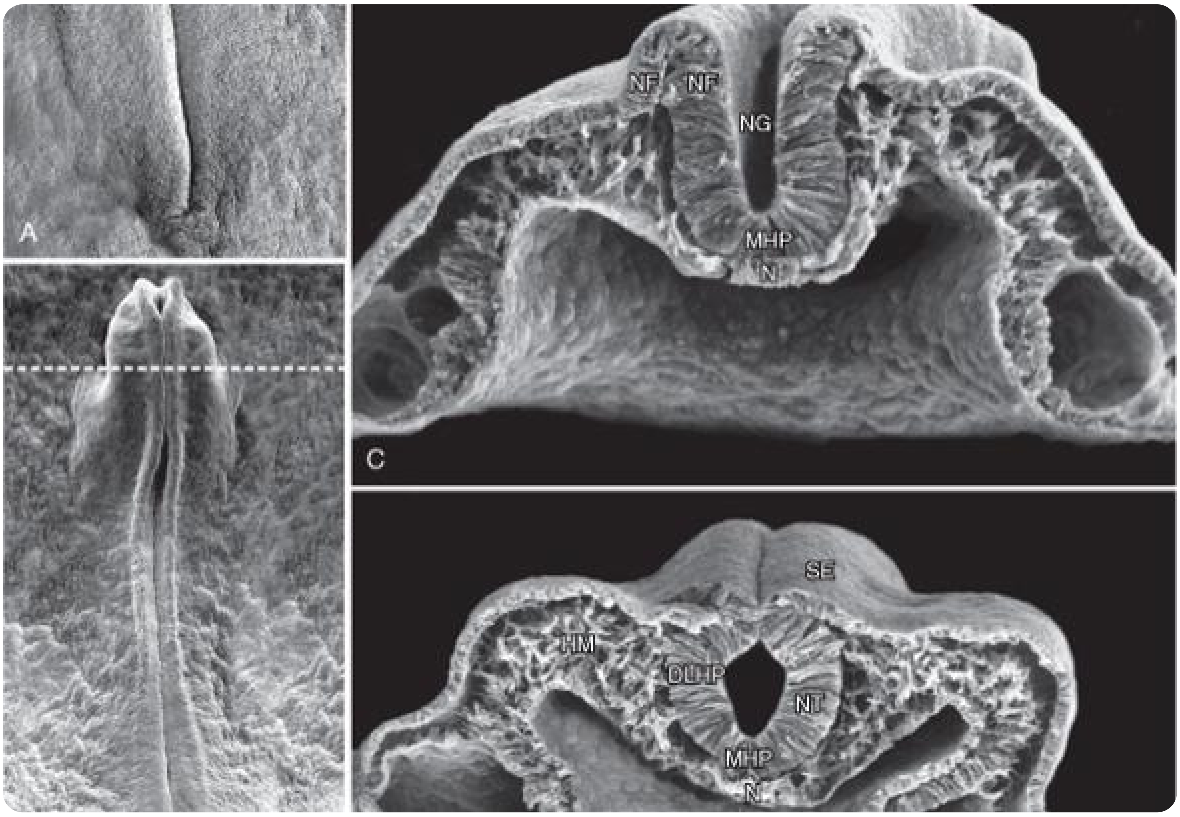


FIG. — Neural groove formation

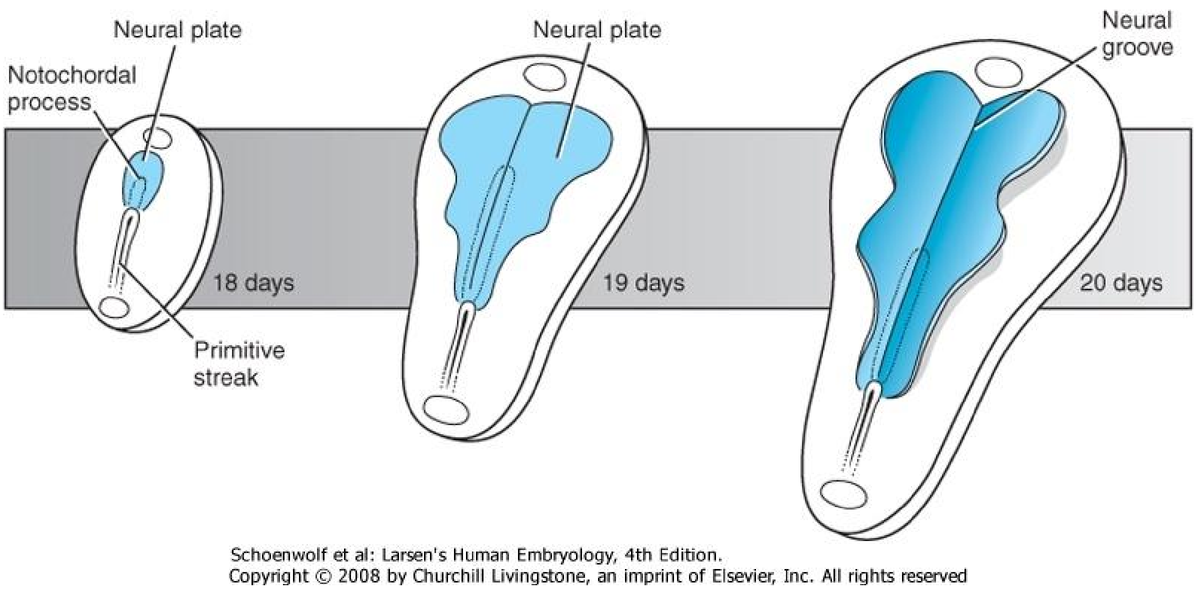


FIG. — Transformation of the neural plate into the neural groove

The notochord is essential for eye development. Without a balanced notochord, it is impossible to have a stable eye. Similarly, a free sacrum is necessary to ensure brain stability. It is therefore fundamental to rebalance the sacrum to ensure good verticality.

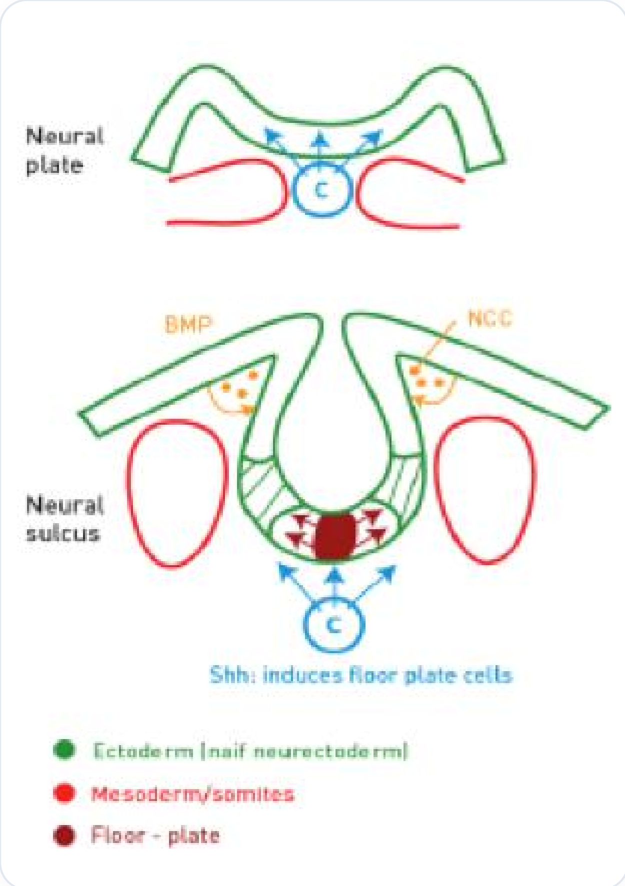


FIG. — Closure to form the neural tube and neural crest formation

The embryo presents the amniotic cavity, the vitelline cavity, and the bilaminar disc, which organise around the notochordal axis and the neural tube. The formation of this neural tube involves the closure of the amniotic cavity and the vitelline cavity.

As the longitudinal axis of the embryo develops, the brain anlage forms and closes at the centre of the embryo, leaving a remnant at the top. It is in this space that the **lens placode** develops, which will form the eye.

Before the closure of the neural tube, an induction phenomenon occurs to form the notochord, which is the basis of cortical development. From this development will emerge the eye, which is an expansion of a ventricular system filled with cerebrospinal fluid. The eye can be considered a fluid sphere, a subtle filter for photons, and a balancer of the surrounding space.

It is possible to rebalance the eye by working on a specific area, often affected by **whiplash**. These whiplash injuries can be emotional or physical. Treating a whiplash involves restoring the movements of **cephalisation**, **cardialisation**, **diaphragmatisation**, and **hepatisation**, which are essential for embryonic development.

The emergence of the eye begins with the first heartbeat, around day 22. The first epiblastic cell transformation occurs in synchronisation with these primitive heartbeats, establishing a significant correspondence between eye development and cardiac activity.

05. NOTOCHORDAL INFLUENCE

DURATION : 03:50 STUDY SHEET

COURSE SUMMARY

This video explores the essential role of the notochord in embryonic development, particularly its influence on the neural tube through S-hatch sonetic systems, crucial elements such as 40A and 40T. Students will learn how the notochord emits electromagnetic signals that act as a GPS for organ formation, utilising nutrients such as beta-carotene and retinoic acid. Through the observation of cellular dynamics and lateralisation, this video highlights the importance of perceiving the notochord not only as a structure but as an essential function for embryonic organisation, playing a key role in the formation of the brain and eyes.

INFLUENCE OF THE NOTOCHORD ON EMBRYONIC DEVELOPMENT

The **notochord** plays a crucial role in embryonic development, particularly influencing the **neural tube**. It is essential to understand that this notochord emits signals, called **S-hatch sonetic systems**, which include elements such as **40A** and **40T**. These induction phenomena are significant, especially concerning **beta-carotene** and **vitamin A**, which are integrated with 40T and are vital for the development of the **eye**. **Retinoic acid** is also important at this stage, involving large genes.

Imagine a central axis around which the neural tube develops. This notochordal axis provides information via an **electromagnetic field** of position, acting like a **GPS** for the surrounding cells, which leads to the reaction of various **proteins**. For example, the eyes will develop at a precise location, while other parts, such as the feet, will form at other locations. The information received in relation to the midline varies according to space and time.

The electrical field emitted by this notochordal axis extends between left and right, as well as between front and back. Upon entering the **nodal flow**, moving cells are observed, similar to falling snow, with cilia rotating around the axis of the notochord. This dynamic allows micro-information to be transmitted to the left, contributing to the **lateralisation** of the embryo.

It is important to note that there is no symmetry, but rather **harmony** in the structuring of genetic information in relation to the notochord. If cells are considered as **molecules**, then **proteins** will be produced and released. This information comes from the **genome**, which activates various genes according to space-time, particularly within the framework of **sonetic edges** and **PAX space-time**.

It is crucial to understand that the notochord is not simply a structure, but a **function**. This distinction is fundamental: perceiving the notochord as a function allows for an understanding of its forces and responses, which is essential for the organisation of the **brain** and the **eye**.

06. INHIBITION AND COLOCALISATIONS

DURATION : 05:42

STUDY SHEET

COURSE SUMMARY

This video explores the complex mechanisms of inhibition and colocalisations in embryonic development, highlighting the importance of left laterality and the crucial role of neural tube closure. Key concepts discussed include the formation of optic vesicles, the interaction between the neural crest and various organ systems such as the hepato-pancreatic system and the heart, and how these interactions influence the evolution of the eye. By integrating principles of electromagnetic theory, the video demonstrates the impact of these processes on embryonic morphology and genomic expression.

INHIBITION AND COLOCALISATIONS IN EMBRYONIC DEVELOPMENT

Lateralised information, sonetic and hutch, is essential in embryonic development, particularly in left-right dynamics. A **left laterality** begins to establish itself, but an inhibitory phenomenon prevents the expression of sonetic and hutch forwards, favouring lateral expression.

This inhibition is linked to the **non-closure of the neural tube**. Without this inhibition, a cyclops eye could develop, as it is the central inhibition that allows the formation of two small lateral vesicles. When the neural tube is not closed, the eye appears around day 22, and the closure of the anterior neuropore occurs around day 24 or 25.

At this stage, so-called **"negative" information** emerges from the neural crest, which shows colocalisations with information from the **hepato-pancreatic** system and the **heart**. The heart receives important information from the neural crest, as does the throat. These colocalisations are expressed at the level of the eye, involving coding genes in different spaces, thus illustrating systems of colocalisation of genetic information.

An **autocorpse** represents a genomic function, acting as a reference for the body. This electrical field, resulting from a polarity, is fundamental. According to **Maxwell's law**, every electrical field generates an electromagnetic field. Thus, each individual receives specific information.

In the anotocorte, the cells at the front receive information called sonetic HH. If this growth is slowed down, an acceleration occurs, leading to an expansion of the epithelium, linked to genetic processes. However, as long as the neural tube is not closed, a small space persists.

Without inhibition, the eye develops centrally, leading to the formation of a cyclops eye. The closure of the neural tube is crucial for the neural crest to express laterally. Once closure is achieved, negative inhibition influences the eye, in connection with **thyroid**, **cardiac**, and **hepatopancreatic** colocalisation phenomena. These large organs have a direct impact on the eye, which is strongly connected to the **choroid**, a vascular mesenchymal tissue.

It is important to consider how the eye interacts in space. The great flexion of the embryo is linked to the appearance of the **optic vesicles**, which form due to growth and this flexion. This flexion is induced by the **primitive aortic vascular axis**, around which the vascular system coils.

07. FORMATION OF THE OPTIC PLACODE

DURATION : 04:28 STUDY SHEET

COURSE SUMMARY

In this video, we delve deeply into the formation of the optic placode, a key event in embryonic development. From the ectoblast and neural tube, the first optic anlage emerges, revealing the importance of cellular communication in various forms, including autocrine, paracrine, and juxtacrine, and its interaction with genetic phenomena such as sonic hedgehog (HH). We explore the formation of the optic vesicle and the transformation of the epiblast into the primitive optic placode, highlighting the influence of surrounding structures such as the notochord and the colobomic fissure in ocular morphogenesis. This content is essential for any student or therapist wishing to deepen their understanding of the embryological development of the eye.

FORMATION OF THE OPTIC PLACODE

In this section, we will explore the **formation of the optic placode** from the surface ectoblast and the neural tube.

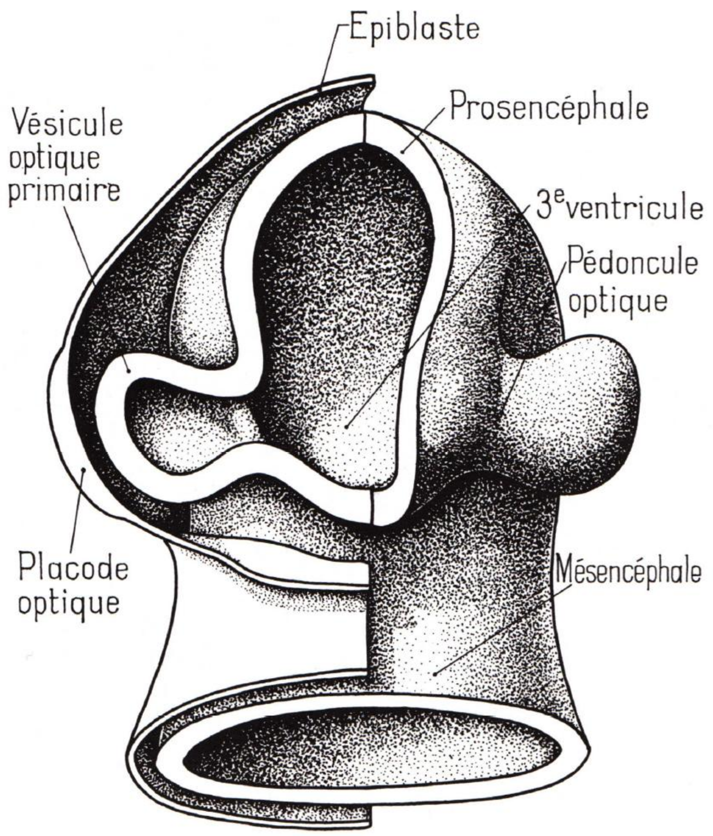


Fig. 1 b. — *Environ 27 jours* (4 mm).

FIG. — Neural tube and forming optical primordia

We first observe the **neural tube** internally, with the **neuropore** and the **cephalic groove** which is not yet closed. It is at this stage that the first **optic anlage** appears. This appearance is due to an **expansion** by points of support outwards.

TYPES OF CELL COMMUNICATION

It is essential to understand the different types of **cell communication**:

1. **Autocrine**: The cell informs itself directly.

2. **Paracrine**: The cell informs its neighbour through direct contact.

3. **Telocrine**: Long-distance communication, as in the case of **endocrine** hormones.

4. **Neurocrine**: Communication via neurotransmitters.

5. **Juxtacrine**: Communication between adjacent cells.

In the case of the optic placode, the information from the ectoblast is linked to a genetic phenomenon, notably **sonic hedgehog (HH)**, which is inhibited at the tip of the prechordal plate. This leads to **lateral expansion** through various genetic phenomena.

FORMATION OF THE OPTIC VESICLE

We observe the **primary optic vesicle** beginning to form. Two lateral vesicles appear and push on the part of the surface epiblast. A **juxtacrine metabolic reaction** occurs, leading to swelling which forms the **primitive optic placode**.

Initially, an epiblastic tissue appears between the **seventh and eighth day**. This tissue evolves into a **neural plate**, which, under the influence of the **notochord**, changes shape to become a **groove**. This groove closes to form a **neural tube**. The upper part disappears, leaving the **neural crest** and the surface epiblastic tissue.

AMNIOTIC FLUID AND CSF

The fluid present above the optic placode is the same as the fluid inside, i.e., the **amniotic fluid**, which will subsequently become the **cerebrospinal fluid (CSF)**.

The small lateral vesicles swell and, thanks to **juxtacrine** molecular contact, the epithelial tissue modifies to form the **optic placode**. This placode becomes a **fulcrum**, giving the impression of sinking in the drawing. This is due to growth from the outside, which creates a point of support.

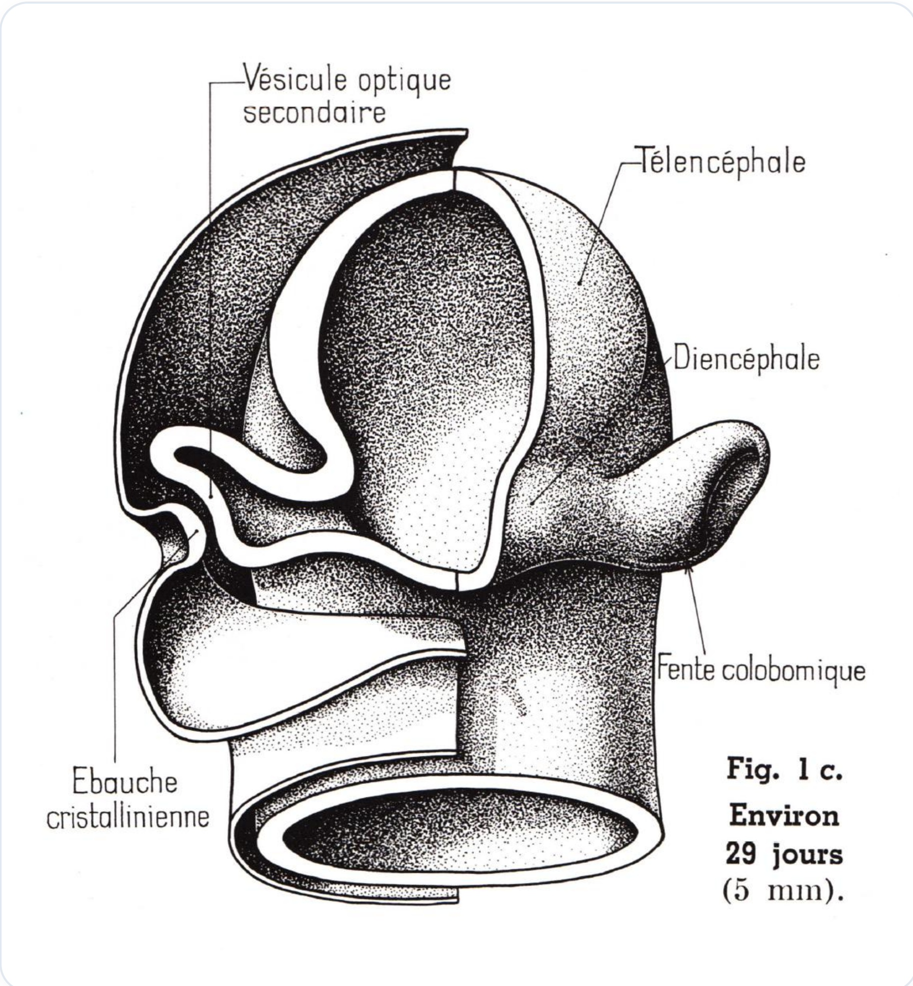


FIG. — Optical placode thickening and acting as a fulcrum

COLOBOMIC CLEFT

It is important to note that there is a **colobomic cleft** through which an artery passes, known as the **hyaloid artery**. This cleft plays a crucial role in the development of the optic placode and surrounding structures.

08. ORIGIN OF THE RETINA AND THE LENS

DURATION : 04:24 STUDY SHEET

COURSE SUMMARY

Examination of ocular embryology, from retinal evolution to the lens. Highlighting juxtacrine communications and interactions between the epiblast, CSF, and amniotic fluid. Study of the primary and secondary optic vesicle, and the transformation of hyaloid vessels into the retinal network (brain extension modulated by chemical gradients). Lens organogenesis is linked to the surface epithelium and the partitioning of ocular chambers.

ORIGIN OF THE RETINA AND LENS

The evolution of the eye begins with the observation of a **space** between the different structures. The **epiblast** attaches and changes shape, while inside, **cerebrospinal fluid (CSF)** and **amniotic fluid** are found, which have a common origin.

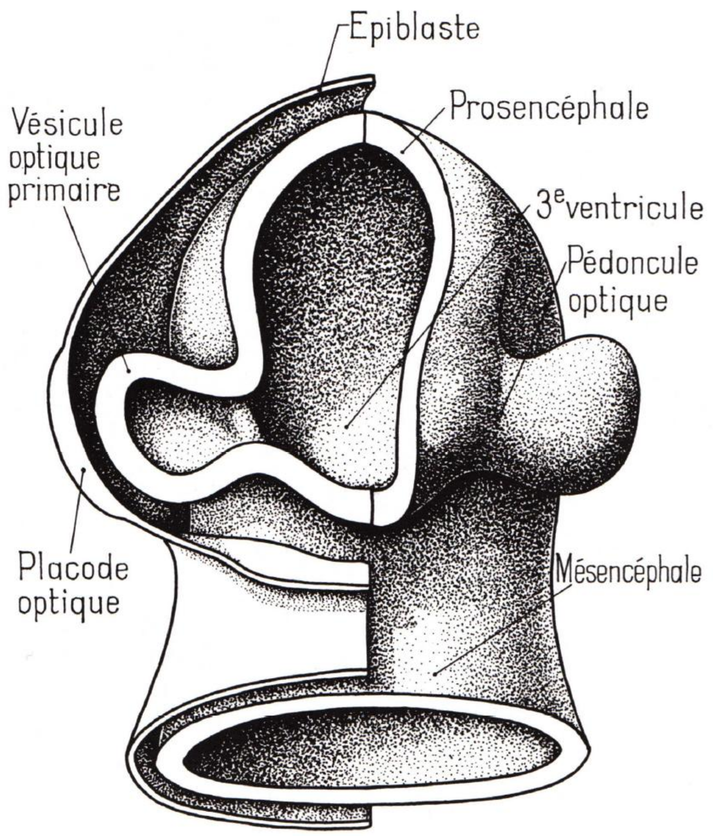


Fig. 1 b. — Environ 27 jours (4 mm).

FIG. — First stages of eye development

Juxtacrine communication is established, allowing for molecular connections. The path of a vessel is also observed, representing the first wave of the **choroid**, the vascular tissue of the eye. This invasion is crucial for the formation of an artery, involving various components such as the presence of **vesicles** and a **chemical gradient**.

The choroidal fissure allows the passage of **hyaloid vessels**, which will later transform into **retinal vessels**. The communication between the epiblast and this area gives rise to the **primary optic vesicle**, which evolves into the **secondary optic vesicle**. This tissue divides into the **inner layer** and **outer layer** of the retina, thus forming the **retinal space**.

Anatomists and embryologists consider the retina to be an extension of the **brain**, influenced by cerebrospinal fluid and information exuded on an intercellular plane. The **lens vesicle**, which will form the **lens**, also originates from the surface epiblast. This encounter occurs through juxtacrine communication.

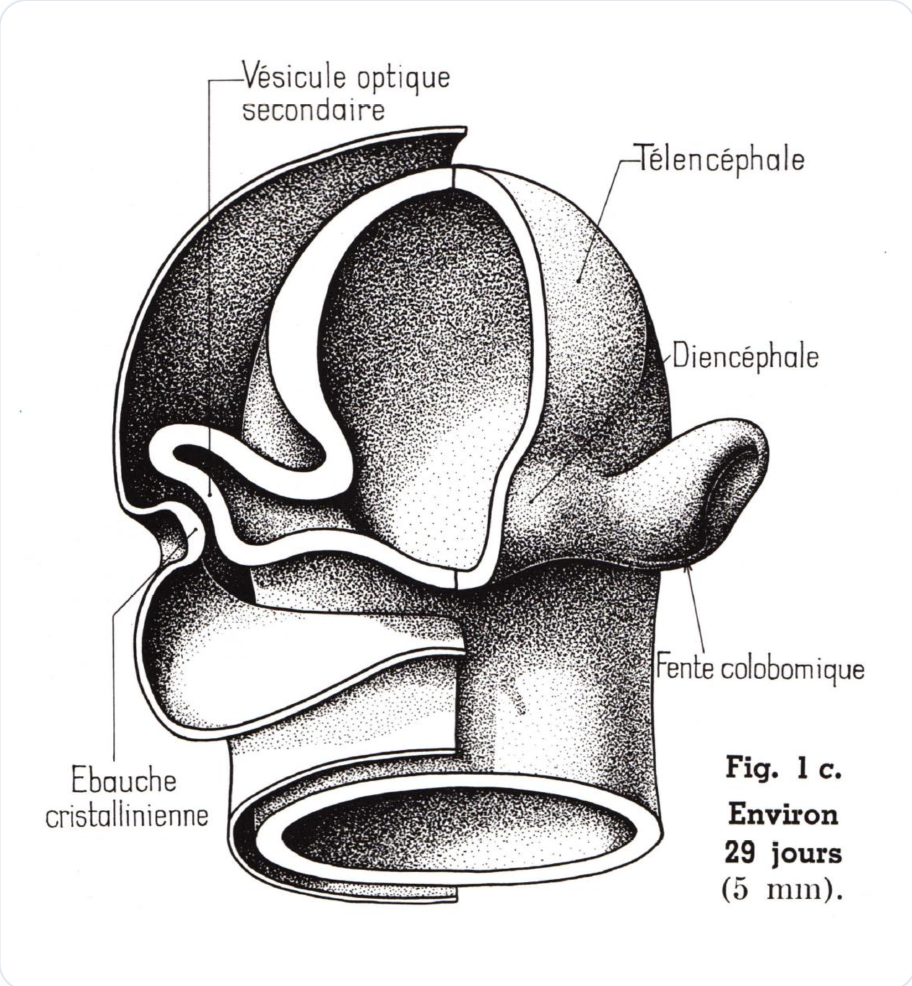


FIG. — Formation of the optic vesicle and retinal differentiation

The lens forms internally, while the surface epithelium remains in place, following a pattern similar to that of **neural tube** formation. The skin, on the surface, divides into different layers. The lens tissue constitutes the first layer of the eye.

The **conjunctiva** is the first membrane that covers the eye, also lining the future eyelids. Inside the lens, a space divides into two chambers: **anterior** and **posterior**. The retina, meanwhile, is extremely thin, similar to cigarette paper.

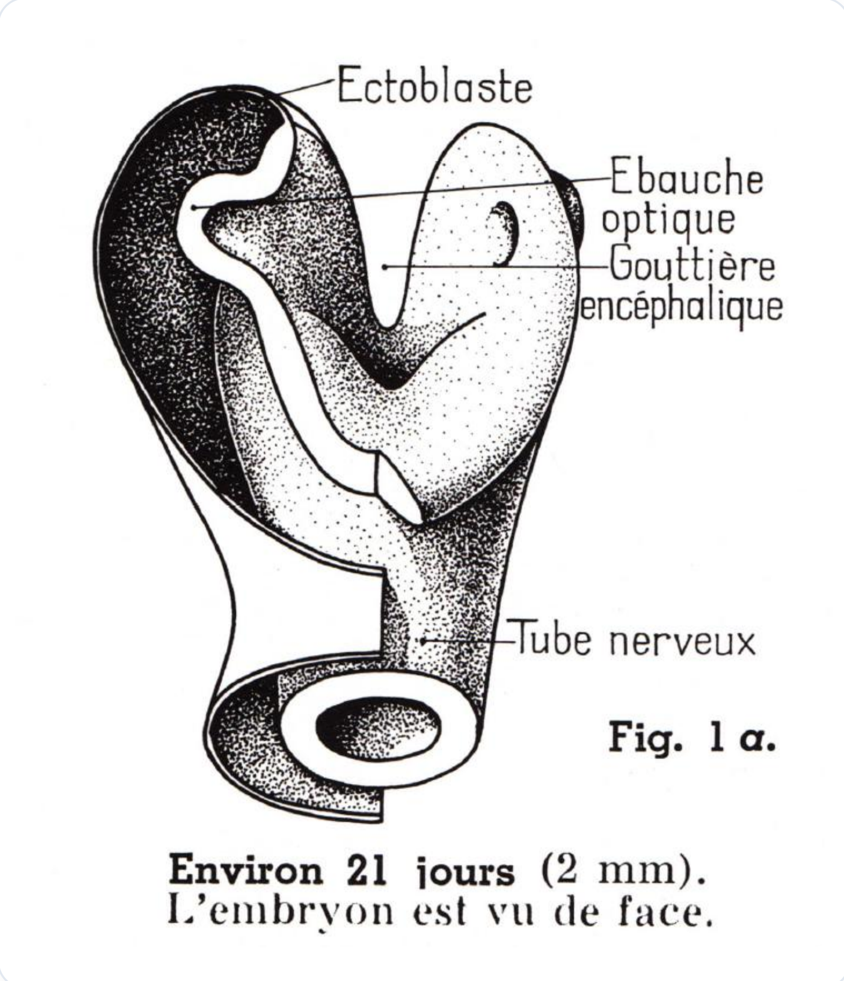


FIG. — Lens development and invagination

09. THE DIFFERENT LAYERS OF THE EYE 1

DURATION : 05:01 STUDY SHEET

COURSE SUMMARY

Analysis of ocular histology (sclera, choroid, ciliary processes, retina) and its functional specificities. Introduction to the principles of iridology as a diagnostic tissue approach. Detailed explanation of photoreceptor and pigment layer activity, illustrating the biochemical mechanisms of visual transduction.

THE DIFFERENT LAYERS OF THE EYE

The **eye** is made up of several layers, often described as superimposed socks.

1. THE SCLERA

The first layer, called the **sclera**, is continuous with the **cornea** at the front. The cornea is a **transparent** structure that allows light to enter the eye. The **limbus** and the white part of the eye, known as the **sclera**, make up this first sock.

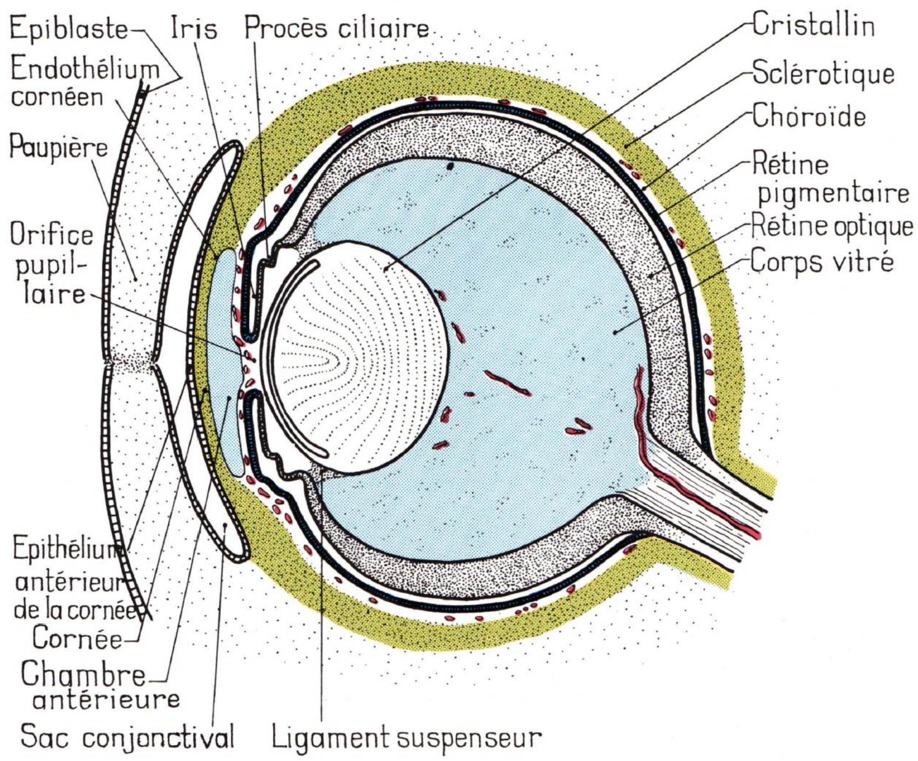


FIG. — General eye diagram

2. THE CHOROID

Upon removing the sclera, the **choroid** is revealed, an extremely **vascularised** layer. The choroid extends forwards with the **iris**. **Iridology** is a practice that involves reading health information through the iris, which can store various elements, such as **sulphur**.

3. THE CILIARY PROCESSES

The **ciliary processes** and the **suspensory muscles of the iris** play a crucial role in the **physiology** of the eye. These structures are essential for the **accommodation** system, allowing the **lens** to change shape so that the image is projected correctly onto the **retina**.

4. THE RETINA

The third sock is the **retina**, which is the deepest nervous tissue and represents an extension of the **brain**. It is formed of two layers: an inner layer and an outer layer. A **detached retina** occurs when the space between these two layers develops.

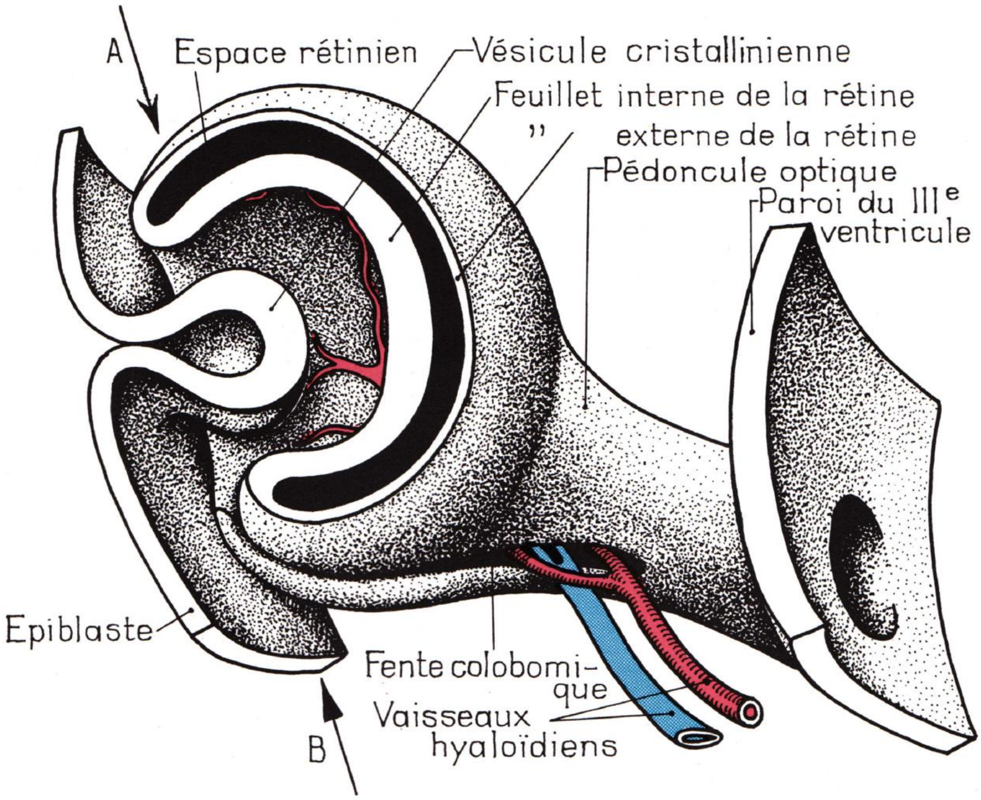


Fig. 3. — *Ebauche oculaire humaine.* Embryon d'environ 33 jours.

FIG. — Eye tunics, ciliary processes and retina

5. THE PIGMENT LAYER AND PHOTORECEPTORS

The retina also contains a **pigment layer** and **photoreceptors**, which are divided into **cones** and **rods**. These cells are essential for **transduction**, the process by which a photon is converted into electrical energy, thereby enabling visual perception.

CONCLUSION

These different layers of the eye work together to enable vision, each having a specific and crucial role in the functioning of this complex organ.

10. REMINDER OF THE DEVELOPMENTAL MOVEMENT

DURATION : 06:13 STUDY SHEET

COURSE SUMMARY

This video offers essential insights into developmental movement, focusing on skull integration and brain development, as well as their impact on ocular function. The concepts of cephalisation, cardialisation, and the importance of the palatine bone are examined to illustrate the synergy between structure and function. By highlighting the role of self-regulation mechanisms and embryonic force, the training provides a deep understanding of developmental mapping and anatomical axes. This leads to a new perspective on treating pathologies such as tendinitis, by considering electrical interactions and dynamic movements within the body's electromagnetic fields. The neurophysiological and therapeutic implications are also discussed, opening new avenues for practitioners.

DEVELOPMENTAL MOVEMENT RECALL

The integration of the **cranium** and **brain development** is essential for understanding developmental movement. The eye, which initially develops very laterally, progressively moves anteriorly. This process is part of an overall body development, involving mechanisms such as **cephalisation**, **cardialisation**, brain **ascensus**, and **visceral descent**.

This development is also influenced by the **palate**, which becomes a fulcrum for the eye on the cranial plane. Each bone of the cranium plays a crucial role in the proper functioning of the eye. Understanding this developmental movement allows us to grasp **functional force** and **therapeutic force**. The relationship between **structure**, **function**, and **form** is fundamental: form is the expression of structure and function. Good structure results in good form.

The body possesses **self-regulating mechanisms** that allow it to balance itself through **allostatic** and **homeostatic** processes. Allostasis is an internal process that helps maintain balance. The embryonic force, or healing force, uses the same energy as that present in the embryo to develop a form. Thus, embryological development and bodily regeneration share a common energy.

To know developmental processes is to understand the **mapping** of this development, which follows a specific organisation. For example, a limb first opens in a certain way before performing a more complex movement. This development occurs simultaneously with other movements in the body.

The importance of the brain is crucial, and it is necessary to revisit the **anatomical** and **structural axes**. Ultimately, all of this manifests in an **electrical field**. When treating tendinitis, it is relevant to think in electrical terms, as interventions on these cerebral axes involve electrical fields rather than simple structures.

The eye is rooted in a sphere of pure energy, seeking an electrical axis. This notion of **polarity** is integrated from the outset in the organism, with nurse and follicular cells, as well as specific metabolic concentrations. These electromolecular levels create dynamic movements.

When working on the eye, it is important to consider these electrical aspects. The axis of the eyes is rooted, and the movements of the roots illustrate how electromagnetic fields work. These movements are not linear, but rather molecular and ionic movements. The **transduction** of a photon into an electrical field involves changes in membrane permeability, which will be explored within the framework of **neurophysiology**, particularly with elements such as **rhodopsin** and **metarhodopsins**. These integration changes show how energetic information modifies membrane shape.

11. PRACTICE: V4 COMPRESSION AND ZONE B

DURATION : 07:04 STUDY SHEET

COURSE SUMMARY

This video explores the practice of fourth ventricle compression (CV4), in relation to the foramen of Magendie and the balance between intra- and extra-cerebral environments. Participants will learn how to work effectively with this technique to promote cerebrospinal fluid circulation and bodily harmony, integrating key concepts such as the attitude of neutrality and the importance of spacious attention. Through clinical examples and precise instructions, the session guides therapists in the art of compression, emphasising the importance of patience and the mind-body connection in the healing process.

PRACTICE: CV4 COMPRESSION AND ZONE B

During this session, we will address **fourth ventricle compression** (CV4). This technique is related to **Mangali**, a key concept representing the exploration of the **extra-encephalic** space.

The connection at the fourth ventricle is essential, as it is linked to the **choroid plexuses** that produce **cerebrospinal fluid**. At the back, there is an opening to the **intra-encephalic fluid** compartment, connecting the fourth ventricle to the third ventricle, the eyes, and the lateral ventricles. This passage is known as the **foramen of Mangali**.

The objective of this practice is to seek a **balance** between the intra- and extra-encephalic environments. The eye plays a crucial role in this balance, and we will work through the CV4 to achieve this state of harmony.

The compression should last at least **seven minutes**, allowing the body to **express itself** and release. At this stage, it is important to listen in order to perform an **EV4**, an expansion of the fourth ventricle towards **Zone B**, thereby filling the surrounding space.

It is essential to adopt an attitude of **neutrality** during this practice. Avoid trying to "do well," as this can lead to internal competition. Allow the space to create itself naturally, as it is in this encounter of **two neutrals** that healing occurs.

An illustrative example is that of Doctor **Jalous**, who treated Becker. After a few minutes of treatment, Becker expressed his impatience, but Jalous explained that the real treatment often begins after the manipulation ends. It is crucial to allow time for the process to settle and for the piece to settle.

To visualise the CV4, position yourself just below the **inion**, at the back of the **cerebellum**. Be careful not to compress the **carotids**; if they turn blue, the compression must be stopped. The thumbs should be slightly loose, and it is important to find a good **fulcrum**.

The CV4 requires the involvement of your entire body, not just your hands. Find a balance between front, back, left, and right, centring yourself on your own **centre of gravity**. Compression must be a conscious act, where your body and mind are at the service of your hands.

It is crucial not to compress with your gaze. Too much focused attention can lead to a loss of consciousness. Seek to maintain **spacious attention**, balancing your Zone B and your internal fluid.

When you feel a response, this indicates an **EV4**, an opening that represents the rebalancing between the extra-encephalic and intra-encephalic compartments, in connection with **Maljandi**.

12. IMPERMANENCE

DURATION : 09:08

STUDY SHEET

COURSE SUMMARY

This video offers a deep exploration of impermanence and its impact on our perception and emotions. The teaching emphasizes the importance of recognizing our own patterns through practices like meditation, teaching us to observe our thoughts and return to a state of neutrality. Furthermore, it addresses the functioning of the visual system and how our prejudices affect our perception of reality. By developing an understanding of the interdependence of experiences and working on our centre of gravity, we can align our vital energy and foster a health-oriented rather than disease-oriented approach. Learn to question your perceptions and free yourself from your false interpretations for a more conscious therapeutic journey.

IMPERMANENCE

Learning gradually to **broaden our perception** is essential. This involves living with our eyes a little more open and learning to spot our own **patterns**. Observe a pattern: if you can observe it, you can **clear** it. Often, we use others as cloths to clean our own windows.

Have you ever noticed that we project our patterns of **satisfaction** outside of ourselves, whether onto a subject or an object? Then, one day, we are disappointed. What's important is that, at some point, we resent the other person. Some go even further: after projecting too much outwards, they resent the whole world.

It is crucial to observe and identify our own patterns. One way to achieve this is to **meditate**. If you want to discover your mind, meditate. After a few seconds, thoughts will arise, and you will often be kidnapped by one of them. Meditation consists of observing these thoughts and returning to a state of **neutrality**. This gradually teaches you to discover your own patterns, which are not neutral.

What do you feel when you see? It is immediate and passive. You open your eyes and the world is there. Often, we think that **seeing** is **believing**. We believe because we have seen. However, what we see is not a universal truth. It is not an objective reality, because we have **prejudices** that influence our perception. Nature does not take these prejudices into account, and neither does the therapeutic process.

We create a **virtual reality** that we believe is unique. It is interesting to note that your visual cortex uses **30%** of your brain, compared to **8%** for touch and **2-3%** for hearing. Every second, your eyes send about **2 billion pieces of information** to your visual cortex, while the rest of your body sends only one billion. This means that sight represents a third of your brain volume, using two-thirds of your resources. It is therefore not surprising that sight is such a significant illusion.

The **Papez circuit**, including the colliculi and amygdaloid nuclei, shows that everything you see influences your **emotions**, and vice versa. This means that sight becomes a **mental composition**. For example, fear can distort reality. I had a traumatised child who had associated fear and danger after watching a film. By dissociating these two experiences, I was able to readjust his perception of reality.

Nothing exists by itself. When we understand this, our vision changes. Even the **ego** does not exist autonomously; it becomes an aggregate. Everything is **interdependent** and **impermanent**. Impermanence is what gives energy and movement, representing our **vital energy**, or **chi**.

To work with impermanence, one must be attentive to every moment, every thought, every detail. This involves learning to see beyond our projections and fears. Gradually, we must free ourselves from our **false interpretations**, which is not always easy. Sometimes, we want to remain fixed in our ideology.

As soon as you think you are in reality, you risk having false emotions. I suggest you harness your **inner strength** and readjust your ideas about luck and success. Having the right intention is paramount.

We will work with your **centre of gravity**. Vital energy is located in the abdomen; this is **ARA**, one of the important centres. This impermanence supports this vital energy. If you manage to adjust yourself with your own centre, it will be much more powerful.

We will explore three main types of energy and movements: **induced movement**, **permitted movement**, and **present movement**. Accepting one's strengths and weaknesses is essential, as is opening one's heart to many possibilities.

When people come with acute problems, I often ask them: "When you are healed, what will you be able to do?" This creates a new direction, because it is important to work with **health**, and not against illness. Directing our attention towards health is crucial.

13. PRIMITIVE CRANIO-SACRAL AXIS

DURATION : 03:34

STUDY SHEET

COURSE SUMMARY

In this video, we delve deeply into the primitive cranio-sacral axis, which emerges during the second week of embryonic development with the formation of the notochord. The teaching highlights the importance of this structure for the cranium and sacrum, particularly by emphasising how a free sacrum is essential for ensuring spinal functionality and flexion-extension of the cranial base. Furthermore, the video discusses the implications these embryonic dynamics have on urogenital pathologies, and how the release of certain emotional or energetic blockages between the sigmoid colon and the bladder can contribute to better overall balance. Students will learn to manipulate these concepts to improve their practice in osteopathy and biodynamic embryology.

PRIMITIVE CRANIO-SACRAL AXIS

During the second week of embryonic development, the **S-shape** of the epiblastic process begins to form the **notochord**. The **Hensen's node**, or primitive pit, becomes a fixed fulcrum and progressively moves posteriorly.

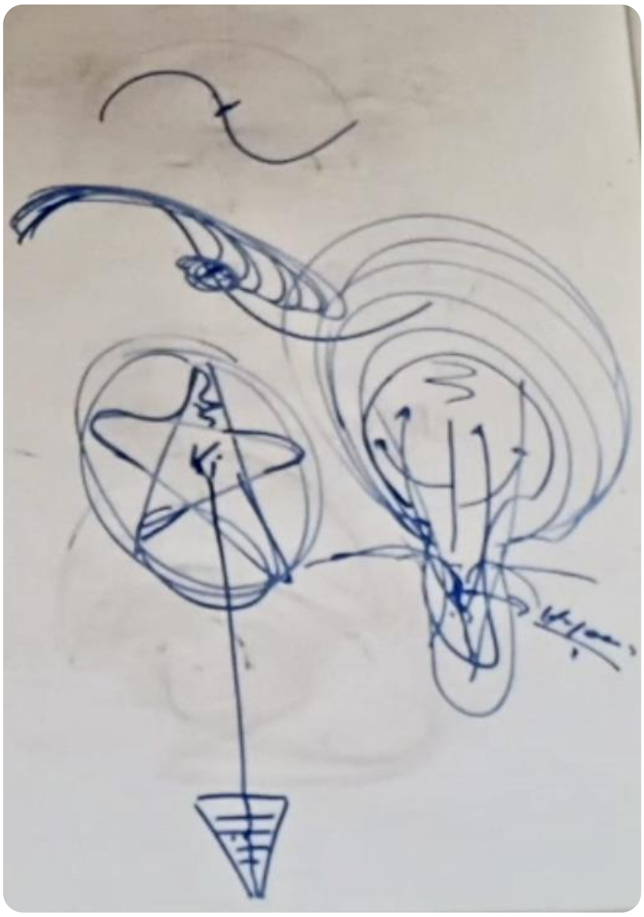


FIG. — Beginning of the epiblastic process

This structure is essential as it forms the base of the **cranium**, integrating elements such as the **pituitary gland**, the **greater wings of the sphenoid**, the **petrous system**, the **occipital system**, and the **ethmoid**. Inferiorly, the development of the **sacrum** begins, with the S2 process representing the vestige of the primitive streak. This process becomes a crucial fulcrum, highlighting the importance of a **free sacrum** to allow for functional unity.

In this developmental phase, the embryo takes on a racket-like shape, with the notochord and the primitive streak. This growth process generates a suction field to form the **mesoderm**, particularly the precardiac field. The anterior growth of the embryo draws material into this area, pulling **bottle cells** towards the centre. These cells release **hyaluronic acid**, which is essential at the mesenchymal level.

The notochordal axis is fundamental because it influences the freedom of movement of the sacrum. If the sacrum is blocked, the spine cannot function correctly, which affects the base of the cranium. A non-free cranial base prevents proper **flexion-extension**, leading to an imbalance between the **neurocranium** and the **viscerocranium**.

This dynamic is linked to the ascending force of the cranium, where the eye settles, and to the establishment of **mesodermal tissue**. Ectodermal growth pulls upwards, while the primitive node descends towards the sacrum. This creates an area of importance for protein information in relation to the mother.

Many blocked **ancestral images** are located between the sigmoid colon and the bladder, where it is crucial to release this information. Urogenital problems can be transmitted by this information, and the sacral downward migratory force plays a key role in this process.

14. DIFFERENT LAYERS OF THE EYE 2

DURATION : 05:25

STUDY SHEET

COURSE SUMMARY

This video deeply explores the different layers of the eye, notably Cloquet's canal, the aqueous humour, the vitreous humour, and the development of the lens from the ectoderm. Key concepts include the importance of invagination in lens formation and the distinction between the anterior and posterior parts of the eye, comprising the lens vesicle and the retina. By revealing the connections between ocular structures, the video highlights complex embryological processes that enrich therapists' practice and their understanding of ocular pathologies.

THE DIFFERENT LAYERS OF THE EYE

There remains a **fine canal** in the eye, called the **Cloquet's canal**. This canal is a remnant in the vitreous humour.

The **aqueous humour** and the **vitreous humour** are two types of humours present in the eye. The vitreous humour can be imagined as a **gelatinous water balloon** that floats in the eye, with a fine canal at its centre that connects directly to the **lens**. This canal is also linked to the **optic nerve** or the **hyaloid canal**.

The lens originates from the surface **ectoderm**. During the fifth week of development, an **invagination** forms a **lens vesicle** surrounded by **vascularised mesenchyme**. The placode invaginates to become the lens, a process similar to that of **neural tube** formation. This movement leaves a space inside, which becomes the lens.

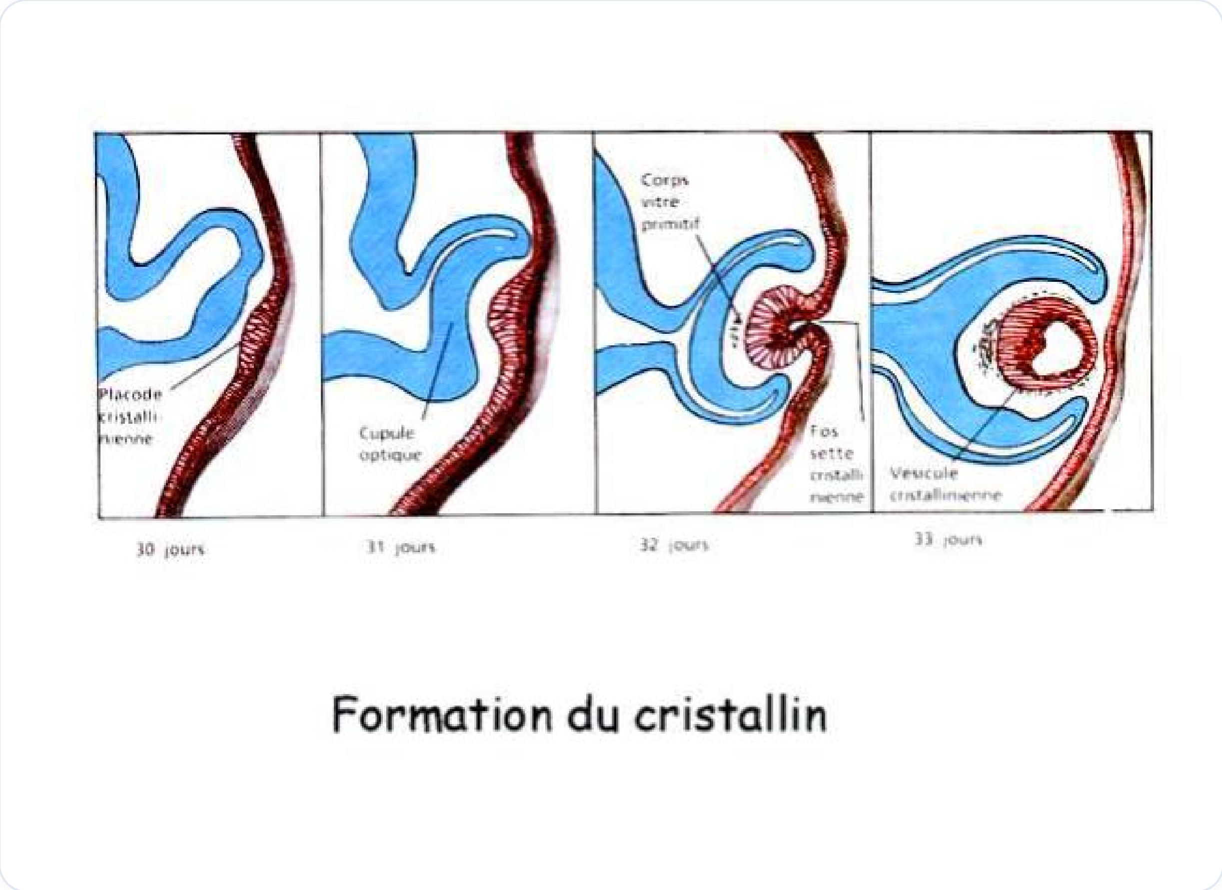


FIG. — Ectoderm invagination

The lens undergoes an evolution that forms **primitive lines**. It is essential to understand how it develops like a neural tube to become the lens of accommodation.

The **coloboma cleft** presents a line which, with the arteries, will give rise to Cloquet's canal. This canal closes with the **hyaloid artery** and **vein** which are located at the centre. The primary lens vesicle evolves to become a secondary lens vesicle with the placode.

On the outside, the layer formed is the **sclera**, which constitutes the first 'sock'. The **choroid** forms the second 'sock', connecting the lens. The eye can be divided into two parts: **anterior** and **posterior**. The anterior part is divided into the **anterior chamber** and the **posterior chamber**.

The **conjunctiva** and the **cornea** are continuous, with the cornea being transparent and continuous with the sclera. The choroid, also called **UV** (from the Latin *uva*, meaning grape), is a skin rich in blood vessels. It is divided into three parts: the choroid, the cornea at the front, and the **iris** at the back, which is a continuation of the choroid.

The **iris** is held in place by a **suspensory ligament** and **ciliary muscles**.

Finally, the **retina** is the innermost layer of the eye. It originates from the first vesicle, the **primary optic vesicle**, which develops by forming a cup-shaped structure. The retina is composed of two layers, inner and outer, and it is here that **retinal detachment** occurs.

15. NEURAL CRESTS

DURATION : 03:29 STUDY SHEET

COURSE SUMMARY

This video focuses on neural crest movement, a key process in embryology that gives rise to various bodily structures after neurulation. It explores types of movement, including prosencephalic, mesencephalic, and rhombencephalic, and explains how the neural crest challenges our understanding of genetic and metabolic interactions during embryonic development. By learning to identify signs in body structures, particularly in the face, therapists can better interpret patients' internal states. The video also emphasises the importance of cell migrations and environmental influences in tissue formation, highlighting the role of micro-vessels and micro-nerves, to provide an integrated view of biodynamic embryology.

NEURAL CREST MOVEMENT

The **neural crest movement** is a complex process involving different types of **differentiation** and **induction factors**. These factors can be **metabolic** or **passive** in nature, and intervene after **neurulation**.

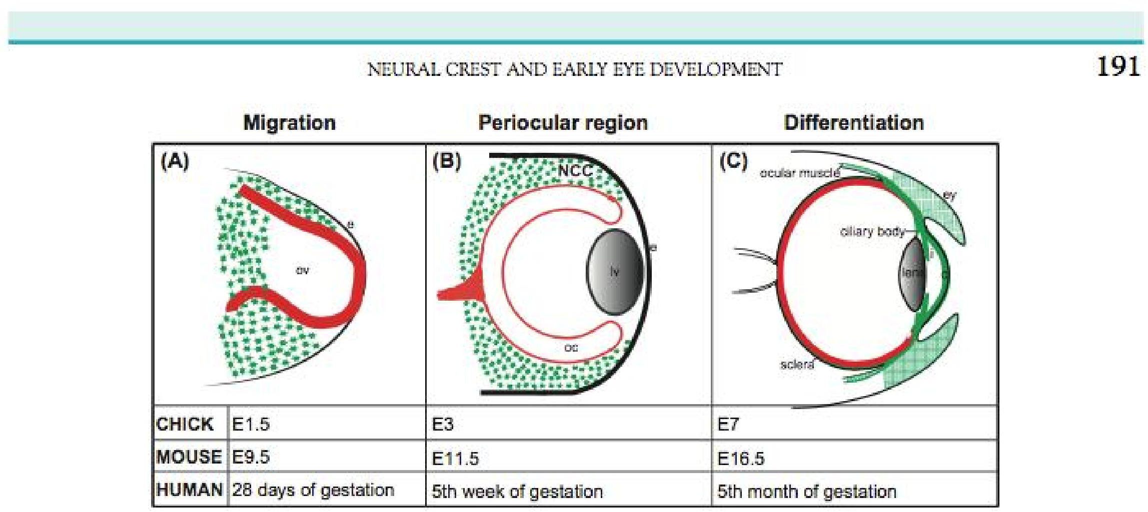


FIG. — Neural crest movement

Blood vessels play a crucial role in guiding the system, while **genetic** elements also influence the orientation of cells towards the appropriate areas.

TYPES OF MOVEMENTS

There are three main types of movements:

- **Prosencephalic**
- **Mesencephalic**
- **Rhombencephalic**

During the **closure of the neural tube**, the cells that will form the neural crest emerge. These cells will give rise to various processes in the body, creating a **flow** that extends from back to front, invading the entire **face**. This means that the face, bones, skin, down to the level of the **platysma**, originate from the neural crest.

This tissue is highly **informed** and **informative**. For example, everything in your abdomen also originates from the neural crest. **Ganglia** and other structures reflect the internal state of the body, allowing one to learn to **read the face** through wrinkles, dark circles, and colours.

MIGRATION AND DEVELOPMENT

The **prosencephalon** develops into the **telencephalon** and **diencephalon**, with significant migration of the diencephalon. The latter, as well as the **diencephalon** and the anterior **mesencephalon**, migrate towards the neural crest. This flow process is essential for development.

Before the closure of the neural tube, negative influences, such as those related to the **hepatopancreatic**, **thyroid**, and **cardiac systems**, provide information to the neural crest, which is also considered **ectomesenchyme**. The latter is a mixture of **mesoderm** and neural crest.

IMPORTANCE OF THE NEURAL CREST

The neural crest plays a fundamental role in **migration** and **colonisation**, contributing to specific characteristics. It also influences the construction of **micro-vessels** and **micro-nerves**, particularly **pericytes**.

This process allows for the creation of support and emergence points, providing an **ambient embryonic force** that restructures growing micro-nerves. It is important to note that these interactions occur at an **electrical** level.

At the level of the eye, a green area encompasses and shapes the anterior tissue, which is **ectoderm**, in relation to this mesodermal flow, particularly around the **sclera**.

16. THE EYE AND MEMORIES

DURATION : 05:02 STUDY SHEET

COURSE SUMMARY

In this video, we delve into the concepts of neuronal and tissue memorisation, detailing how our body records experiences and the consequences for physical health when these memories remain unintegrated. Through concrete examples, such as the emotional impacts on children, we explore the crucial role of glands like the adrenals and thyroid in this process. Therapeutic techniques such as Eye Movement Desensitisation and Reprocessing (EMDR) and heart coherence are presented as tools to treat these buried memories and restore balance between the cognitively discordant dimensions of our existence.

THE EYE AND MEMORIES

Neuronal memorisation and **tissue memorisation** are essential concepts in understanding how our body records experiences. It is possible to store **memories** in different parts of the body, influenced by the eye. If these memories are not integrated, they can **somatise**. In other words, what is not integrated manifests physically.

Each time a problem remains unresolved, the body symbolically memorises it, according to various **spaces**: mineral, plant, animal, or human. This memorisation also depends on **timings**, **coordination** and **synchronicity** processes. For example, a three-year-old child, left alone at home, may experience panic and develop physical symptoms such as chronic sinusitis. This phenomenon illustrates how an emotional problem can translate into physical symptoms.

The **adrenal glands** play a crucial role in this dynamic. They react to stress and, if blocked, this can lead to health problems. It is important to release the adrenal glands to treat symptoms. Similarly, the **thyroid** is linked to vocal expression and communication, especially around the age of seven, when the child begins to assert themselves.

At the age of twelve, the **pituitary** phenomenon manifests, marking the onset of puberty and the maturation of the **gonads**. This hormonal process is also linked to spaces of **synchronicity**.

There are people who can be stuck in time, reliving childhood experiences. These memories can be reactivated by visual or auditory stimuli. To treat these memories, techniques such as **Eye Movement Desensitisation and Reprocessing (EMDR)** are used. This method does not remove the trauma but allows for dissociation and reorganisation of memories.

Eye movements help to restore balance between the **rational** and the **irrational**, as well as between the **intuitive** and the **rational**. An imbalance in these areas can lead to pituitary problems. The pituitary gland plays a key role in transmitting information between different areas of the brain, particularly during eye movements.

Dreaming and eye movements are therefore essential for the **rationalisation** and reorganisation of lived experiences. Practices such as **heart coherence** can also contribute to this balance.

17. THE EYE AND THE CORTICALISATION MOVEMENT

DURATION : 08:22 STUDY SHEET

COURSE SUMMARY

This video explores the dynamic link between the eye and the corticalisation movement in brain development. The process of cephalic and cervical flexion is examined, as well as the reorganisation of brain tissues in relation to facial and visceral structures. The eye is presented as a central player in this movement, influencing the formation of the falx cerebri and the tentorium cerebelli, while integrating psychic and emotional aspects through its link with the hormonal system. The concepts of the diaphragm of equilibrium and the terminal fulcrum are introduced, highlighting the importance of stabilising the cruciform suture to rebalance the eyes and promote bodily harmony. This lesson offers practical and theoretical keys for practitioners wishing to better understand embryological development and its therapeutic implications.

THE EYE AND THE CORTICALISATION MOVEMENT

The **brain**, initially perceived as a flat tissue, undergoes a significant transformation during its development. Imagine an individual lying on a table, with their brain in hand. The first piece of information to remember is that of **expansion** and **neural tube formation**.

This process begins with a **cephalic flexion movement**, followed by **cervical flexion**, then **pontine** development and a **telencephalic** process. As the brain straightens, the smaller structures follow this developmental movement, indicating a change in orientation.

The **eye**, in its developmental spiral, contains the information of **corticalisation**. During this straightening, all the tissue reorganises in a rotational movement, moving closer to the **midline**. Below, the tissue pulls towards the heart and digestive system, while above, it straightens. This movement is created by the **ascension of the brain** and the **descent of the cardio-visceral**.

The eye plays a crucial role in this phenomenon of **corticalisation**, contributing to the development of the **falx cerebri** and the **tentorium cerebelli**, which inserts onto the clinoid processes. Nearby is the **optic chiasm**, an essential element to consider.

Another important aspect is the **diaphragm of balance** and **Tenon's capsule**, which connects the eye to the entire facial system and all the body's fascia. This means that the eye constitutes an integral **unit**, containing all facial information. It also receives emotional and psychic information, conveyed by hormonal substances. Thus, the psychic state is intrinsically linked to a hormonal and emotional state.

Treating the eye therefore means addressing **peritoneal**, **peripheral**, and **cortical** development simultaneously. The eye is related to the totality of the body in its expression and development. Each organ has its own point of view, and each organ is a symbol.

Around the **26th day**, an **optic sulcus** appears, linked to an internal connection. This sulcus is stimulated by a vesicle that touches the superficial wall. Cephalic and cervical flexion lead to a meeting between the brain and the heart, with the eye settling on the heart. This creates a **posterior fold**, known as the **pontine flexure**, where the sacrum and the tentorium cerebelli are located.

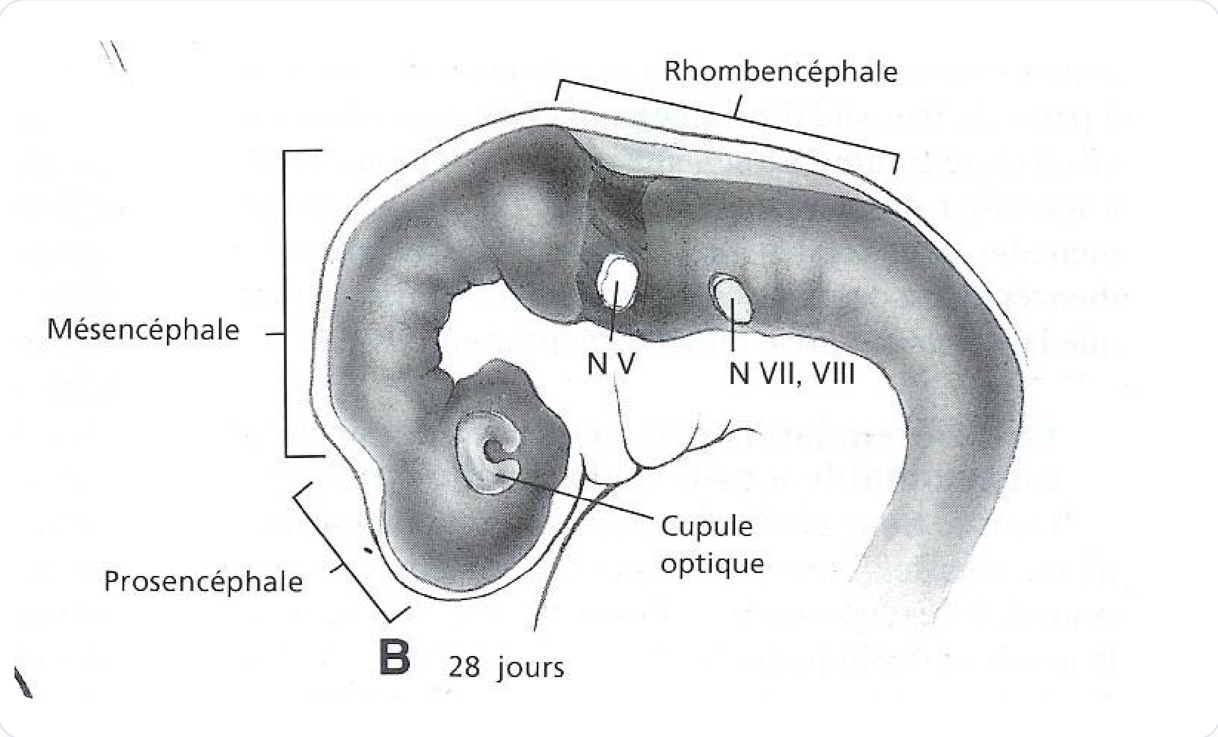


FIG. — Optic sulcus and vesicle

The **tentorium cerebelli** is paramount, as it establishes synchronicity with the peritoneum and the brain. The **submesencephalic flexure** is related to the tip of the **notochord**, serving as a fulcrum. The **supraocciput** is also linked to cervical flexion, while the pontine flexure is associated with the occipital base.

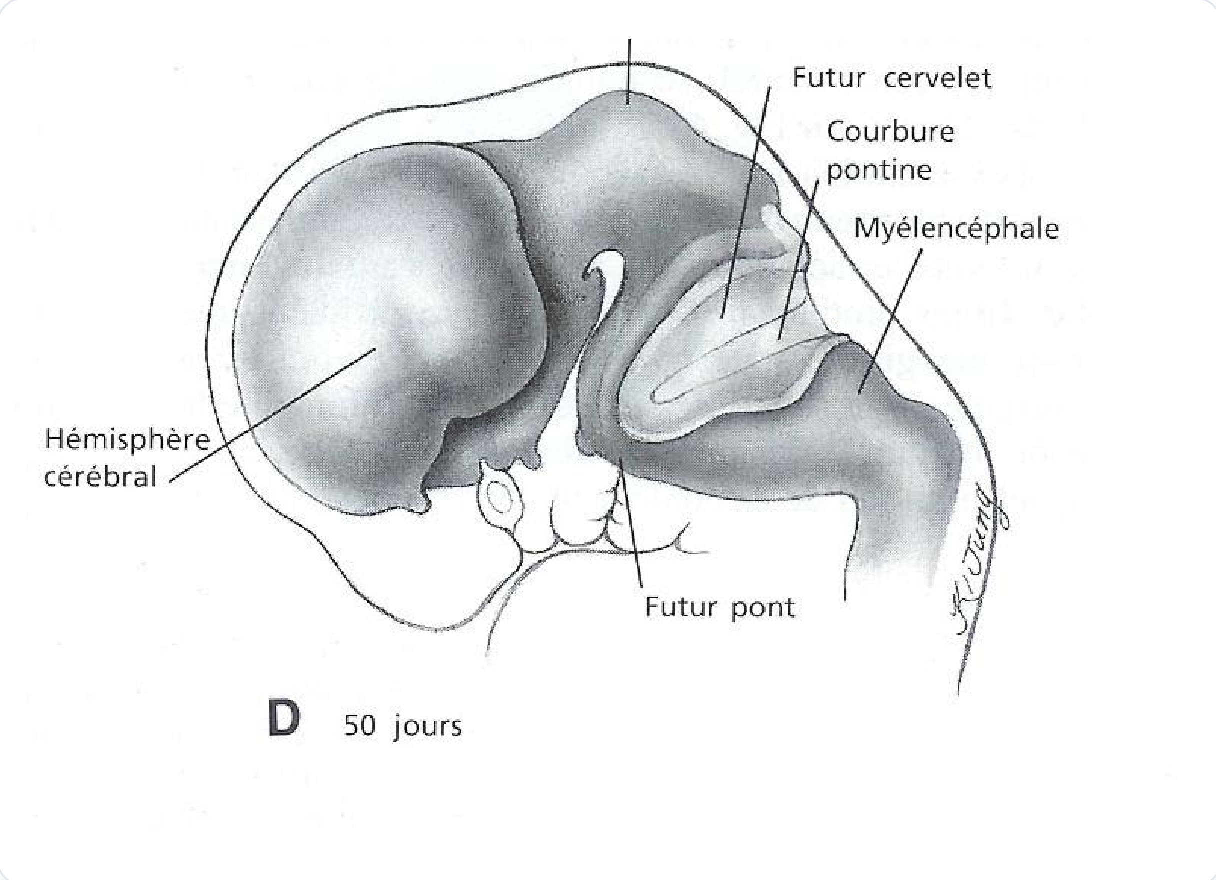


FIG. — Meeting of the brain and heart

The ascension of the neural tube influences the development of the digestive and cardiac systems, as well as other developmental spaces. **Colocalities** and **synchronicities** during morphogenesis help to understand the fulcrums. The ascension of the **ectoderm** and the descent of the **endoderm** are essential for brain integration.

The eye follows this developmental phase, moving closer to the midline to form the **crista galli**, which brings the membranes together to create the falx cerebri. The tentorium cerebelli also develops during the pontine phenomenon.

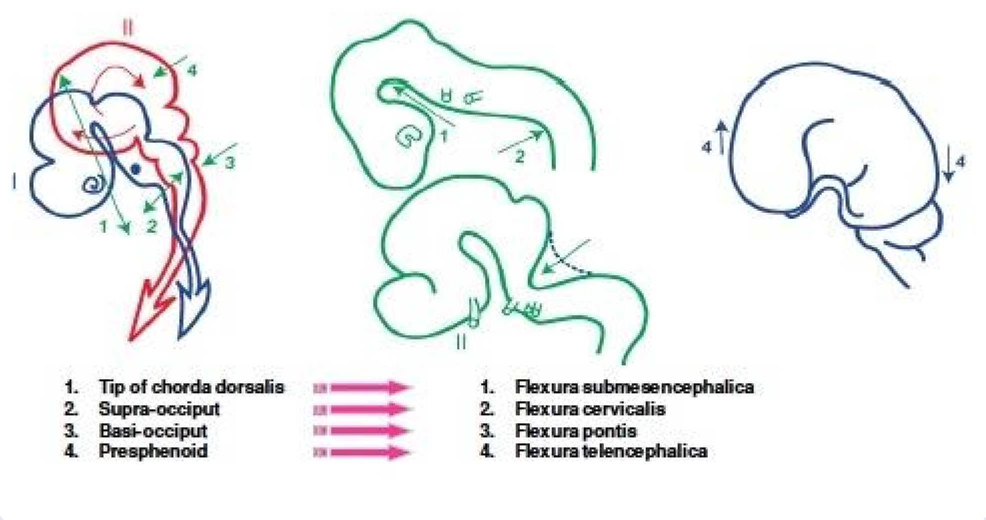


FIG. — Support points

It is crucial to note that the fulcrum, or **terminal fulcrum**, for the placement of the eyes is essential. To rebalance the eyes, it is recommended to place a finger in the mouth, at the level of the small **cruciform suture**. This allows both eyes to find their space when this suture stabilises.

The **frontal margin**, the **zygomatic margin**, and the **maxillary margin** are important fulcrums. The ascent and descent of the structures contribute to the closure of the midline, which is also linked to the closure of the palate. When the palate finds its fulcrum, the eyes also find their place, marking an **arrival fulcrum** in development.

18. NOTE: WHIPLASH

DURATION : 01:12 STUDY SHEET

COURSE SUMMARY

This video explores the impact of ocular tensions on cranial and emotional development. It highlights the concept of ascensus as a method to rebalance eye-related tensions, while revealing the links between the state of the eyeballs, emotions, and mental health. You will learn how these tensions can reflect imbalances in the nervous system and how they can be linked to physical manifestations, such as torsions and dissociations. By studying these aspects, this video paves the way for a better understanding of cerebral and cranial development, thus offering a valuable approach for therapists and students in osteopathy and biodynamic embryology.

OCULAR TENSIONS AND THEIR IMPACT ON CRANIAL DEVELOPMENT

The **brain** can be affected by different types of **tensions** that manifest in the **eyes**. These tensions can be rebalanced through a process of **ascensus**. While this may seem complex, it is important to understand that our memory and emotions are expressed through our body, particularly in **Zone B**.

When discussing **ascensus** and **descensus**, it is essential to note that some individuals present with deeply sunken **eyeballs**, which can be associated with depressive states or poor connection between the left and right hemispheres of the brain. These manifestations can also be linked to significant **torsions** and **dissociations**.

The eyes play an exceptional role in expressing our emotional and physical state. They reflect the position and state of our **nervous system**. Furthermore, there is a significant relationship between the development of the **cranium** and that of the eyes.

Thus, by studying the eye, we can better understand brain development and, by extension, cranial development.

19. LINES OF FORCE AND ELECTRICAL

DURATION : 03:44

STUDY SHEET

COURSE SUMMARY

This video explores the impact of lines of force and electrical charges on bone structure and movement. The instructors explain how the eye, as a shaper, plays a crucial role in bone formation, while also emphasising the counter-movements and cephalisation that occur in response to electromagnetic charges. Beam complexes, including the canine-nasal frontal beam and the maxillary frontal line, are presented with relevant anatomical details. Furthermore, the importance of sensing frequency variations and the balance between the left and right sides of the body is highlighted, paving the way for effective adjustments to treat micro-lesions and improve body dynamics. Practitioners will also learn to assess the impact of ocular axes on overall body balance, in order to optimise their therapeutic interventions.

LINES OF FORCE AND ELECTRICAL

The **lines of force** are formed by the charging of **electrons** on the convex side at the positive level, thus creating charges at the bone level. This reverberates into deep lines of force, called **beams**, which play a crucial role in bone structure.

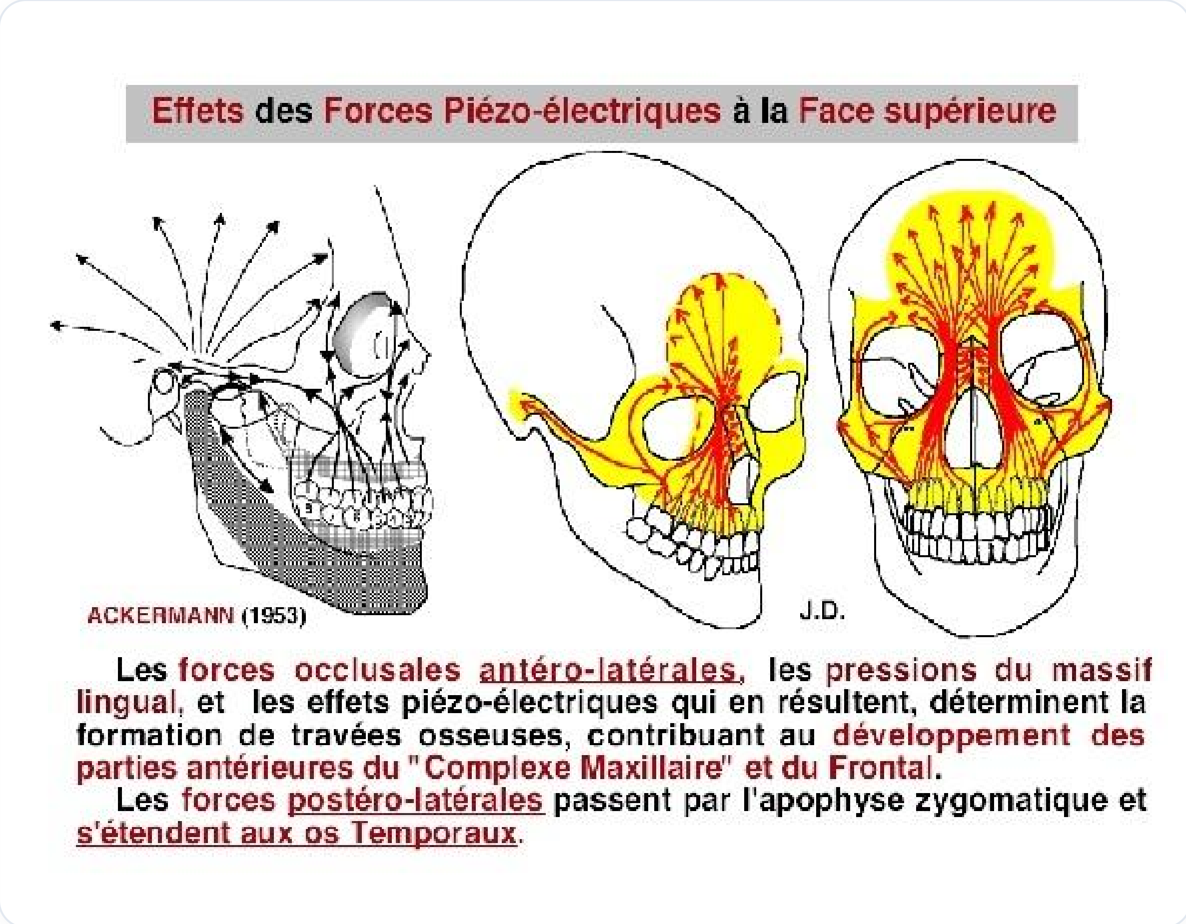


FIG. — Electrical loading of bone

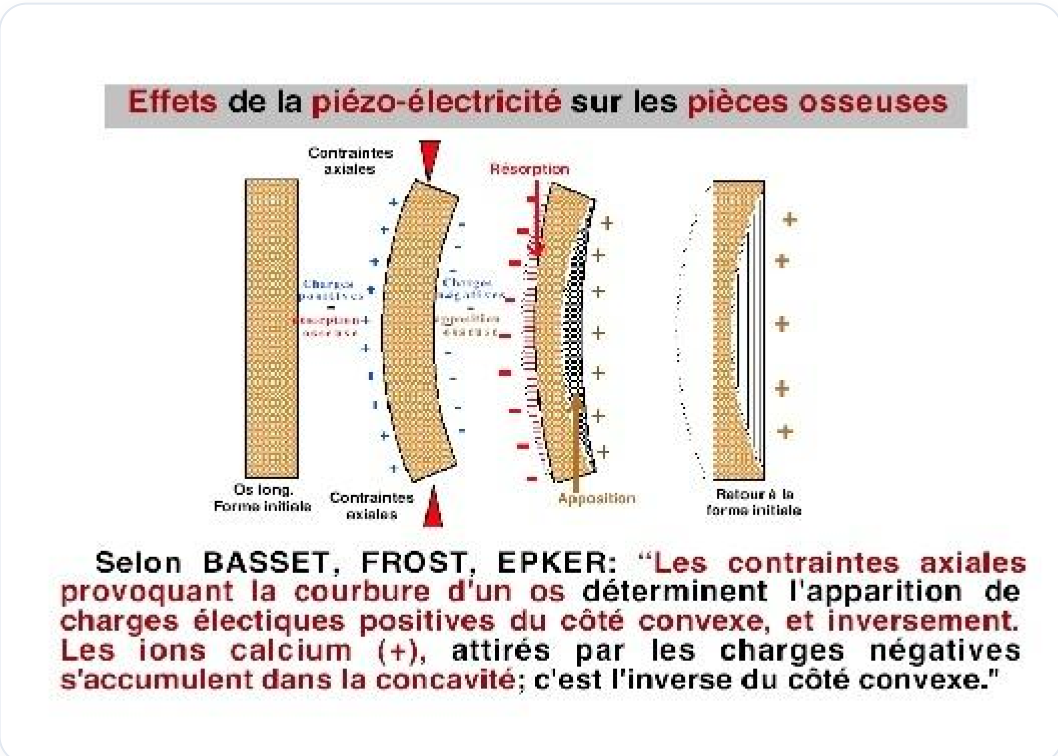


FIG. — Creation of charges on bone

The eye acts as a **conformator**, influencing the formation of the bone that builds around it. In this downward movement, the bone performs a **counter-movement**. The latter is associated with **cephalisation**, where the bone exhibits a **counter-rotation** movement. A particularly interesting beam is the **canine-nasal frontal beam**, which develops contralaterally.

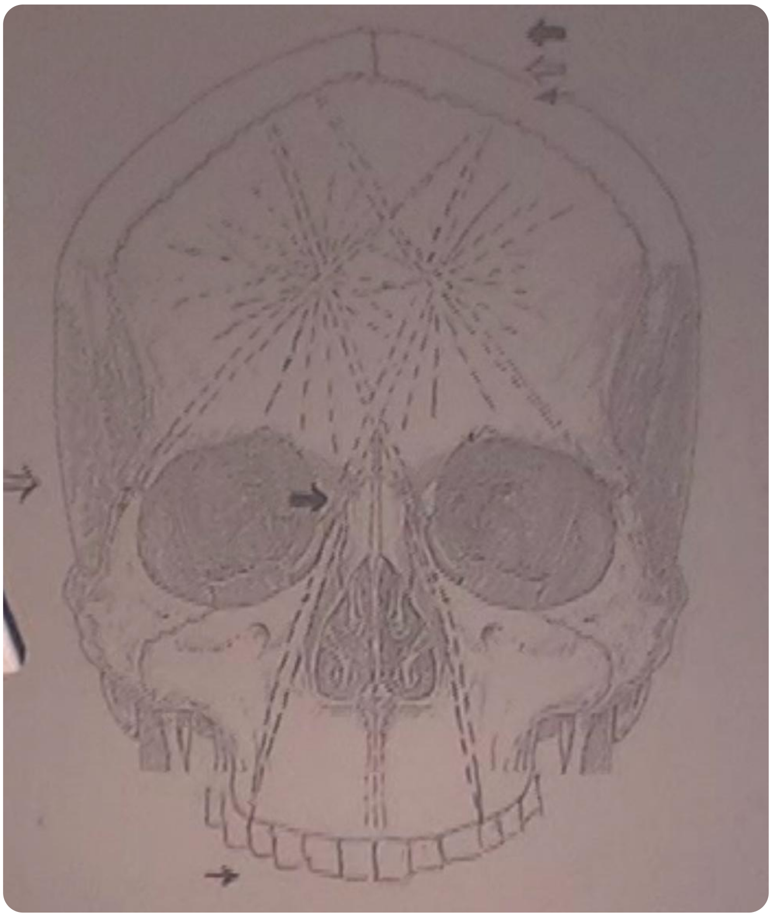


FIG. — Lignes de force (poutres)

At the base of the skull, lines of force are also present, representing the major axes of this region. On the facial plane, a line of force extends from the canine to the **frontal boss** and the **coronal suture**. Sometimes small **micro-lesions** form at the coronal suture, leading to noticeable changes. For example, a tooth may show a change in bevel, and an adjustment can cause a significant **electrical change**.

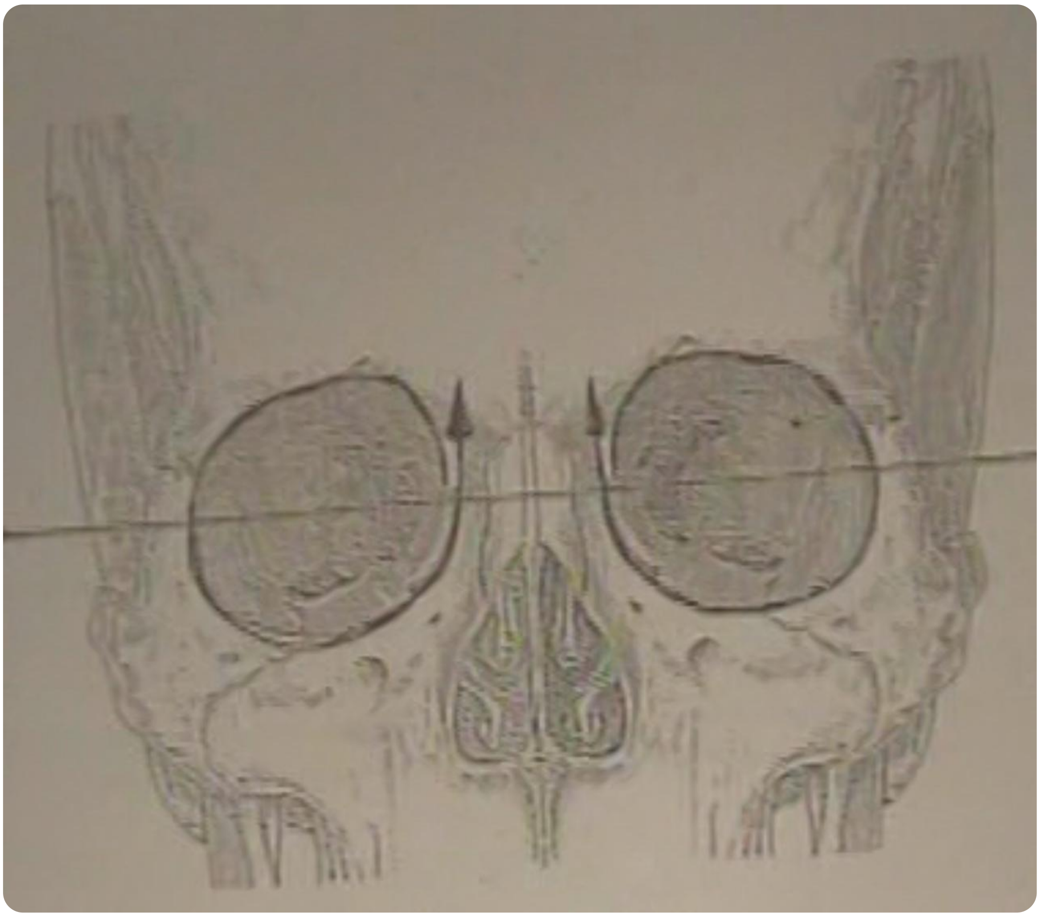


FIG. — Frontal canino-nasal beam

It is essential to work on the **frequency** of these lines of force, feeling the variations from left to right to try and restore balance. Understanding **synchronicities** is also crucial, as what is worked on at one level can have repercussions at another.

Another interesting line of force is the **maxillary frontal line**, which passes at the level of the ingus. In addition, the **zygomatic line** is noteworthy, connecting the same side at the frontal level and the opposite at the coronal suture.

Finally, at the level of the eye and orbit, it is important to add the axes of the eye to assess their balance. Changes can occur, requiring particular attention to these structures.

20. THE EYE AND CRANIAL DEVELOPMENT

DURATION : 04:35 STUDY SHEET

COURSE SUMMARY

In this video, we explore in detail the complex links between eye development and skull formation. The focus is on embryological processes, such as primitive cephalic flexion and their impact on bone structure, particularly the central role of the sphenoid in eye stability. The concepts of cephalic growth and orbital spasm are discussed, highlighting the importance of maintaining harmony between the eye, nose, and palate formation. In addition, the video illustrates the developmental spirals that influence cranial structure and their repercussions on visual function, thus providing practical tools for students and therapists wishing to deepen their understanding of osteopathy and biodynamic embryology.

THE EYE AND CRANIAL DEVELOPMENT

The development of the **eye** is intimately linked to the formation of the **cranium**. During **primitive cephalic flexion**, the **sub-mesencephalic mesenchyme** is compressed between the anterior and posterior parts of the **cerebral vesicle**. This process leads to water loss and the transformation of the mesenchyme into **cartilaginous tissue**, thus forming the base of the skull.

At the top, we find the **ethmoid**, followed by the **body of the sphenoid**, and then the **basilar part of the occipital bone**. The **wings of the sphenoid** play a crucial role as a fulcrum for the eye. The latter is supported by the sphenoid, which ensures its **stability**. As the cranium grows, the bony tissue of the eye develops in response to the growth of the **brain**, engramming itself around the eye and closing with specific lines of force.

It is essential to understand that the eye and the cranium are interconnected. A slight impact can destabilise the **orbit** and affect visual **reflection**. Thus, it is crucial to restore the stability of the eye, both through the brain and the cranium.

As the nose develops, the eye appears on the side. Simultaneously, the **palate** forms. The closure of the **palatine** coincides with the alignment of the eye in its vertical and horizontal axes, marking an important terminal point in bone development.

Cephalic growth is paramount. It is at this moment that the eyes move closer to the **midline**. A fulcrum forms at the **nasion**. As cortical development progresses, the eye orientates into its final position.

Two spirals are observed: an eye spiral and a bony counter-spiral, visible in the shape of the cranial **sutures**. The orientation of these sutures is essential for studying the three-dimensionality of the **premaxilla**, which has a complex shape, contributing to the orbital and ocular structure.

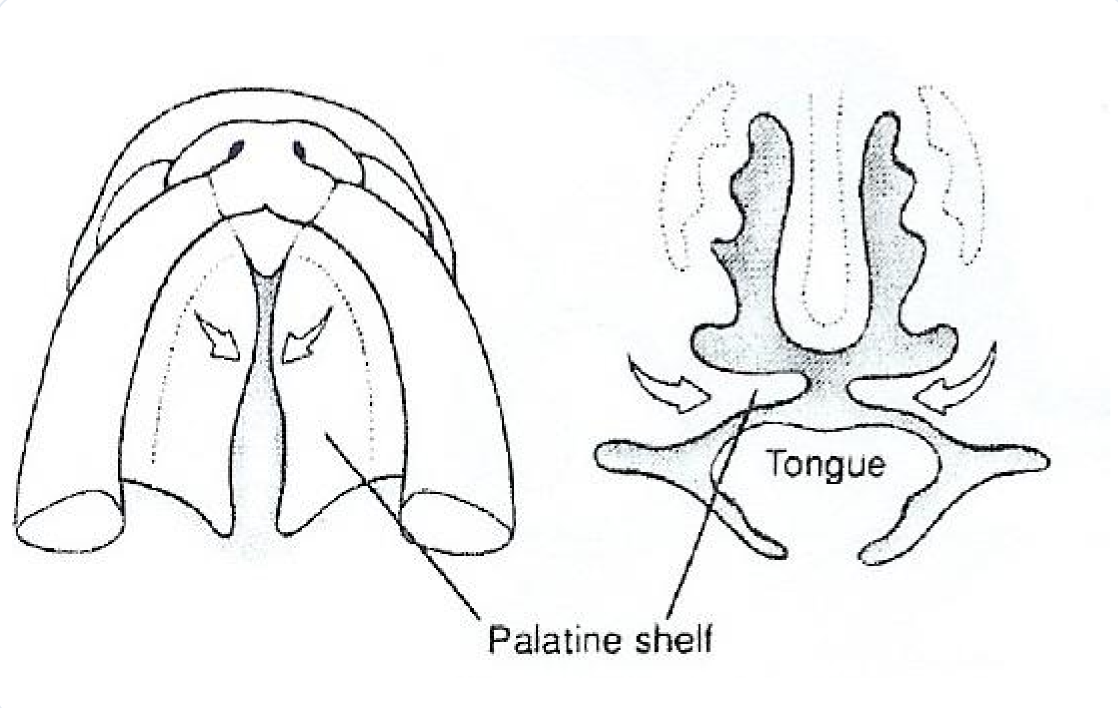


FIG. — Explanatory diagram

21. LIGAMENT OF LILIEQUIST

DURATION : 07:37 STUDY SHEET

COURSE SUMMARY

This video explores the anatomy and mechanisms of the Liliequist ligament, an essential structure that acts as a diaphragm at the level of the third ventricle. It explains how this ligament interacts with the oculomotor nerve and the basilar trunk, thus influencing the ventricular system and the pituitary gland. The concepts of membranous tensions, cranial movement axes, and their impact on the facial and hormonal systems are discussed, while highlighting the importance of cranio-sternal-sacral and mandibular movements. The analysis of tensions can also help diagnose occlusal dysfunctions, offering therapeutic keys to restore bodily harmony.

THE LIGAMENT OF LILIEQUIST: ANATOMY AND MECHANISMS

The **ligament of Liliequist** is an anatomical structure that acts as a small diaphragm, commonly known as the **membrane of Liliequist**. This ligament inserts at the level of the **third ventricle** and descends to the **sphenoid**. Here, it plays a crucial role in relation to the **oculomotor nerve** and arterial passage, particularly the **basilar trunk**, which is responsible for the formation of the **cerebral arteries**.

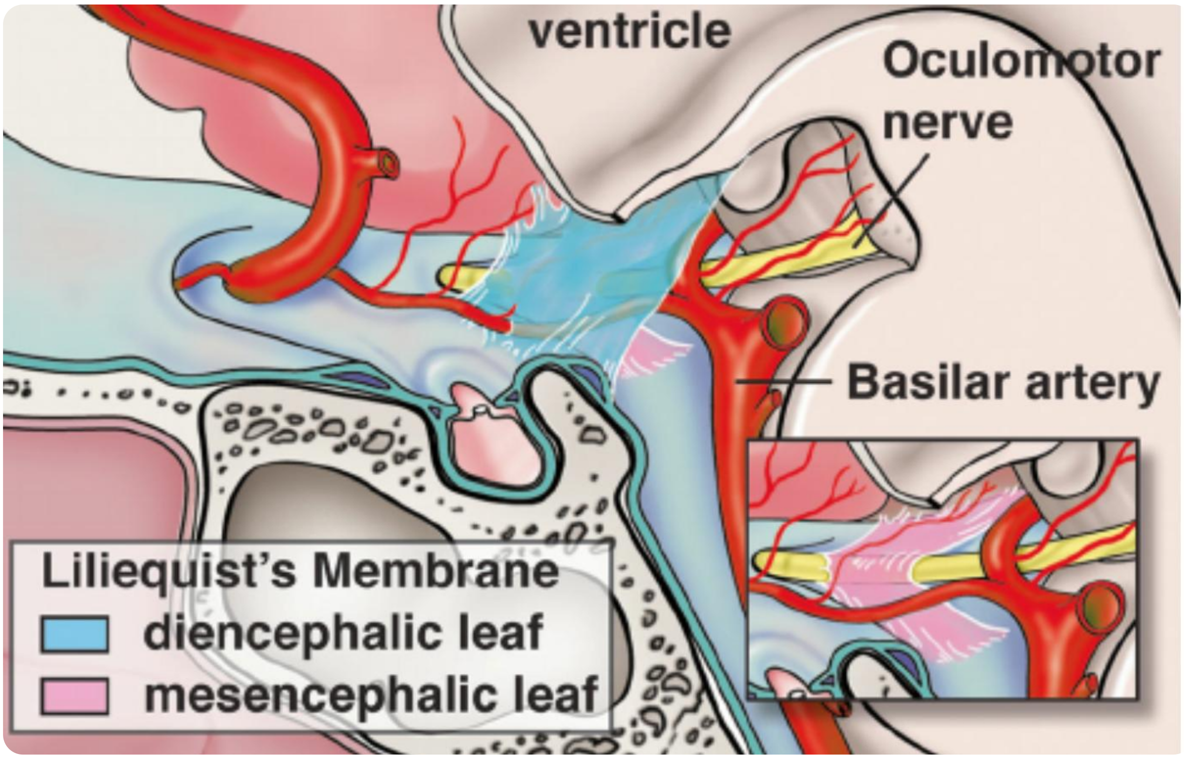


FIG. — Liliequist's ligament

As the ligament descends, it creates an **axis of tension** that influences the **ventricular system**. The third ventricle, the fourth ventricle, as well as the **lateral ventricles** are all interconnected by this dynamic. A tension pattern is established, having a significant impact on the space where the **pituitary gland** is located.

Above the pituitary gland is the **optic chiasm**, which is oriented at approximately 30 degrees, with a postural movement of 5 degrees. This movement is essential for **drainage** between the third ventricle, the sphenoid, and the surrounding **microvessels**. The **pericytes**, present in the microcapillaries, interact with the nerves, creating **blood-brain spaces** that act as a deep fulcrum.

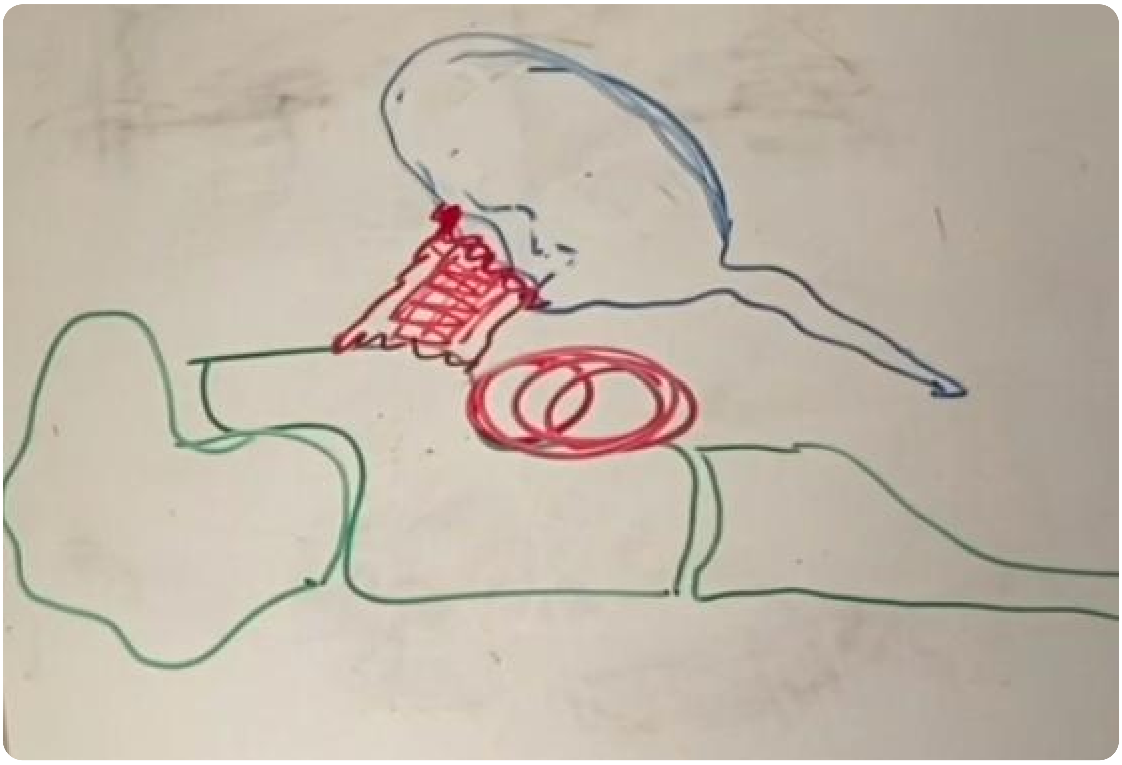


FIG. — Ventricular system

Examining the **presphenoids**, which correspond to the **occiput**, reveals a balance space on the membrane of Liliequist. This membranous tension affects both nerves and blood vessels, and is located near the pituitary gland, where the nervous system communicates with the hormonal system via the **hypothalamic-pituitary-thyroid-adrenal-gonadal axis**.

Cranial movement is also important to consider. By observing a skull, one can note the different movements at the level of the **parietals** and other structures. These movements follow specific lines of force. For example, the movement of **flexion** at the frontal level, associated with the **metopic suture**, represents two balance points that influence eye movement.

It is crucial to understand that the **brain** conditions the **ventricular system** and the **facial system**, both in terms of the **chondrocranium** (cartilage) and the **desmocranium** (membrane). During development, when the **occiput** descends, the **sacrum** also descends, and the **temporals** play a key role in this dynamic.

The **sternum** rises in response to these movements, converging towards the anterior midline. All of this is inscribed in the eye, and it is essential to add elements concerning the facial system and the capsular system of the eye, which insert around and organise the whole with a counter-rotation at the level of orbital development.

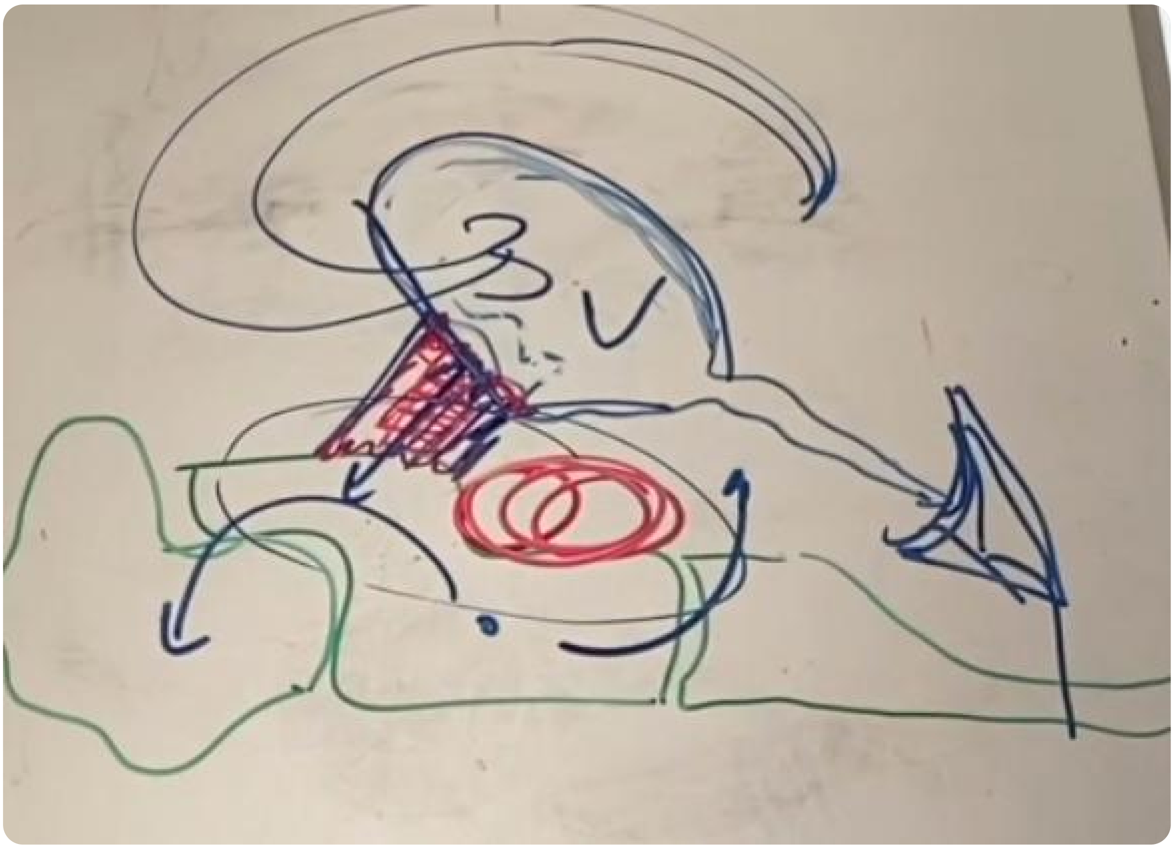


FIG. — Cranial movements

Craniosterno-sacral and mandibular movements are interconnected. The canine axes are also important, as they influence **occlusion**. If the upper and lower canine meeting is not correct, this leads to a loss of information at the brain level, because these electrical fields must unite.

Occlusion is therefore a reflection of the tensions present on the **sphenobasilar synchondrosis**. By observing the teeth, it is possible to determine the presence of **strain**, whether vertical, lateral or otherwise. This allows for an understanding of the constraints and the adjustment of necessary treatments to restore balance.

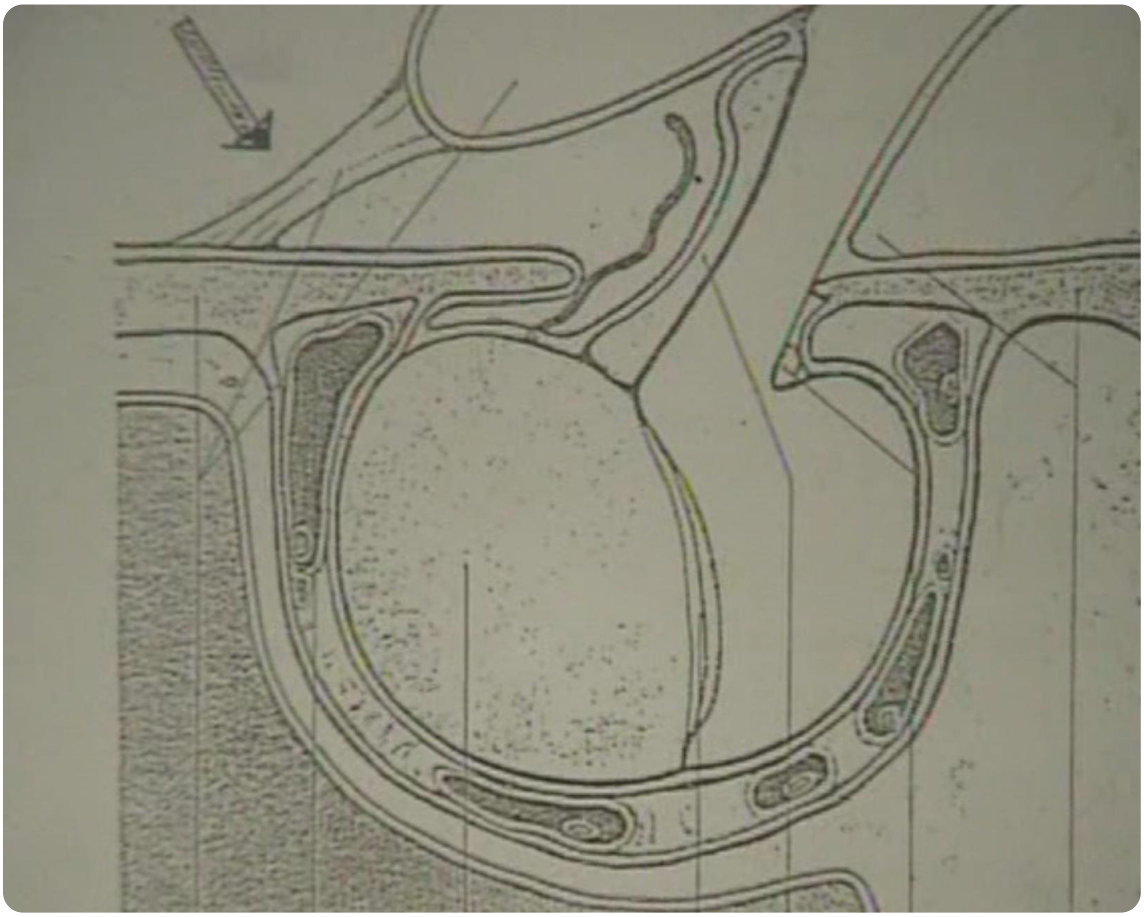


FIG. — Coordinated movements of the skull, sternum and sacrum

22. PRACTICE: LINES OF FORCE

DURATION : 08:41 | STUDY SHEET

COURSE SUMMARY

In this video, we delve into the study of lines of force through the coronal dural sheath, the symphysis, and the maxilla, while adopting an electrical approach to effectively treat tendons and tendinitis. The connections between the neurocranium and the viscerocranium are explored, emphasising the importance of the coronal suture and its role in the body's vitality. Participants will learn to palpate and detect energy losses, work on cardiac and endothelial systems, and use induction techniques to restore balance. This session teaches how to harmonise different structures to improve energy circulation and promote patient well-being.

PRACTICE: THE LINES OF FORCE

In this session, we will explore the **lines of force** in relation to the **coronal dural sheath**, the **symphysis**, and the **maxilla**. It is essential to place oneself on an **electrical level** to understand how these fields interact. This approach is also applicable to the treatment of **tendons** and **tendinitis**.

CONNECTION BETWEEN THE NEUROCRANIUM AND THE VISCEROCRANIUM

The establishment of the cranial base, which includes the **neurocranium** and the **viscerocranium**, is crucial. One can imagine that the entire visceral system is supported at the base of the skull, while the brain is above. The **dural membranes** of tension attach to this base, forming a complex network.

We will start with the **coronal suture** and the part of the **maxilla**. By placing oneself on the **maxillo-zygomatic** level, one can observe how the lines of force meet. These connections must be in harmony, like **rings** that align.

IMPORTANCE OF VITALITY AND ENERGY

It is important to note that the coronal suture is relatively low, and it is essential to work on this area, particularly at the location of **Bregma**. Vitality is linked to thermal exchanges, and a felt warmth indicates good energy circulation.

By palpating this area, one can detect points of **energy loss**. By connecting to these areas, one can observe a difference in vitality, which may indicate fatigue or a struggle at the level of the **adrenals**. It is therefore crucial to **recharge** these areas to improve the patient's general condition.

ENERGETIC APPROACH AND CONNECTED SYSTEMS

During the consultation, it is possible to work on the **cardiac** and **endothelial systems**. Four entry points are to be considered: the **internal carotids** and the **vertebrals**, as well as the **basilar system**. By touching these areas correctly, one can release tensions and improve circulation to the heart.

We use different energies, notably that of **weight** and **gravity**, to re-establish a correct contact. Some patients may be too heavy or too light, which affects their energy balance.

PALPATION AND INDUCTION TECHNIQUES

By palpating the **coronal suture**, it is possible to feel variations in warmth and vibration, which indicates a good balance. By working on the **frontal boss** and the **zygomatico-frontal boss** line, one can identify areas of imbalance.

It is essential to observe the patient's breathing during these manipulations. By using **induction** or **facilitated movement** techniques, one can accompany the body in its natural movements. Observation of **expiratory** activity can also reveal changes in polarity, indicating an **oxidation-reduction** process.

CONCLUSION

These techniques help to restore energetic balance and promote healing. By working on the lines of force and establishing connections between different structures, we can improve the patient's vitality and well-being.

23. ANATOMY OF

DURATION : 21:58 STUDY SHEET

COURSE SUMMARY

This video offers an in-depth exploration of the anatomy of the eye and neurocranium, explaining the relationships between different structures such as the neurocranium, viscerocranium, and their integrated functions. You will learn the key characteristics of the tunics of the eye, including the sclera, uvea, and retina, as well as the importance of the extraocular muscles and their innervation. The discussion also integrates the complex vascularisation of the eye, essential for maintaining its health. This knowledge is vital for healthcare professionals, particularly osteopaths, who seek to better understand the anatomical foundations of vision and their impact on overall health.

ANATOMY OF THE EYE AND NEUROCRANIUM

The anatomy of the eye and neurocranium is a fascinating subject that involves a deep understanding of structures and their function.

NEUROCRANIUM AND VISCEROCRANIUM

The **neurocranium** is a metabolic field of detraction, while the **viscerocranium** is made up of embryological derivatives. The three-dimensional shape of the **zygoma** is particularly interesting, with its small facets and grooves, notably the **greater groove** above, called the **sphenoidal fissure**, through which the **oculomotor nerve** (nerve III) and trigeminal branches pass. The **optic canal** is crucial, as it houses the optic nerve and ophthalmic artery.

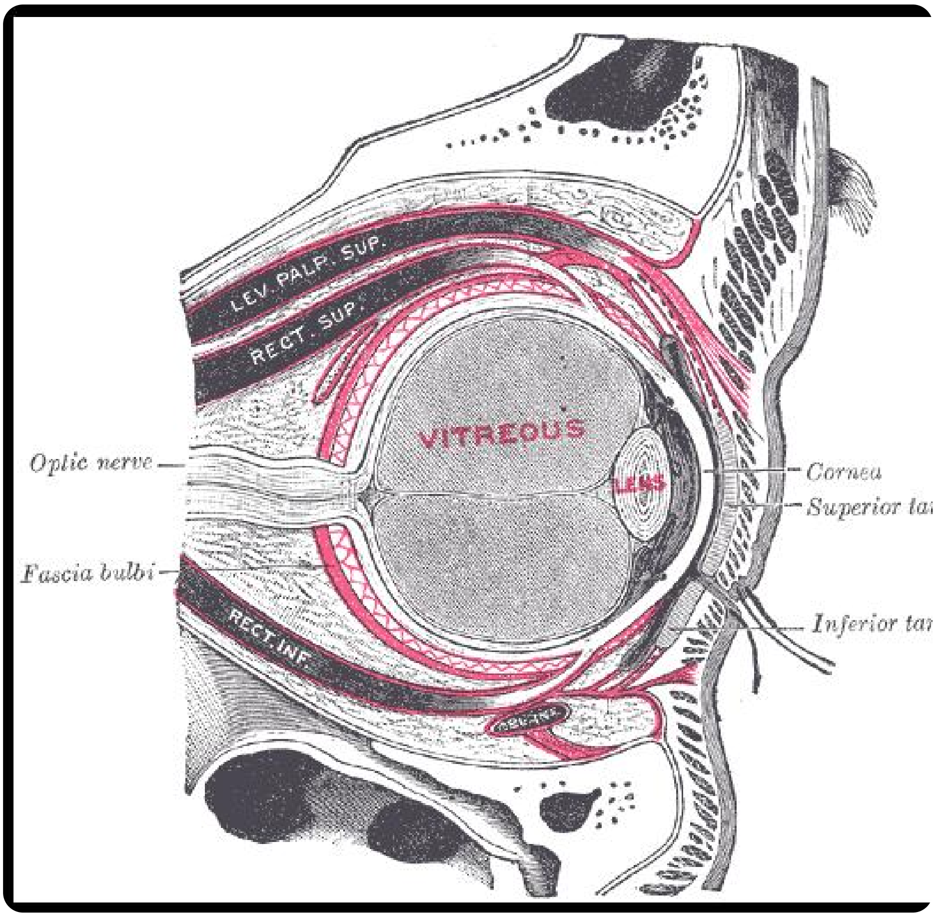


FIG. — Sphenoidal fissure

The **palatine**, although a small process, plays an important role in balancing the flexion movement of the embryo during palate formation. It is associated with the **zygomatic** and **maxilla**, which are also three-dimensional bones. The lacrimal fossa, lacrimal bone, **ethmoid**, **nasal bone**, **frontal**, and **sphenoid** form a concentration of bony information.

TUNICS OF THE EYE

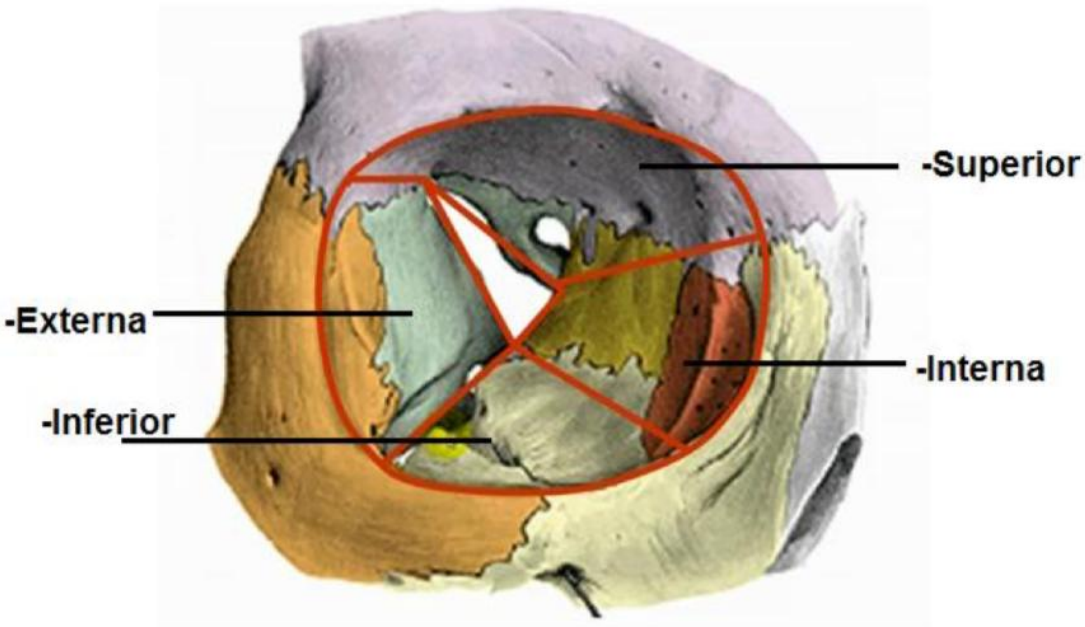


FIG. — Neurocranium and viscerocranium, including the importance of the palatine and zygoma mentioned

The eye is composed of three main tunics:

1. **Sclera**: The sclera is the outer layer, which extends forward to form the **cornea**, a transparent sclera. The white of the eye is a non-transparent sclera.

2. **UV (Uvea)**: Composed of several parts, including the richly vascularised **choroid**, the **iris** at the front, and the **ciliary body** in between. The ciliary body contains ligaments and muscles, called **suspensory ligaments**, which hold the lens.

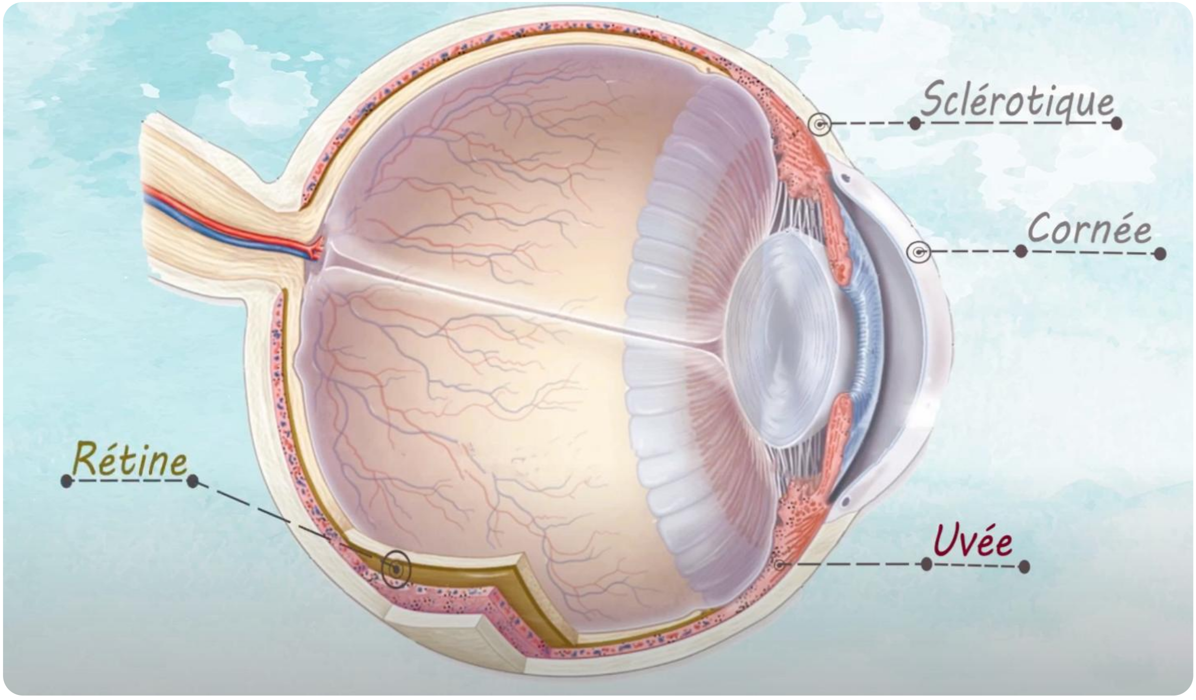


FIG. — Overview of the eye tunics

3. **Retina**: The retina contains the **canal of Cloquet** or the **hyaloid canal**, as well as the **ora serrata**. It is rich in photoreceptors, notably **cones** (day vision) and **rods** (night vision).

La cornée

TRANSPARENTE ET ANTERIEURE

1) Epithélium de revêtement pluristratifié pavimenteux épidermoïde :

Nombre variable de couches, nombreuses terminaisons sensitives, forte capacité de régénération

2) Membrane de BOWMAN (=membrane basale)

Epaisseur : 10-12 µm

Fibrilles de collagène IV

3) Stroma cornéen : Tissu conjonctif dense régulier LAMELLAIRE, NON vascularisé

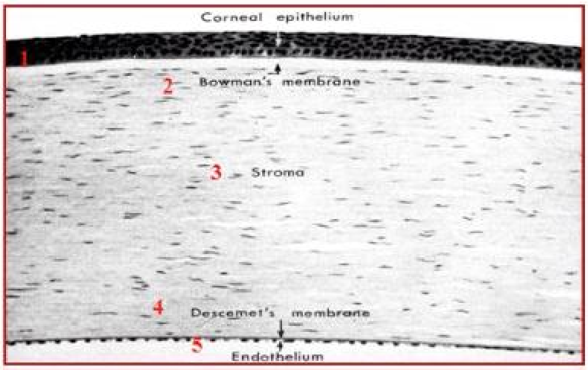
- Faisceaux de fibres collagène I disposés perpendiculairement l'un à l'autre d'orientation régulière

- Fibroblastes

- Substance fondamentale riche en chondroïtine- et keratan-sulfate (hydrophilie)

4) Membrane de DESCEMET (=membrane basale)

5) Endothélium (=ERU pavimenteux)



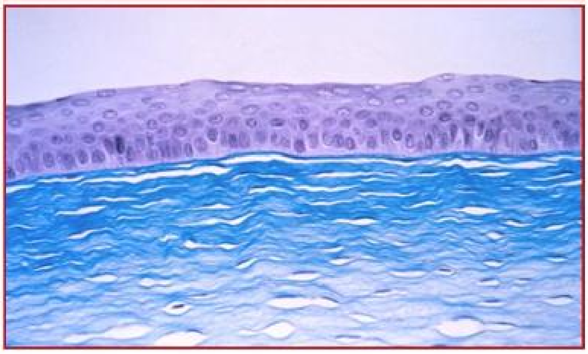


FIG. — The structure of the cornea and its frequencies

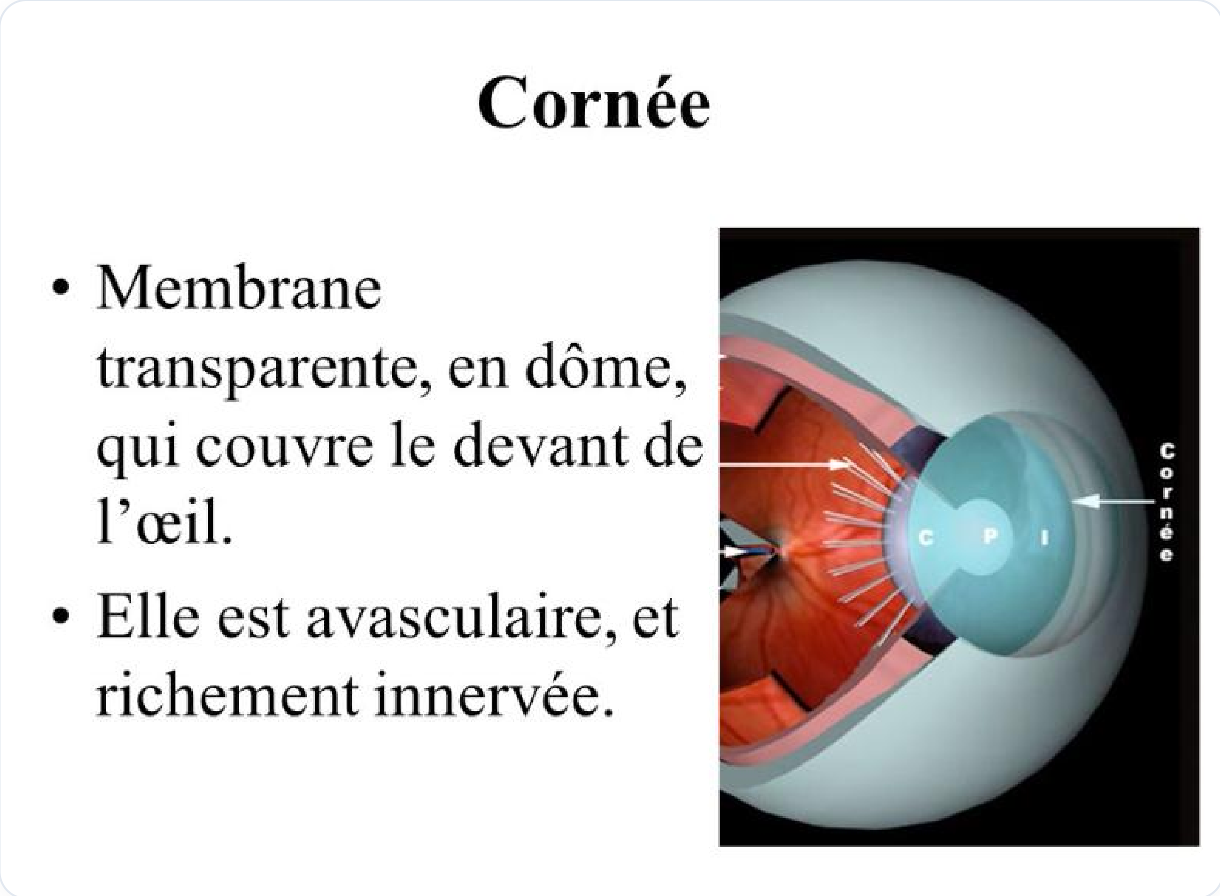


FIG. — Eye cross-section

ANATOMY OF THE CORNEA AND SCLERA

FIG. — The vascularization related to the choroid and the integration of information in the eye

The **limbus** is the junction between the cornea and the sclera, also called the **corneoscleral imbus**. Schlemm's canal is essential for the resorption of fluid from the anterior chamber. The **anterior chamber** and **posterior chamber** of the eye are separated by the pupil.

The **aqueous humour**, a nutritive fluid, is secreted by the ciliary processes, filtered by the trabecular meshwork, and reabsorbed by the scleral venous sinus and Schlemm's canal. It maintains balance and nourishes the cornea and iris.

OCULAR MUSCULATURE

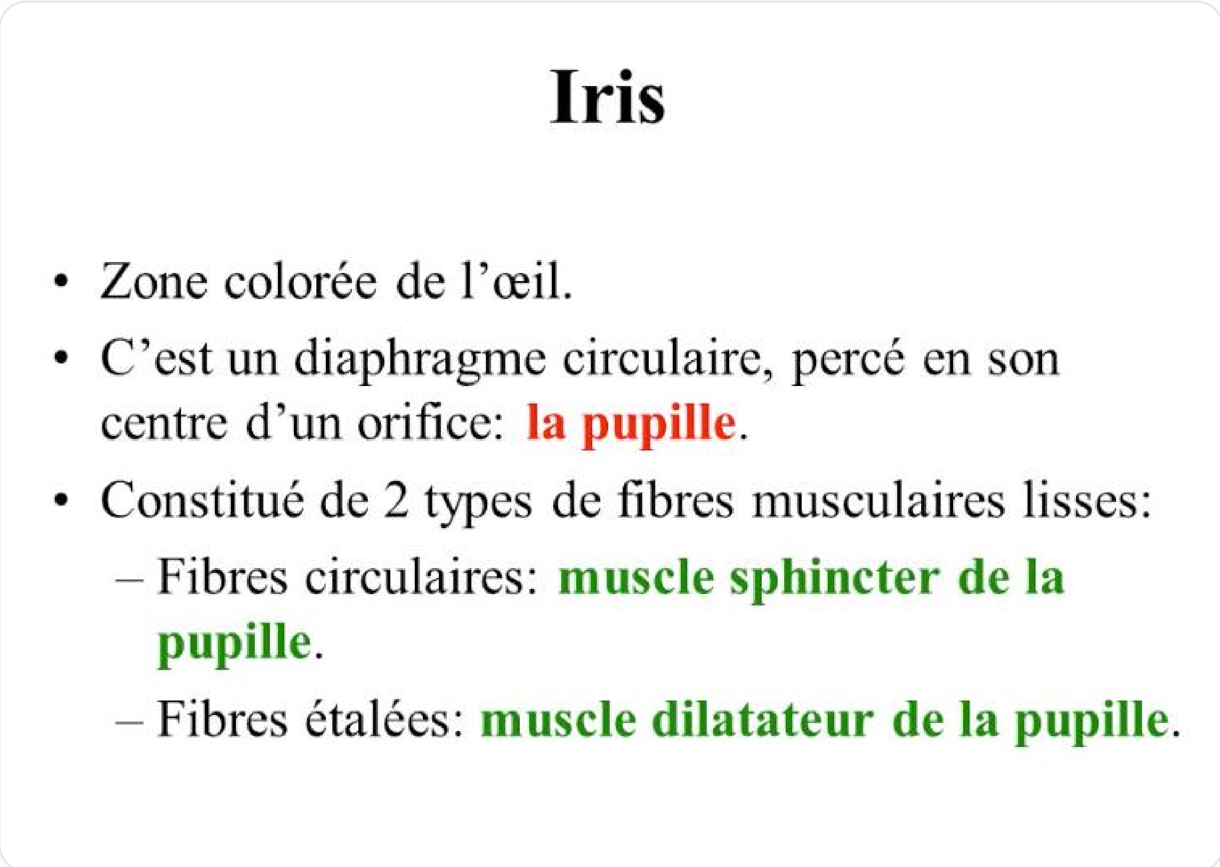


FIG. — Pupil and its light regulation function

The oculomotor muscles, such as the **superior rectus**, **inferior rectus**, and oblique muscles, are essential for eye movement. The **levator palpebrae superioris muscle** is also important, and eyelid ptosis can indicate health problems.

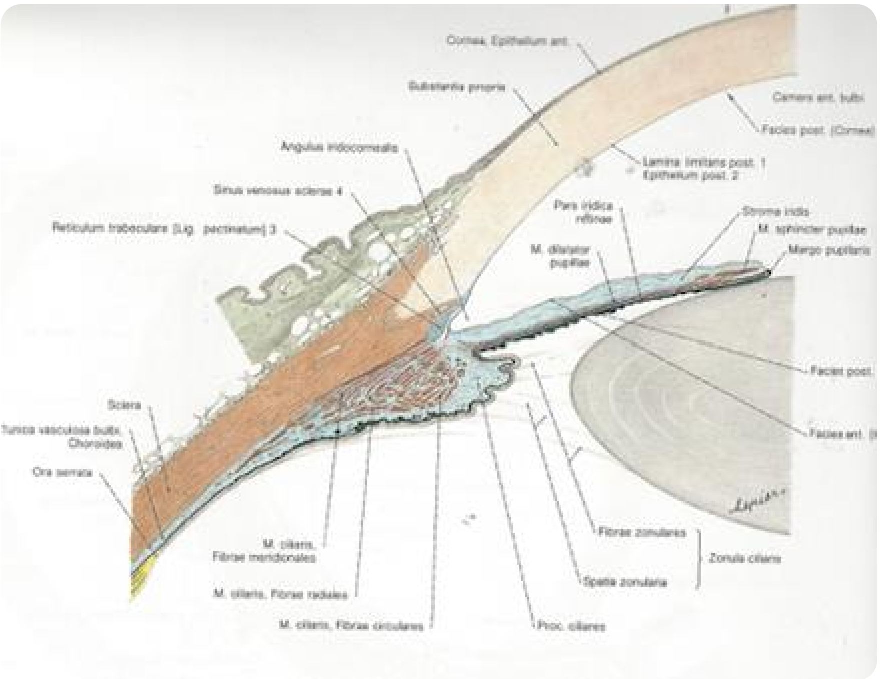


FIG. — This image can show a more detailed view of Schlemm's canal and aqueous humor regulation

Zinn's tendon is the attachment point for the oculomotor muscles, which are connected to the neck muscles. The optic nerves and ophthalmic arteries play a crucial role in the vascularisation and innervation of the eye.

VASCULARISATION AND INNERVATION

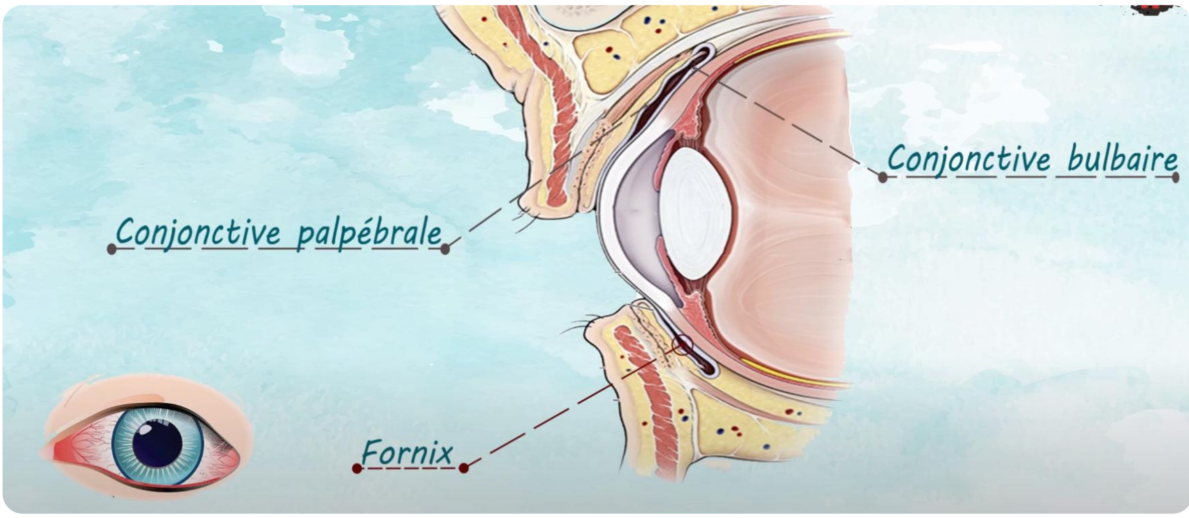


FIG. — Conjunctival junction

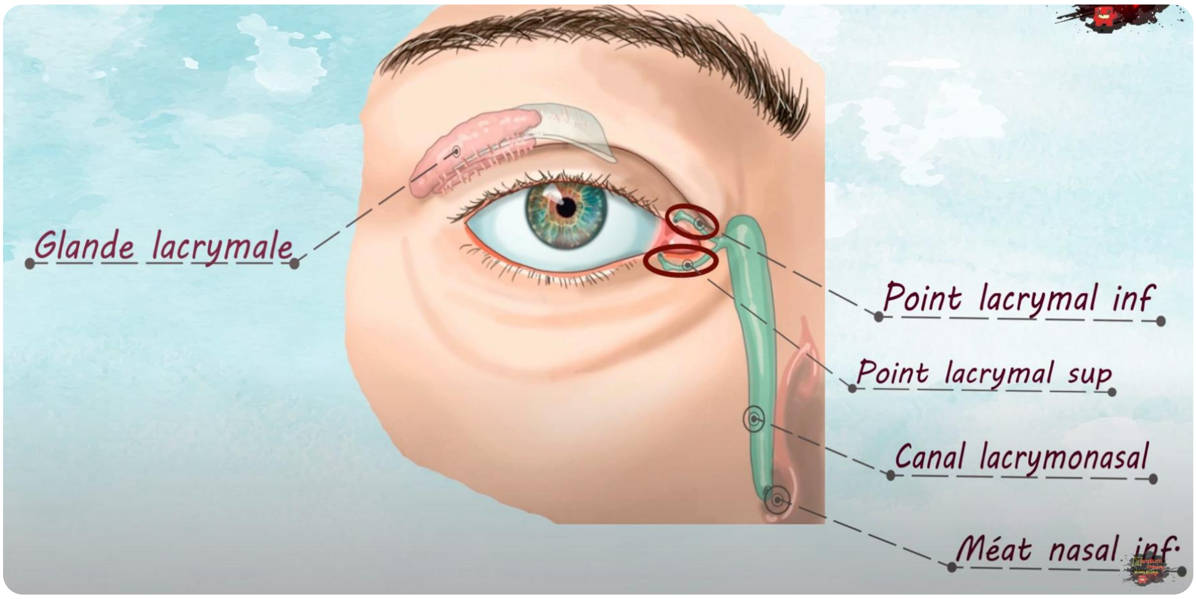


FIG. — Lacrimal gland and its drainage system

Secretory glands and the continuity of the tarsus with Muller's muscle

FIG. — Secretory glands and the continuity of the tarsus with Muller's muscle

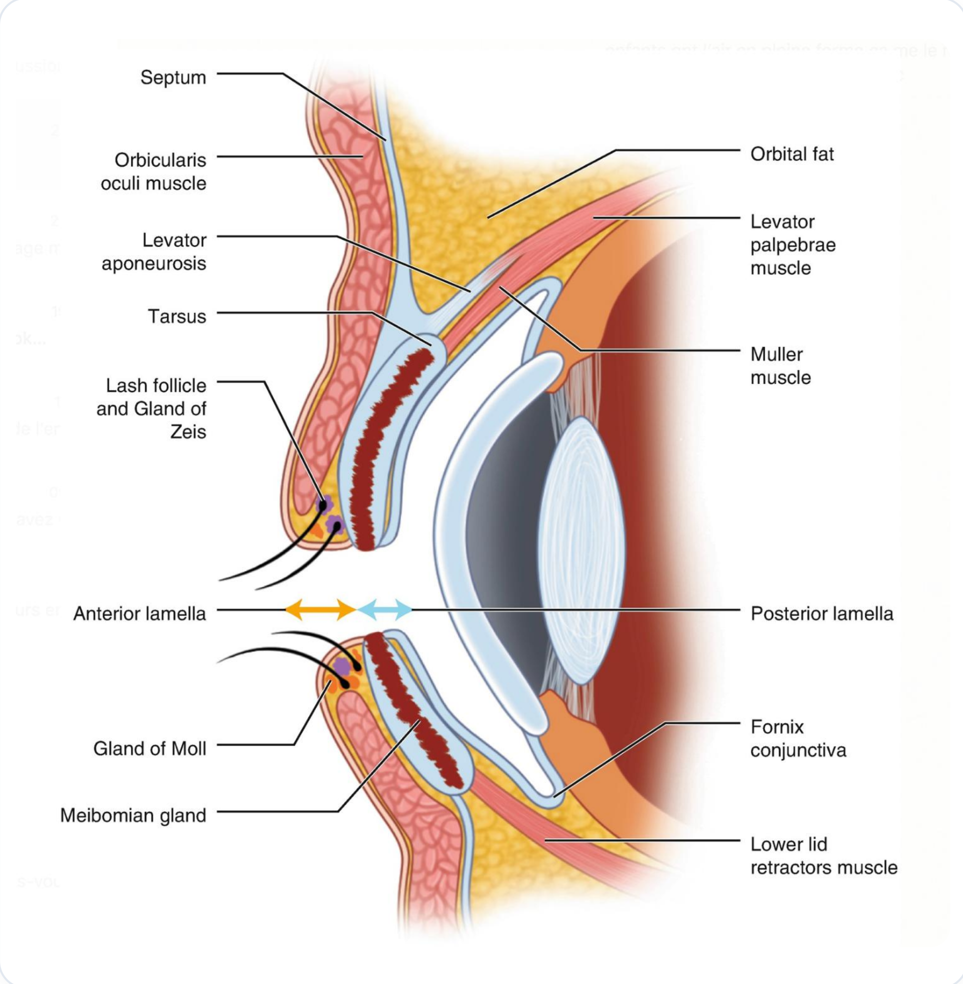


FIG. — Face muqueuse de la paupière (conjonctive)

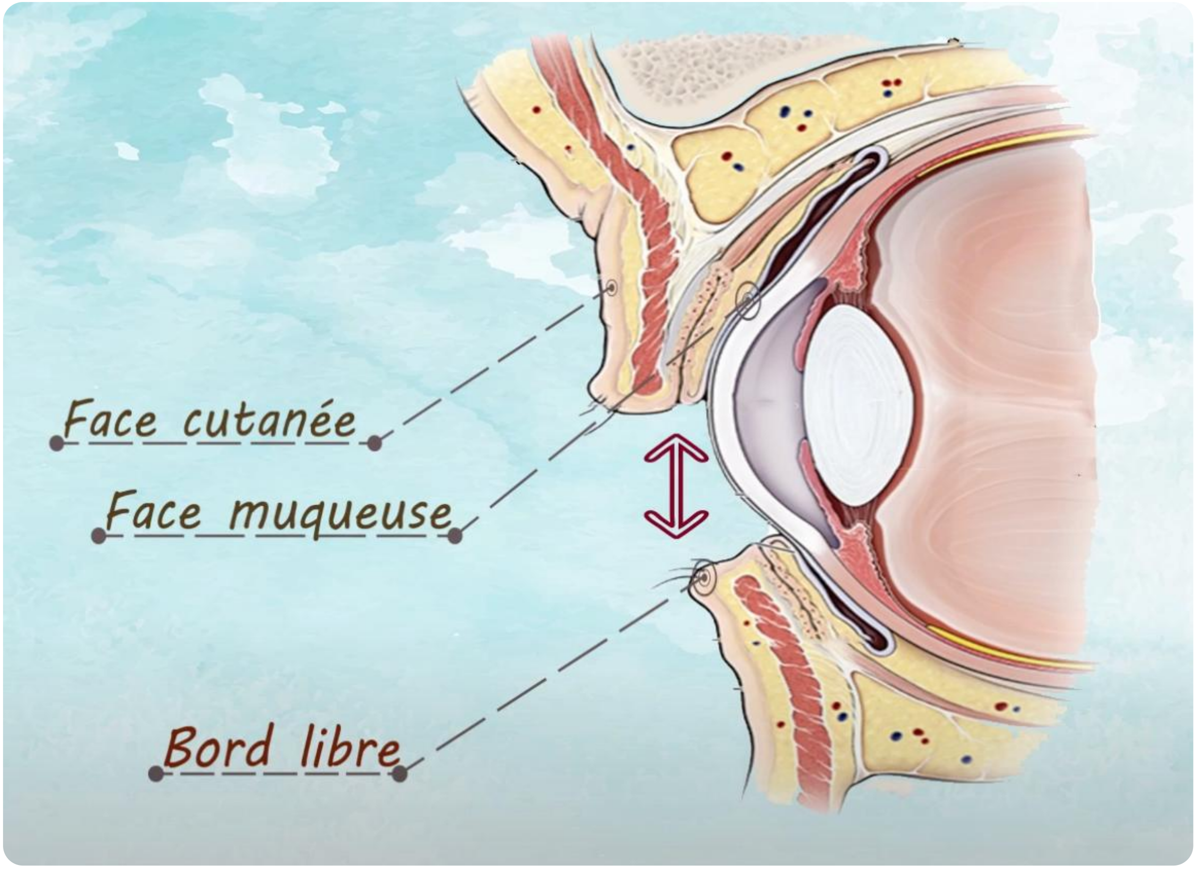


FIG. — Eyelid anatomy

The vascularisation of the eye comes from the carotid artery, with ophthalmic arteries and branches for retinal vascularisation. **Pericytes** or Rouget cells modulate information at the capillary level, particularly in the retina.

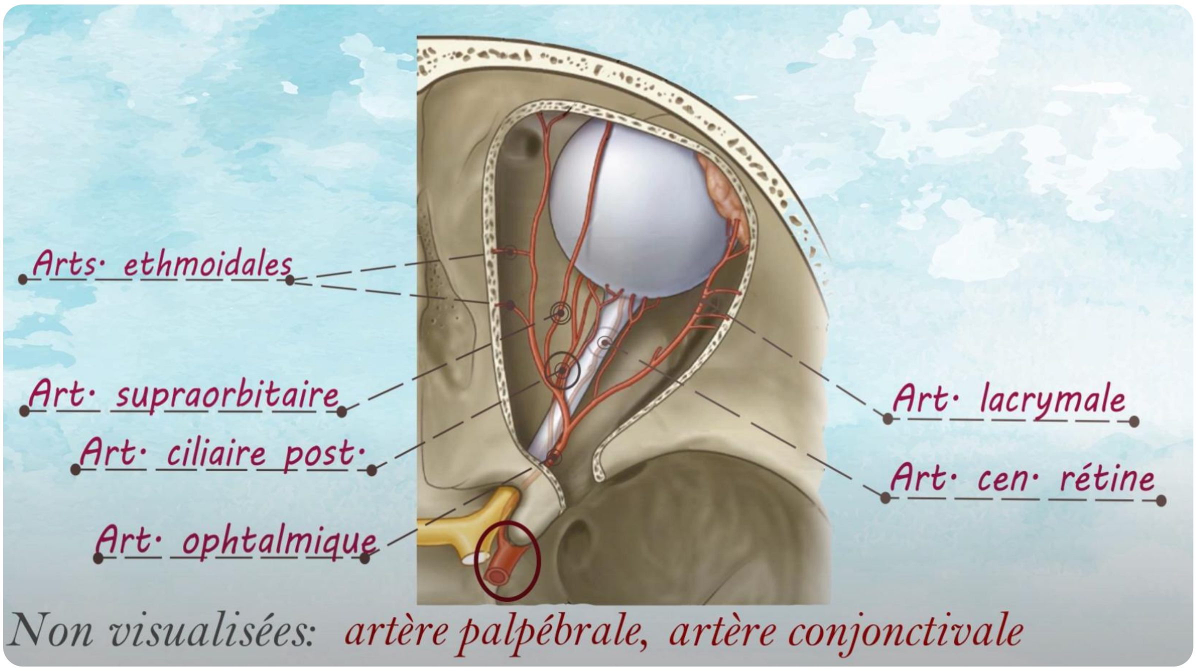


FIG. — Relationship of Tenon's capsule with body fascias

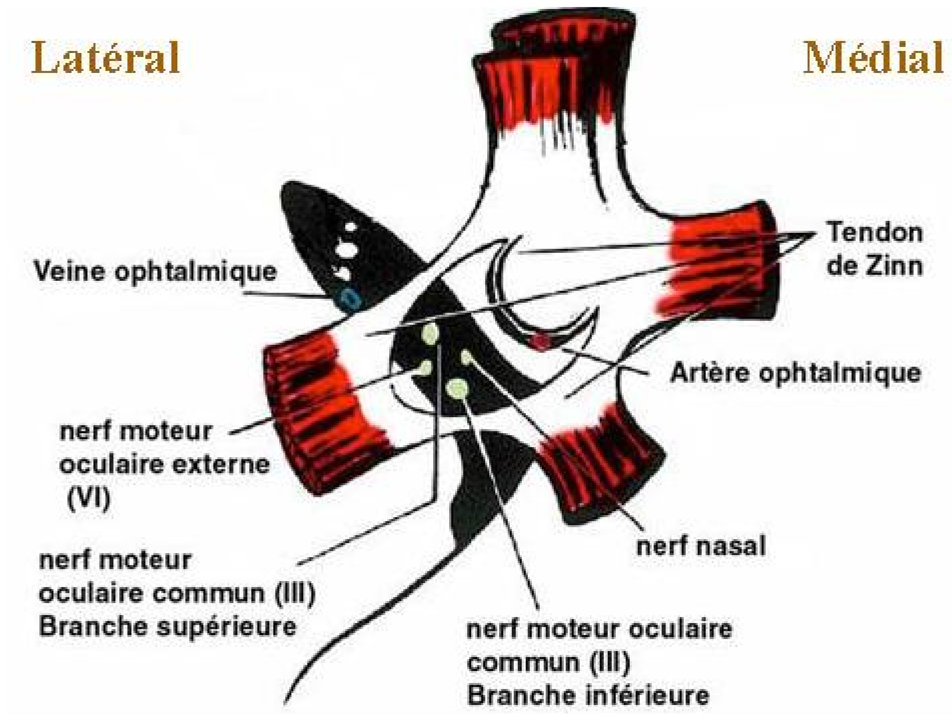


FIG. — Oculomotor muscles

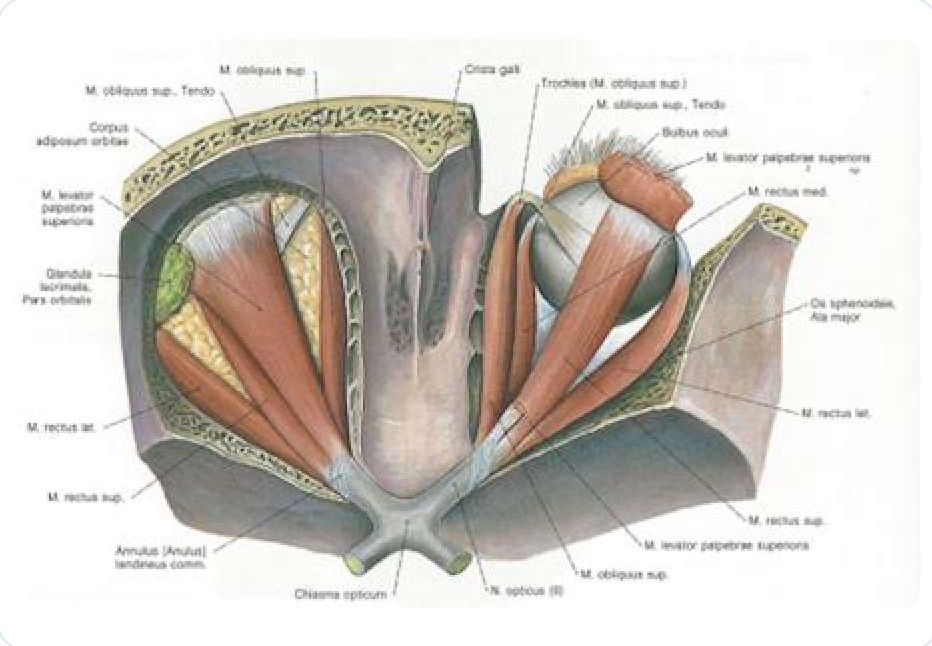


FIG. — Oculomotor muscles

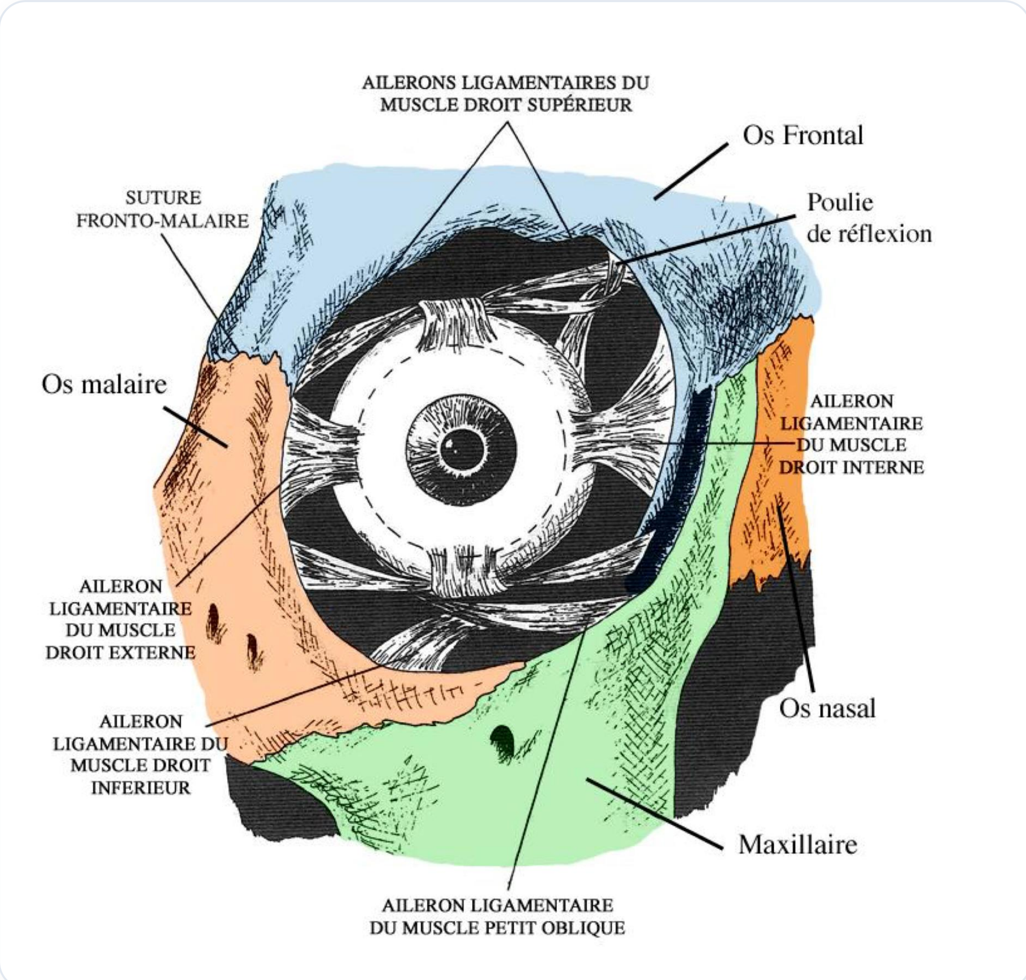


FIG. — Oculomotor muscles

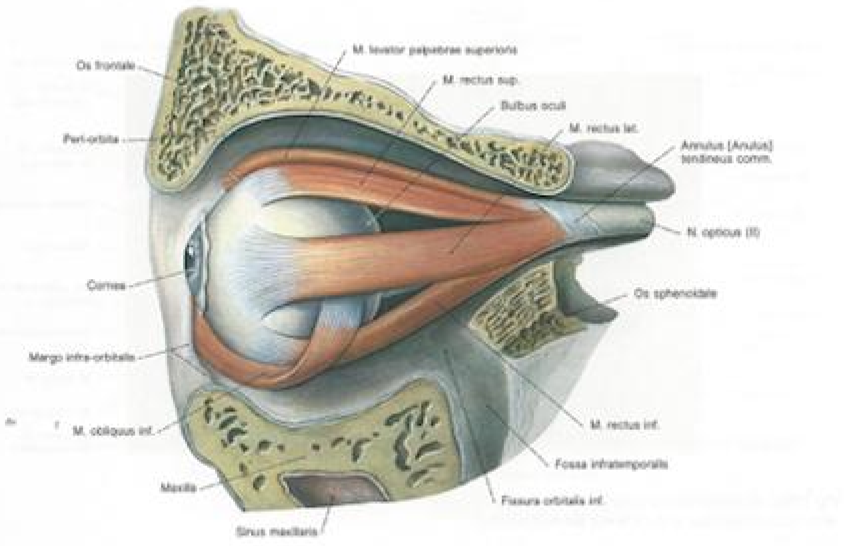


Fig. 176. The musculi bulbi, left lateral view. The paries lateralis of the left orbita has been removed.

FIG. — Oculomotor muscles

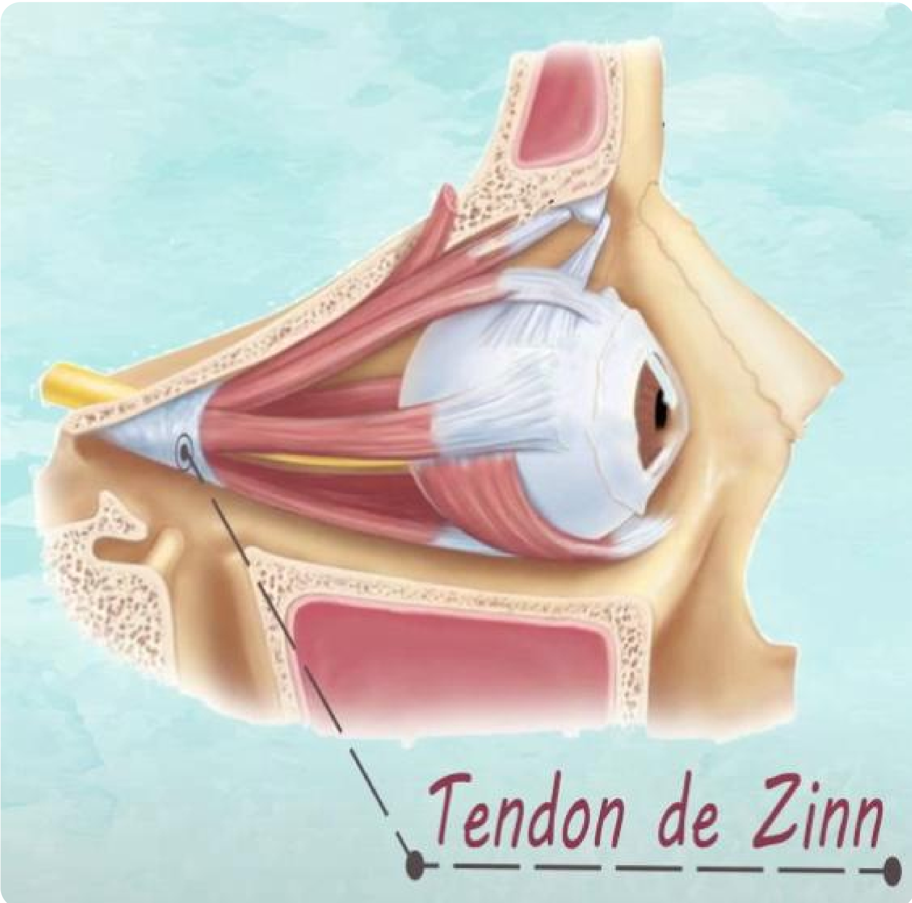


FIG. — Oculomotor muscles

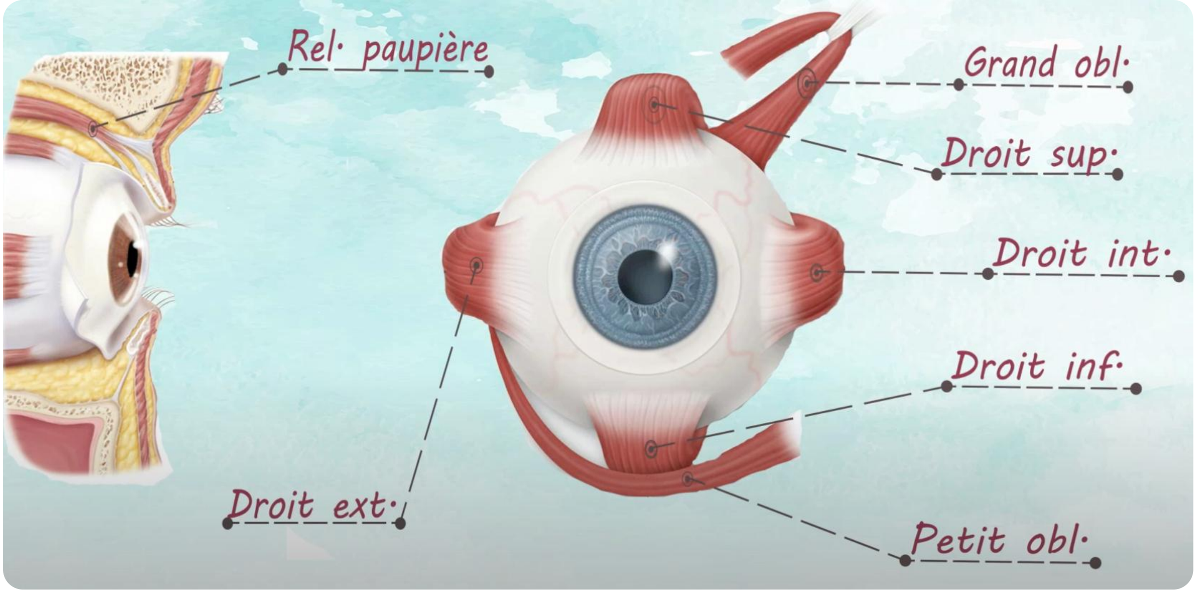


FIG. — Ocular musculature and the attachment point of Zinn's tendon

The innervation of the eye includes sensory, motor, and autonomic pathways, with the optic nerve capturing visual information. The meeting of these systems at the retinal level forms the optic nerve, which transmits sensory information.

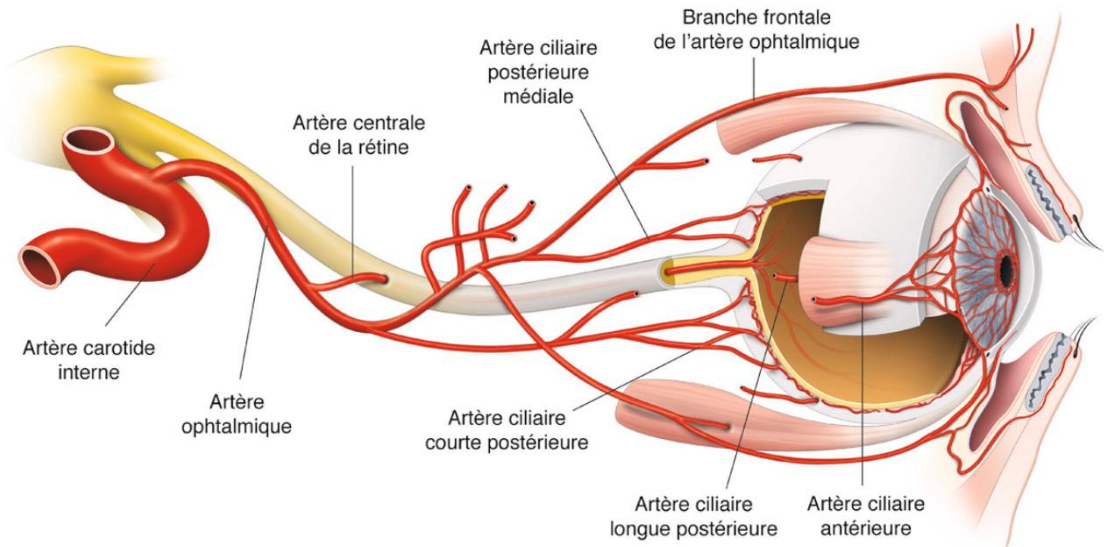


FIG. — Relationship between carotid and optic nerve

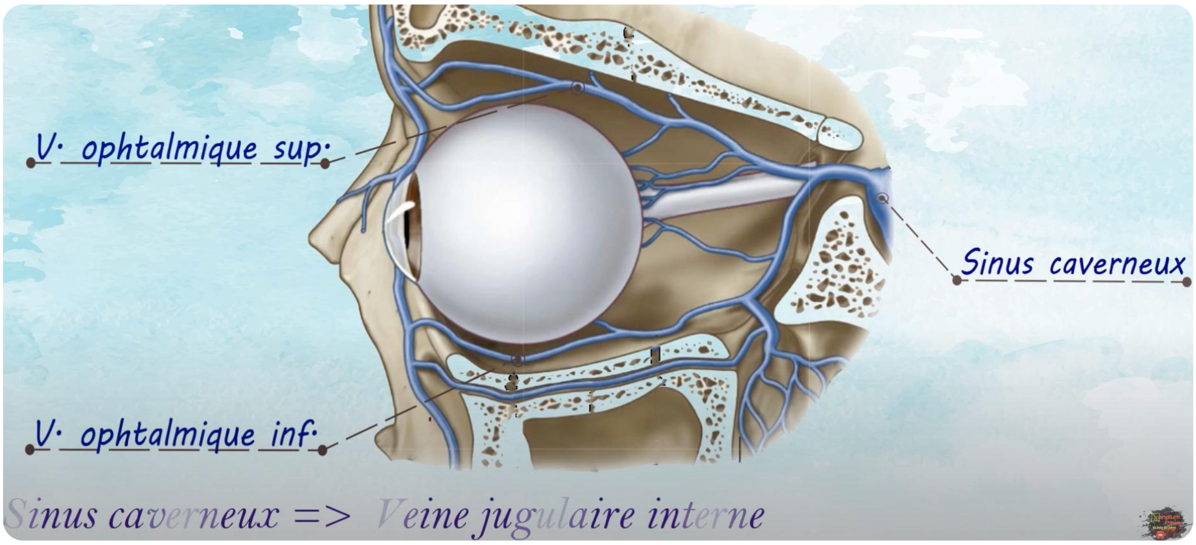


FIG. — This image may illustrate a venous sinus or an aspect of vascularization

CONCLUSION

The anatomy of the eye and neurocranium is complex and interconnected, involving various structures that play essential roles in vision and neurological function. Understanding these elements is fundamental for healthcare professionals, particularly in medicine and osteopathy.

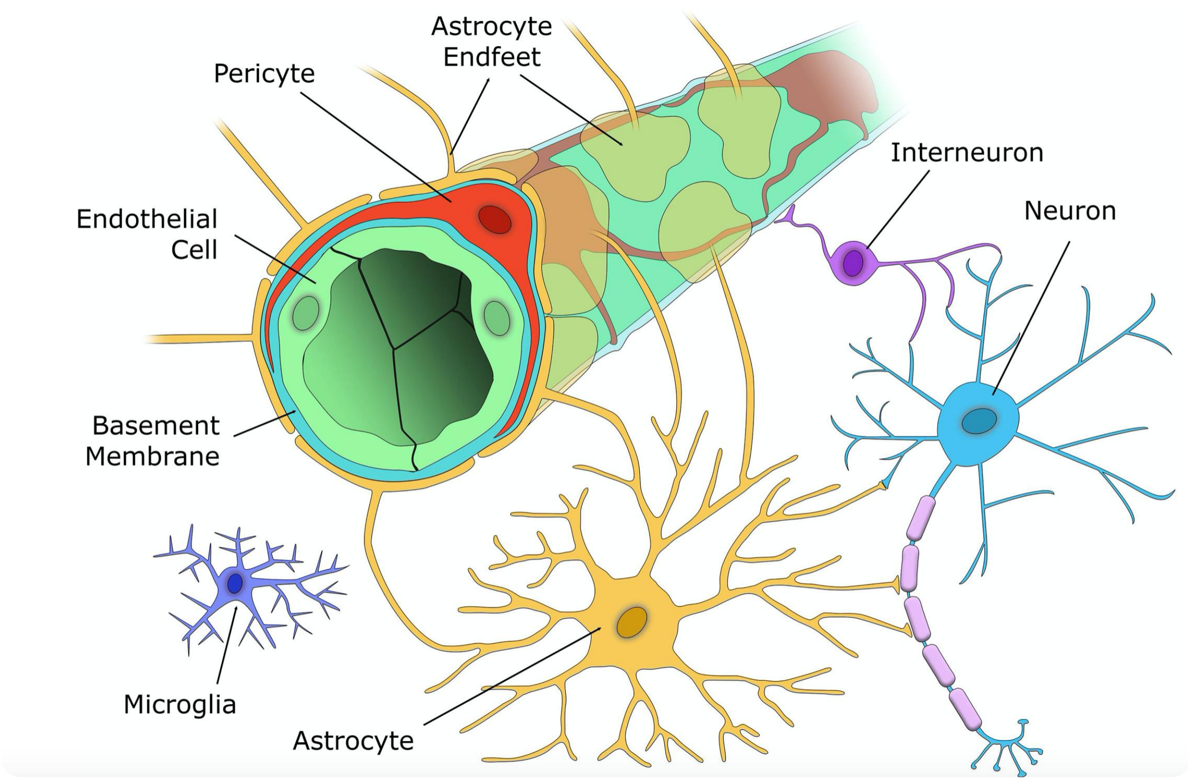


FIG. — Eye vascularization, starting with the carotid

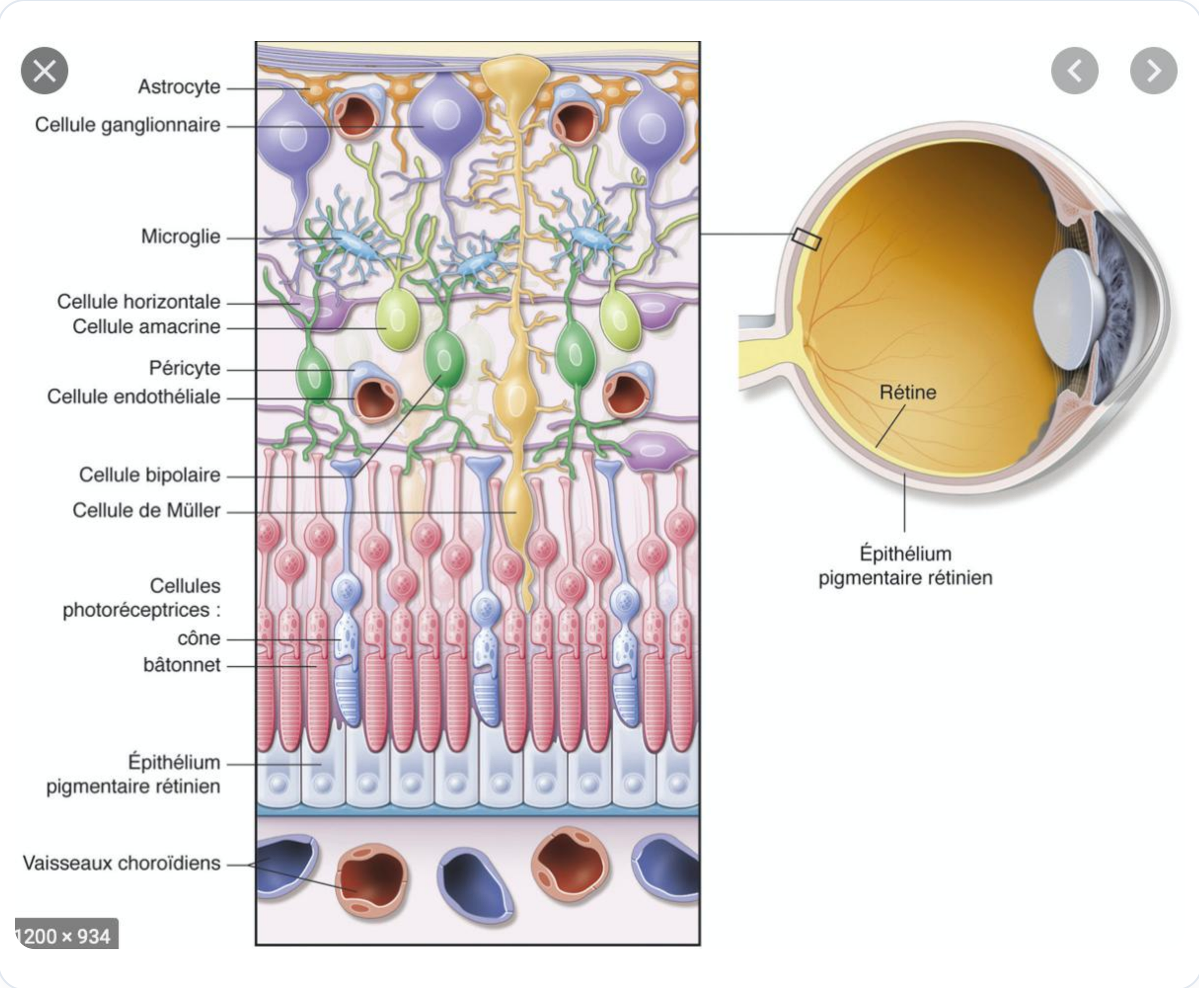


FIG. — Pericytes and their role in information modulation

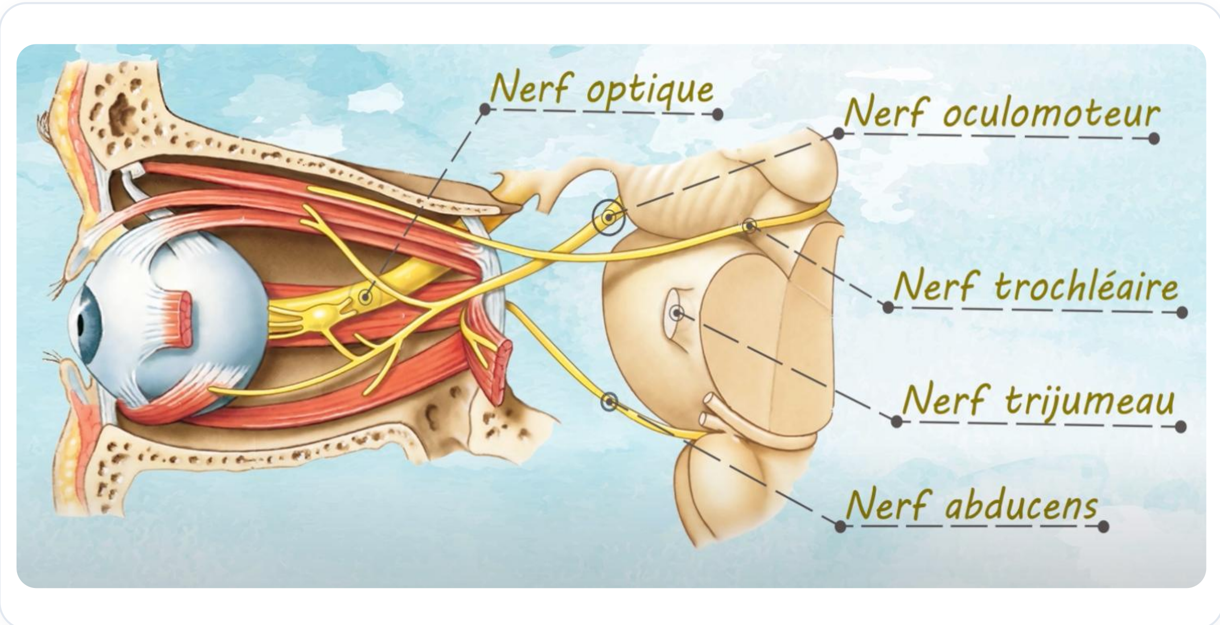


FIG. — Innervation and information gathering in the optic canal

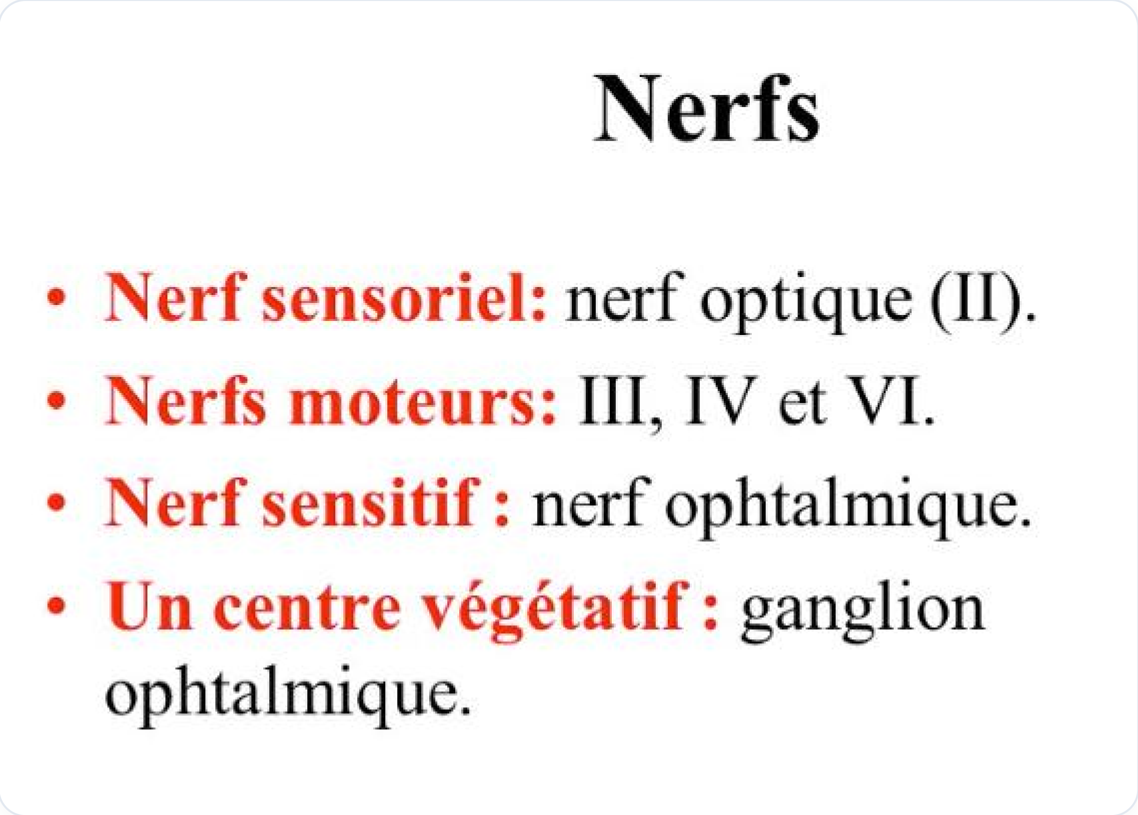


FIG. — This image can complement the illustration of innervation and information transmission via the optic nerve

24. THE WORLD OF INTENTION

DURATION : 08:26 STUDY SHEET

COURSE SUMMARY

In this video, the teacher addresses the notion of suffering linked to fixation and the importance of freedom of being. The Five Aggregates – Form, Sensations, Perceptions, Volition, and Consciousness – are examined in depth, each aggregate representing a dimension of our existence and subjective experience. By becoming aware of these elements, students will learn to let go and better manage their sensations and emotions, avoiding the attachment that leads to suffering. Key concepts such as impermanence, interdependence, and personal responsibility are also explored, while emphasising the importance of reflecting on one's intentions to create a more aligned and harmonious existence.

THE WORLD OF INTENTION

Suffering is often linked to our tendency to freeze things, thus preventing the natural **movement** of life. It is essential to become aware of the need to let **freedom** express itself.

THE FIVE AGGREGATES

1. FORM

The first aggregate is **form**, which encompasses **phenomenology**. It is the physical manifestation of our existence.

2. SENSATIONS

The second aggregate concerns **sensations**. Once the body is present, it is invited to feel. The Buddha taught that the body is **impermanent** and **interdependent**. We can imagine our existence as a vast game, as big as the Earth, which implies a certain **responsibility**.

Sensations can be pleasant or unpleasant. Sometimes, a pleasant sensation can become unpleasant over time, and vice versa. The Buddha illustrates this with the image of **soap bubbles**: they appear, then disappear. We tend to attach ourselves to these sensations, thus creating an **attachment** to what no longer exists.

It is common for people to reactivate unpleasant past events, as if opening a **dome** to relive a bad smell. This process of attachment or rejection is a form of fixation. **Letting go** should not be confused with abandonment; it is about accepting that sensations do not last.

3. PERCEPTIONS

The third aggregate is that of **perceptions**. These can be compared to a **mirage**, where we project our own patterns of satisfaction onto others. For example, during a meditation, the late arrival of some participants can provoke reactions of frustration. By becoming aware of these perceptions, we can choose to consider them as **noise**.

4. VOLITION

The fourth aggregate concerns **volition** or **mental factors**. It is crucial to become aware of our **intentions**, as they shape our karmic world. Before acting or speaking, it is essential to reflect on our intentions. The Buddha illustrated this by separating his intentions into two bundles: those that are beneficial and those that are less so.

5. CONSCIOUSNESS

The last aggregate is that of **consciousness**. This consciousness can be perceived as a **magician**, representing the great illusion. It often becomes the ultimate refuge of the ego.

CONCLUSION

The aggregates – foam, bubbles, perceptions, the banana tree, and consciousness – are interconnected and constantly superimposed. It is beneficial to take the time to reflect on each of these aggregates, even dedicating a week or a month-long retreat to their study. This allows us to deepen our understanding of ourselves and our experience.

25. REVISION

DURATION : 01:05 STUDY SHEET

COURSE SUMMARY

This video delves deeply into the anatomy of the retina, explaining its envelope, which includes the uvea, choroid, iris, and ciliary bodies. You will learn how the retina, composed of the outer and inner retina, develops from the primitive optic cup and its juxtacrine interaction with the epiblast. One of the key points discussed is retinal detachment, a phenomenon essential to understand both theoretically and in a clinical context. Furthermore, it is highlighted that the retina is a brain-derived tissue, a notion that enriches your embryological understanding and opens up perspectives on ocular pathologies.

REVISION OF RETINAL ANATOMY

The **envelope** of the eye comprises the **uvea**, which consists of the **choroid**, the **iris**, and the **ciliary bodies**. By removing this envelope, we access the **retina**, which is composed of two main layers: the **outer retina** and the **inner retina**.

It is at the level of the retina that **detachment** occurs, an important phenomenon to understand. At this stage, the **primitive optic cup** forms and establishes a **juxtacrine** connection with the surface **epiblast**. This interaction leads to a thickening of the epiblast, thus creating the **outer optic placode**, which is held in place while the surrounding structure develops.

This dynamic gives rise to a **primitive depression**. The retina, as it develops, surrounds itself and forms what is known as the **outer retina** and the **inner retina**. This is where retinal detachment occurs, a problem many people have already encountered.

It is essential to note that the retina is actually a tissue derived from the **brain**, more precisely from **ventricular tissue**.

26. NEUROPHYSIOLOGY OF THE EYE: INTRO

DURATION : 04:43 STUDY SHEET

COURSE SUMMARY

This introduction to the neurophysiology of the eye explores the interconnections between suffering, perception, and the anatomical specificities of the eye. You will learn how eye movements and iris geometry can signal emotional states, as well as the distinctions between seeing, clairvoyance, and polyvoyance. The video then details the different structures of the eye, from the conjunctiva to the retina, and explains how light is converted into electrical signals. By integrating concepts such as the importance of eye orientation and photon transduction, this presentation enriches your theoretical and practical approach to osteopathy and biodynamics.

NEUROPHYSIOLOGY OF THE EYE: INTRODUCTION

When a person experiences **suffering**, whether **physical**, **emotional**, or of another nature, it can be reflected in their eyes. The eye can perform specific movements, and there is a **geometry** in the iris that can manifest in different ways.

The eye is also a point of initiation in relation to the **Christagalie**. Initially, this structure is flat, but through the process of **telencephaly**, it evolves. This point of concentration is often referred to as the **eye of clairvoyance**.

There are several levels of perception:

- **Clairvoyance** (Voyance): Using the emotion of others.
- **Clairvoyance** (Clairvoyance): The ability to perceive the akashic world of another, allowing one to understand their functioning.
- **Polyvoyance**: Vision of the divinity present in everything.

The eye receives **light information** in the form of photons. The first layer traversed is the **conjunctiva**, followed by the **cornea**, which is the transparent part of the sclera. Then, light passes through the **pupil**.

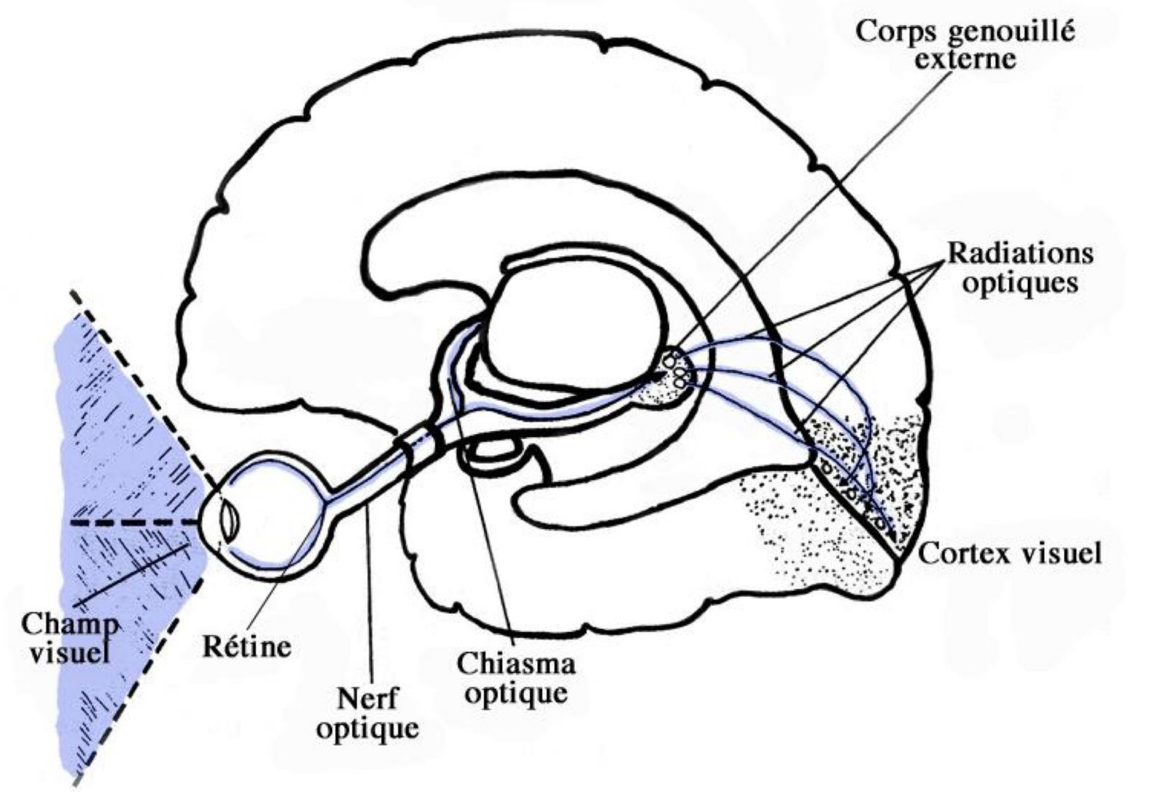


FIG. — Layers of the eye

The anterior chamber, which contains **aqueous humour**, is followed by the posterior chamber, which also contains this fluid. The **ciliary bodies** at the front produce fluid, while the **canal of Schlemm** at the back reabsorbs this fluid. This process allows for a continuous circulation of fluid that nourishes the eye.

Light then passes through the **lens** and the **vitreous humour** to reach the **retina**, composed of several layers. The eye is capable of capturing a small amount of light, even a single photon, while having both central vision and peripheral vision.

It is essential to learn to use these two types of vision. The frequency, wave, and light spectrum vary, and the orientation of the eye is often **23 degrees**, a value that resonates with the orientation of the heart, the Earth, and the inner ears.

This specific orientation influences the **retina**, where the phenomenon of **transduction** occurs. Photons are converted into electrical signals, sent as nerve impulses to the left and right. Each eye transmits this information to the **lateral geniculate body (LGB)**, which interprets it through **parvocellular** and **magnocellular** cells.

Once integrated, this information travels along the **optic radiations** to reach the **visual cortex**, located above and below the eye, near the **calcarine sulcus**. This pathway of the eye and the interpretation of visual signals are crucial for understanding the neurophysiology of vision.

27. MECHANISM OF TRANSDUCTION

DURATION : 08:39 STUDY SHEET

COURSE SUMMARY

This video explores in depth the mechanism of visual transduction, detailing the path of light through the different layers of the retina, from the photoreceptors to the optic nerve. The various cells of the retina, including cones, rods, bipolar cells, and ganglion cells, are examined to understand their role in visual information processing. The chemical transformation of rhodopsin and iodopsin upon light absorption, as well as the importance of vitamin A, are also discussed, highlighting the consequences of a deficiency on vision.

MECHANISM OF TRANSDUCTION

Light rays first pass through all layers of the retina to reach the **photoreceptors**. These photoreceptors are active due to their location in an area called the **pigment layer**, which constitutes the first layer. Photoreceptors are divided into two types: **cones** and **rods**.

After passing through this first layer, light reaches the **bipolar cells**, then the **ganglion cells**. The latter group their axons to form the **optic nerve**, where transduction occurs.

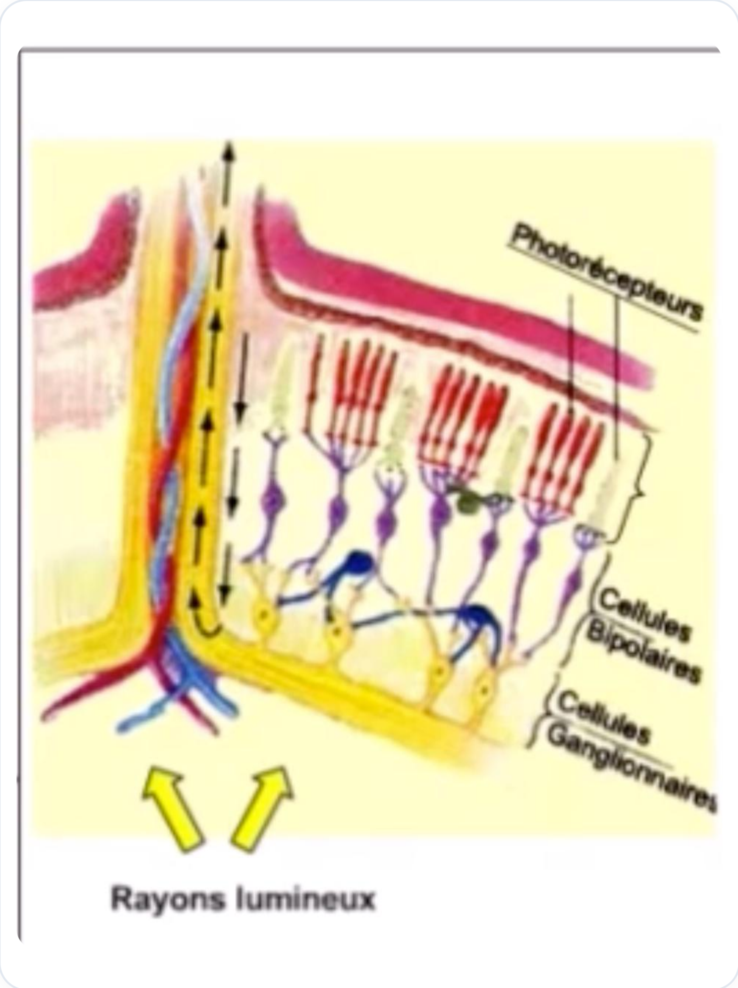


FIG. — Light pathway to the optic nerve

Delving deeper, we also observe **horizontal cells** that capture and redistribute information to the bipolar cells. Below these are the **amacrine cells**, which gather information before directing it to the ganglion cells. This complex system allows for the processing of a large amount of visual information.

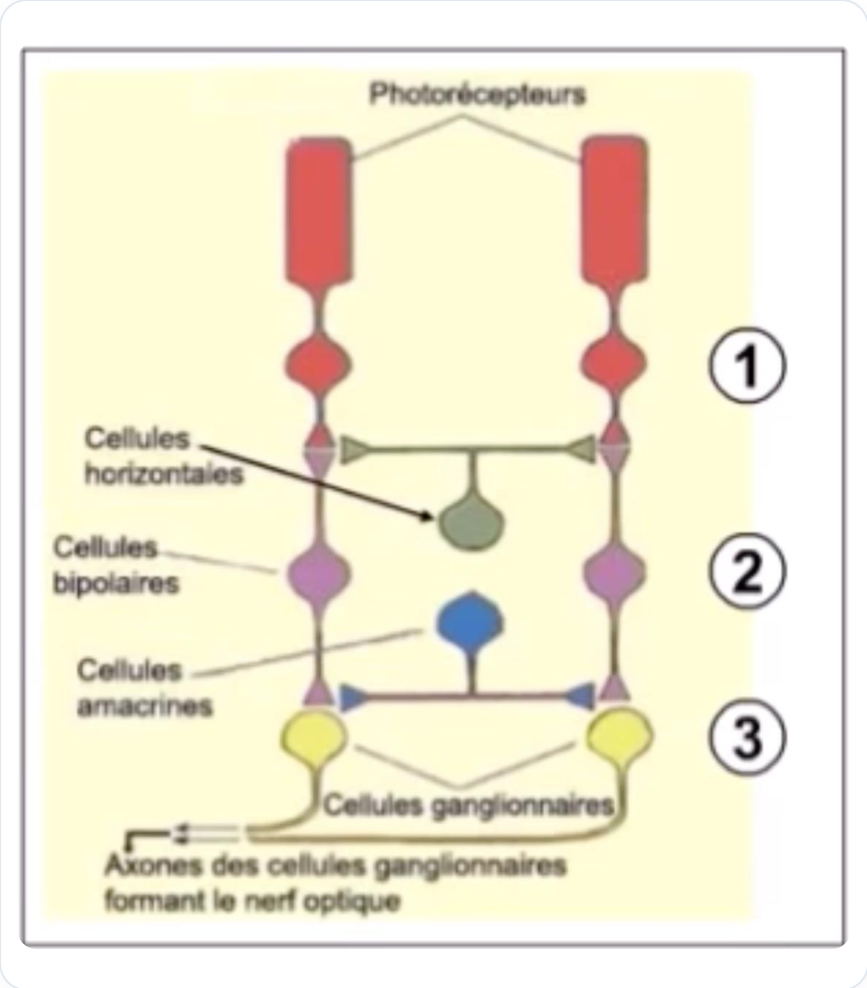


FIG. — Role of horizontal and amacrine cells

Let's take the example of a cone photoreceptor. It consists of an **outer segment**, an **inner segment**, a **nucleus**, and a **synaptic terminal**. In the outer segment, we find **plasma membranes** and a stack of small discs containing a protein called **opsin**. For rods, this opsin is called **rhodopsin**, while for cones, it is called **iodopsin**.

In rods, a small molecule called **retinal** is attached to rhodopsin. When light strikes these discs, retinal changes shape, transforming rhodopsin into **metarhodopsin**. This transformation triggers a series of chemical cascades at the cell membrane, preventing sodium entry and creating an ionic change.

This change in membrane permeability leads to a **membrane potential** that generates a **nerve impulse**. Thus, light, by passing through the layers, activates a pigment, beta-carotene, present in vitamin A, essential for activating rhodopsin and iodopsin.

Rods, located mainly in the peripheral retina, provide low-detail vision but are very sensitive to light, without colour perception. In contrast, cones offer detailed and coloured central vision, with specific types for blue, red, and green, corresponding to different wavelengths.

Rods consist of three parts: the outer segment, the inner segment, and the synaptic terminal. The outer segment contains small opsin discs, which, in the case of rods, is rhodopsin.

When light reaches these discs, rhodopsin transforms into metarhodopsin, leading to a chemical cascade that modifies the membrane's polarity, making it impermeable to sodium. This change creates an electrical current, essential for the transmission of visual information.

The **transduction** of light energy occurs in the photoreceptors through the absorption of light by opsin, which changes shape according to the light received. This process is crucial for vision, and vitamin A plays a fundamental role in retinal synthesis. A deficiency in vitamin A can lead to vision loss, particularly night blindness.

28. THE KNEELING BODY, LATERAL

DURATION : 17:52

STUDY SHEET

COURSE SUMMARY

Study of the role of the lateral geniculate body in processing visual information, via its parvocellular and magnocellular layers (processing of colour and movement). Analysis of synaptic transmission via the optic nerve to the cortical areas and interaction with the pulvinar. The presentation illustrates multisensory integration and its links with Papez's circuit, demonstrating how visual perception is modulated by memory patterns.

THE LATERAL GENICULATE BODY: INTEGRATION AND PROCESSING OF VISUAL INFORMATION

The **lateral geniculate body** plays a crucial role in processing visual information. It receives nerve impulses from the retina and sorts this information before transmitting it to the visual areas.

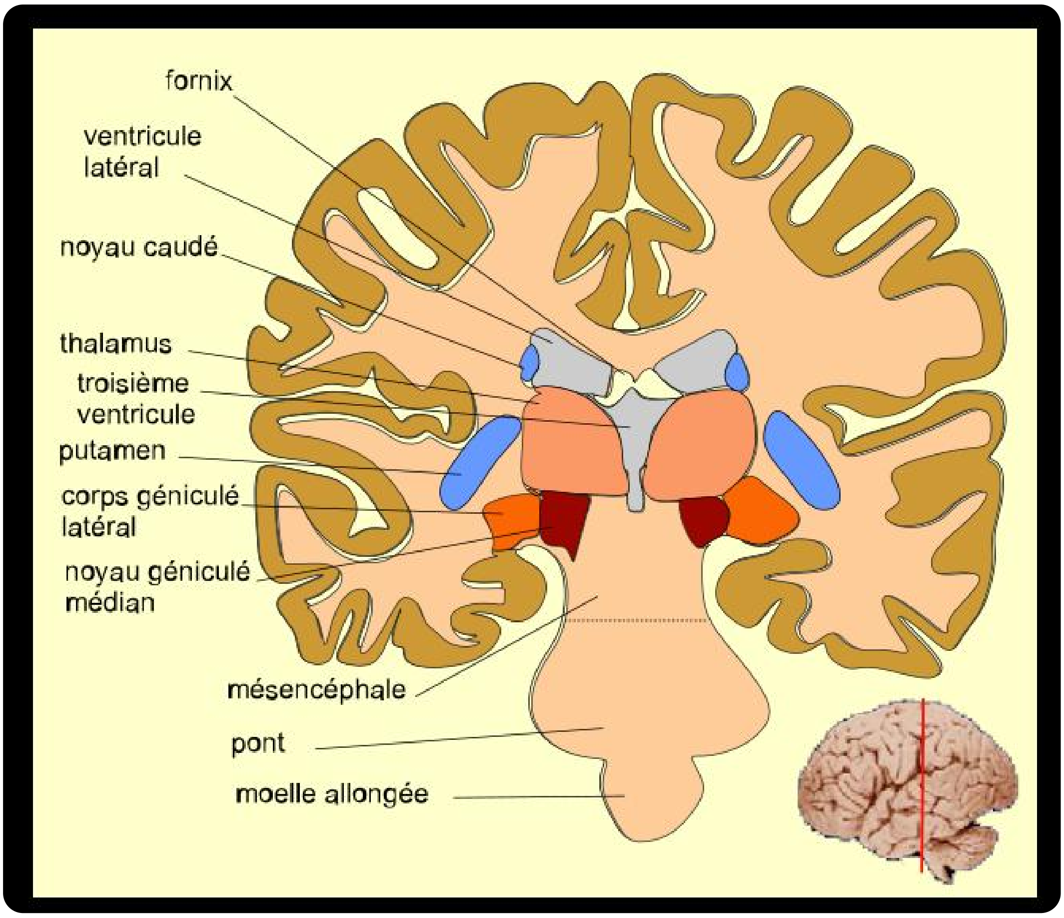


FIG. — Corps Genouillé Latéral (CGL)

STRUCTURE AND FUNCTION

The lateral geniculate body is composed of several layers, each specialised in processing specific information. There are mainly three types of cells:

- **Parvocellular cells**: These process information related to **colour** and **form**.
- **Magnocellular cells**: These are responsible for detecting **movement** and **orientation**.
- **Koniocellular cells**: These focus on **spatial frequencies**.

Each layer of the lateral geniculate body contributes to the formation of an image by integrating these different pieces of information.

INFORMATION TRANSMISSION

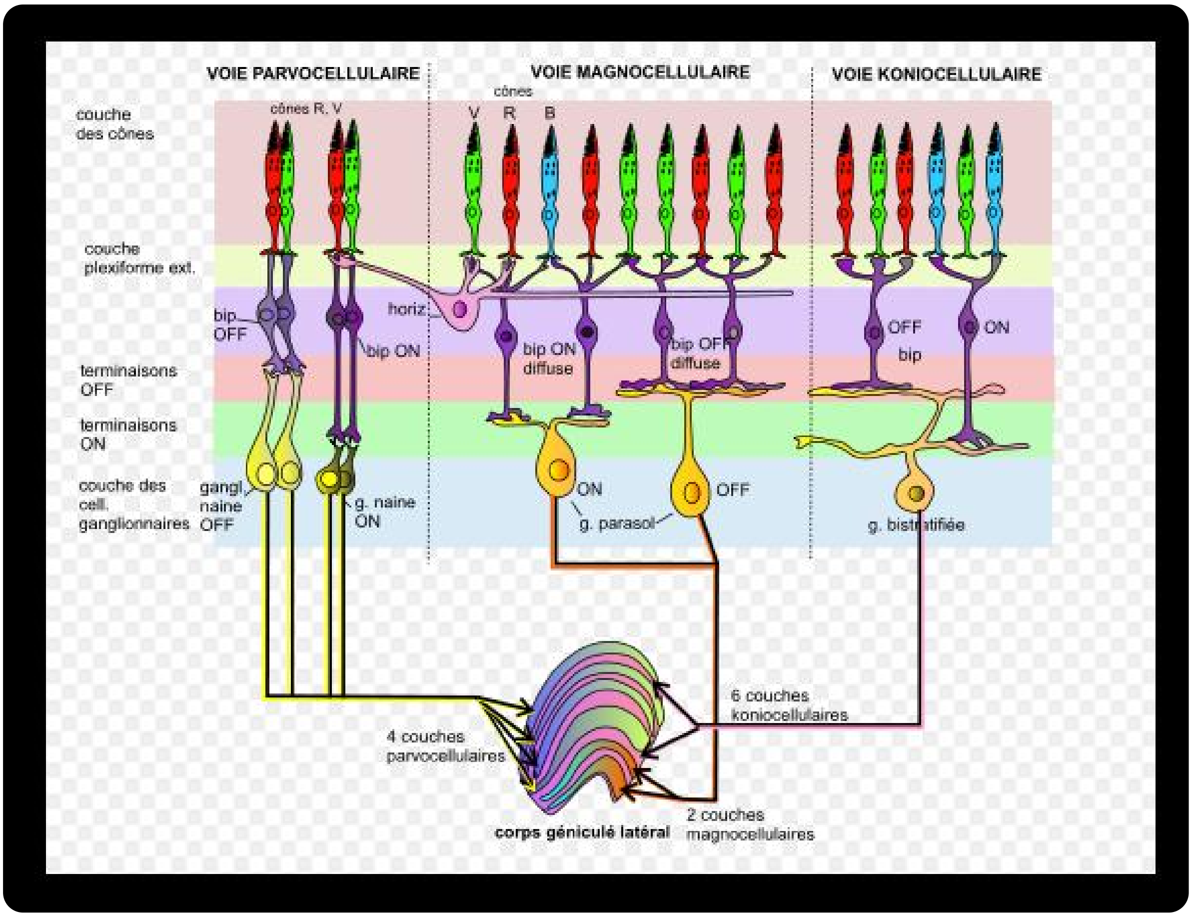


FIG. — Parvocellular and magnocellular cells

Visual information is transmitted through the optic nerve and arrives at the lateral geniculate body, where it is sorted before being sent to the thalamus and visual areas. Approximately **80%** of the information processed by the lateral geniculate body is directly sent to the visual areas, while **20%** passes through the pulvinar, another thalamic structure, for more subtle processing.

The pulvinar also receives information from other nuclei, such as the **superior colliculus** and the **olivary nuclei**, which are essential for integrating auditory and visual information.

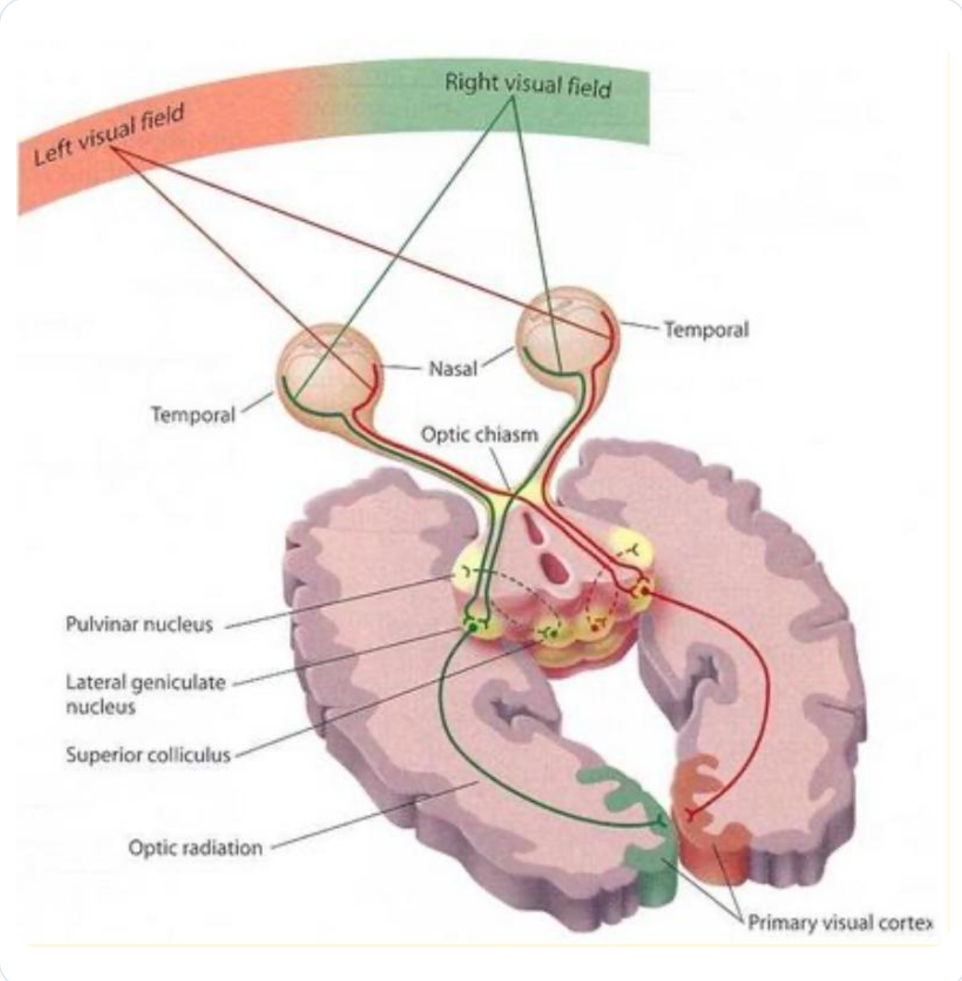


FIG. — Information distribution from LGN to pulvinar

MULTISENSORY INTEGRATION

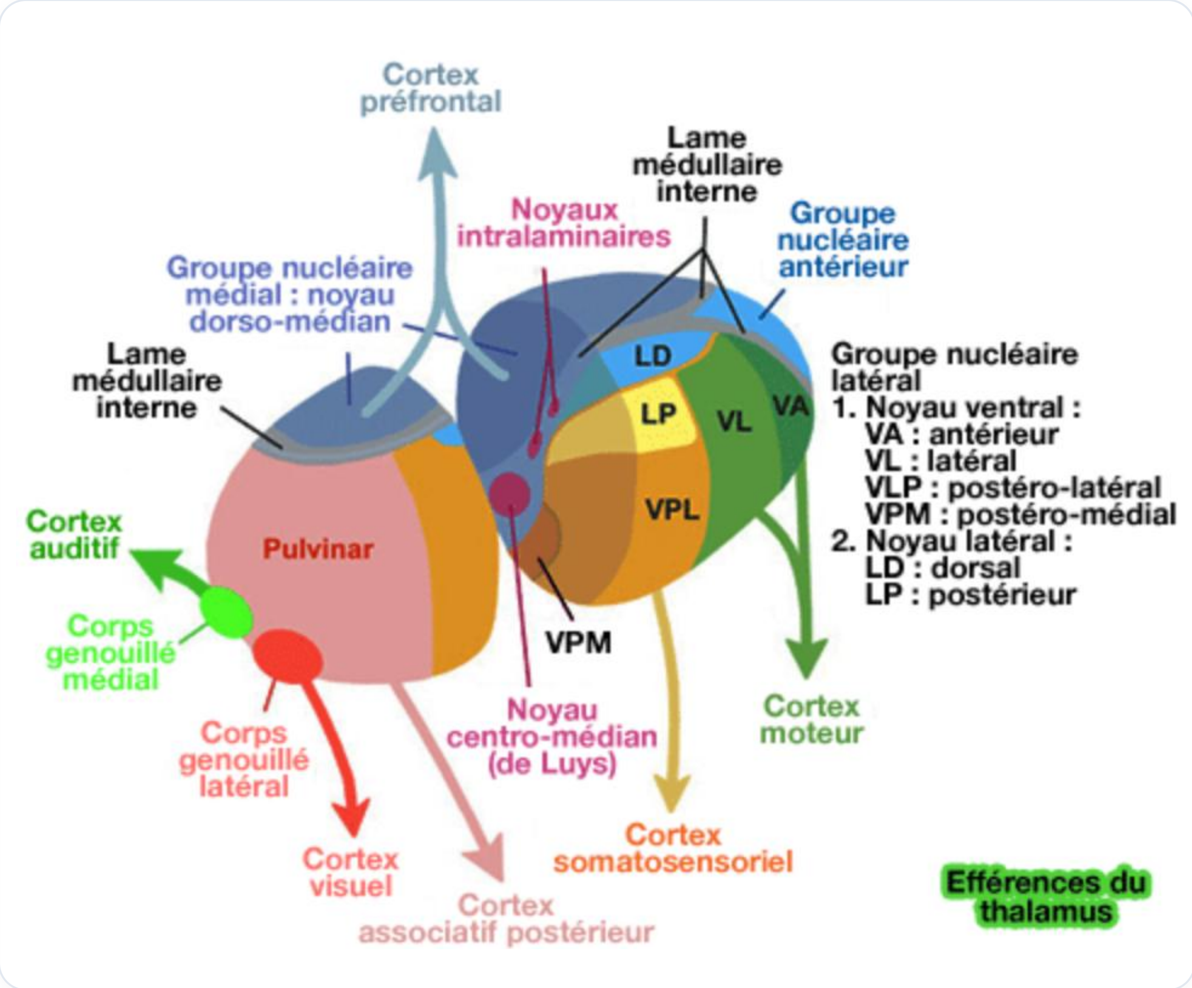


FIG. — Pulvinar connections

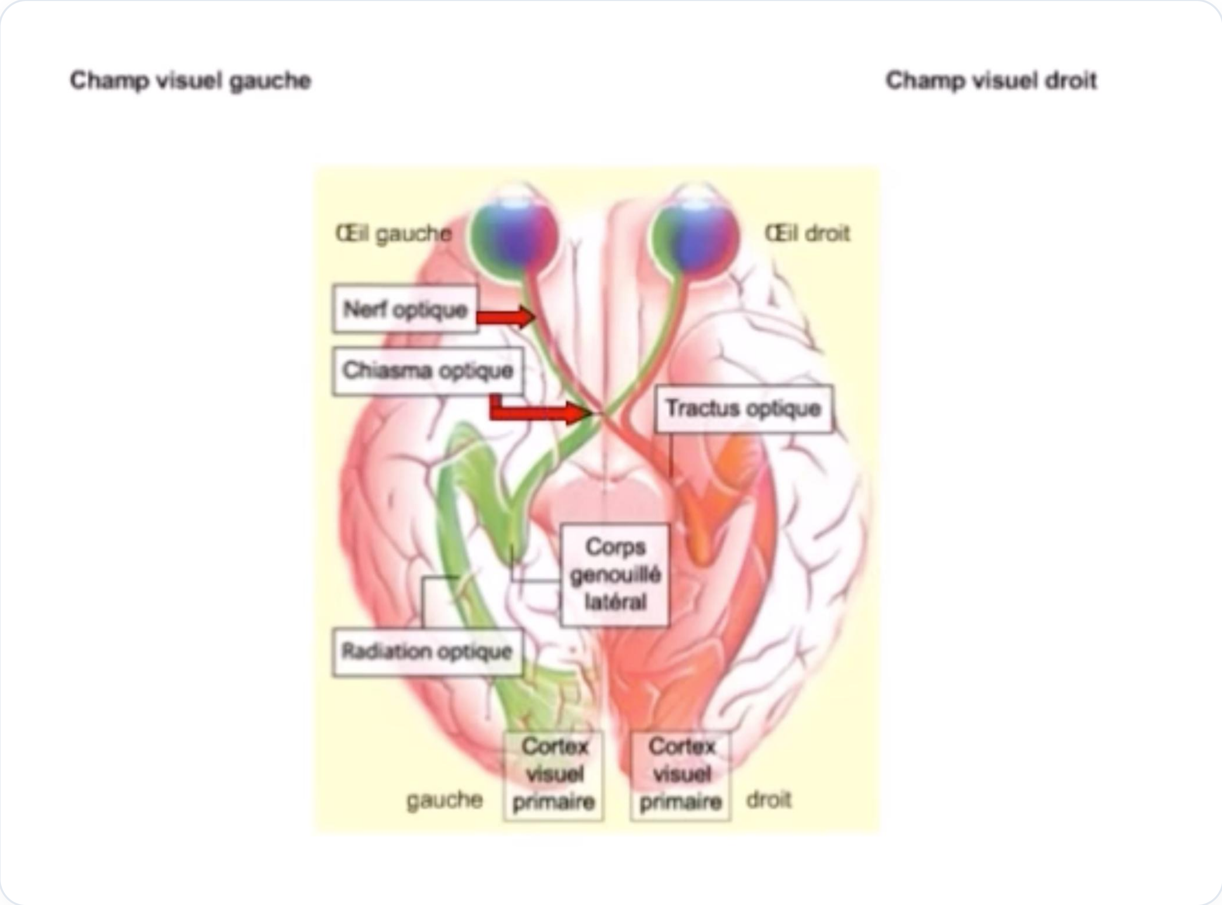


FIG. — *Visual information pathway to the LGN*

The lateral geniculate body not only processes visual information but is also connected to auditory and vestibular systems. This allows for the integration of information from the eyes, ears, and the body's position in space. This process is essential for actions such as ocular **saccades** and head orientation.

PAPEZ CIRCUIT AND MEMORY

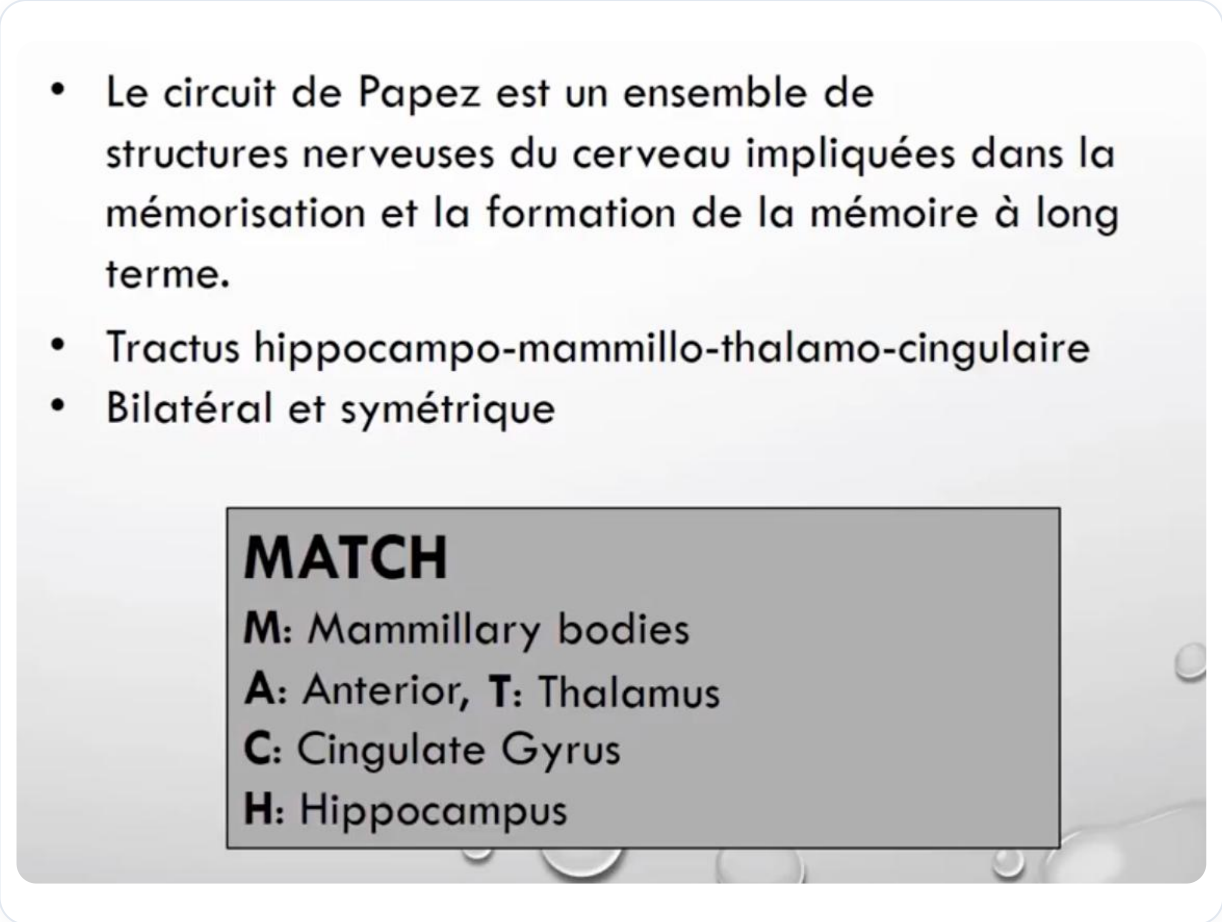


FIG. — Intégration multisensorielle (auditive et vestibulaire)

The lateral geniculate body is also linked to the **Papez circuit**, which plays a role in memory. This circuit includes structures such as the **mammillary bodies**, the **anterior thalamus**, and the **cingulate cortex**. It is involved in the storage and retrieval of memories, linking visual information to past experiences.

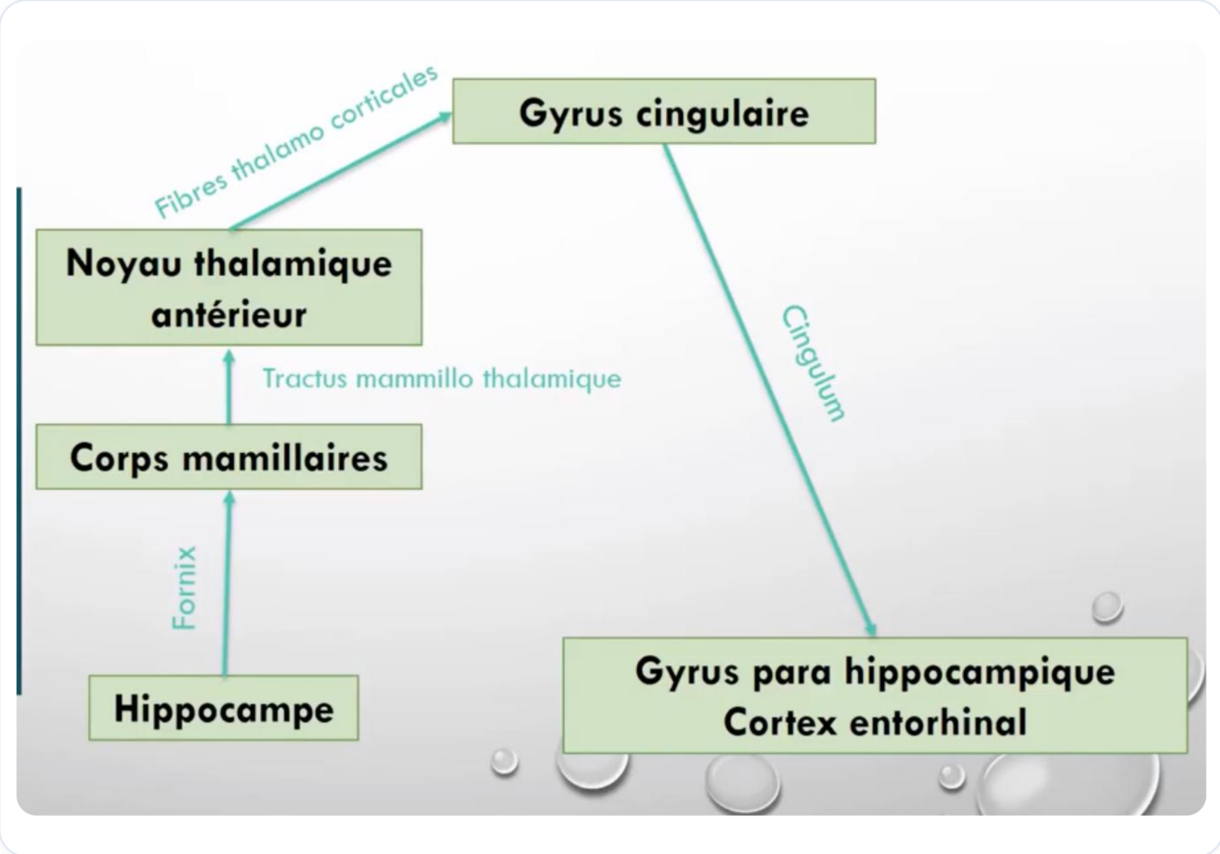


FIG. — Role in saccades and head orientation

INTERPRETATION OF REALITY

It is important to note that when we perceive information, we only receive a fraction of reality. Indeed, only **20%** of perceived information is real, the rest being a reinterpretation based on our experiences, emotions, and prior knowledge. This interpretation process is influenced by the connections between the visual areas and the lateral geniculate body.

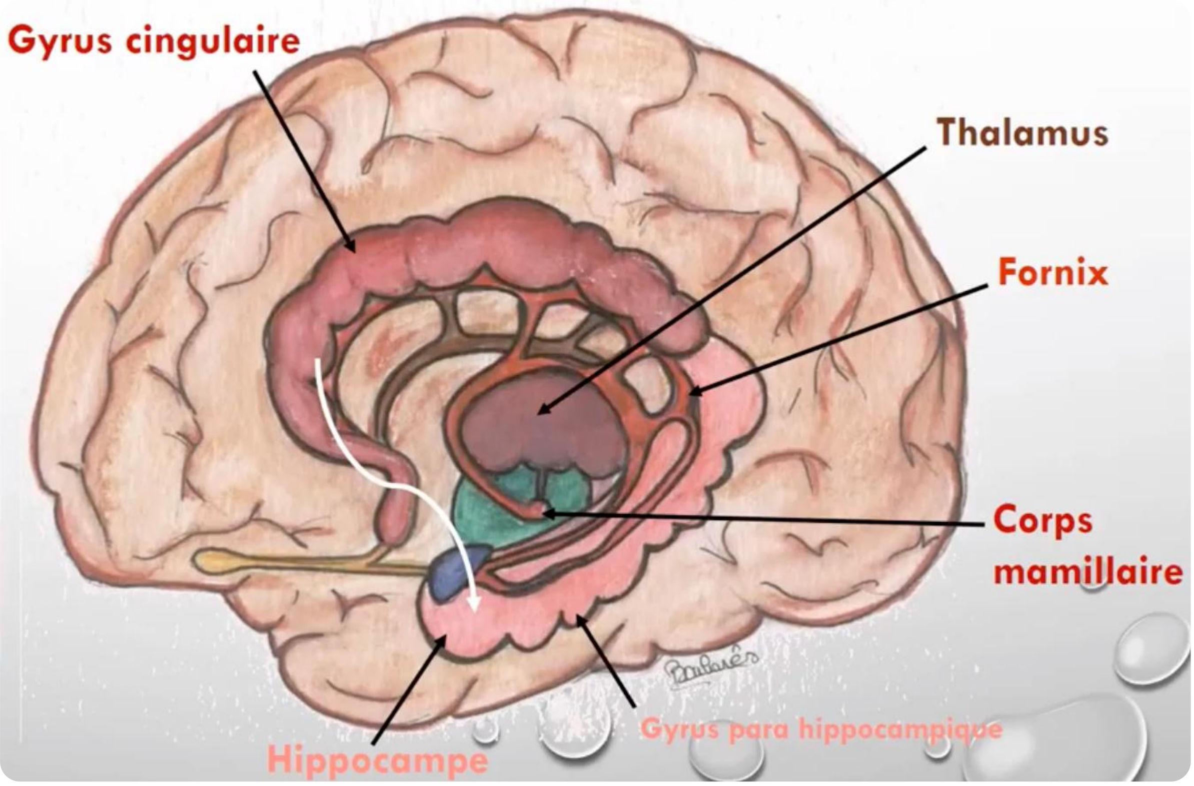


FIG. — Papez circuit and memory

CONCLUSION

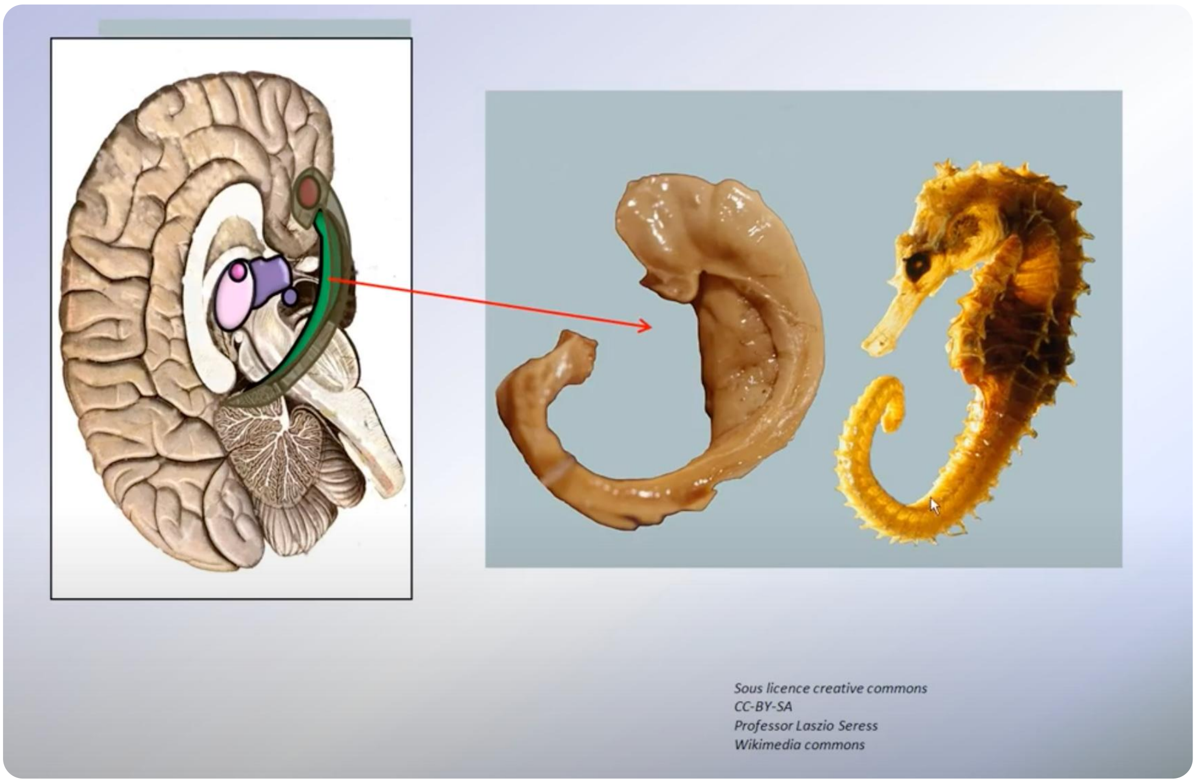


FIG. — Papez circuit: memory consolidation

The lateral geniculate body is a key structure in the processing of visual information, integrating data from different sensory sources and playing a crucial role in our perception of reality. Its ability to sort and interpret information is essential for our interaction with the world around us.

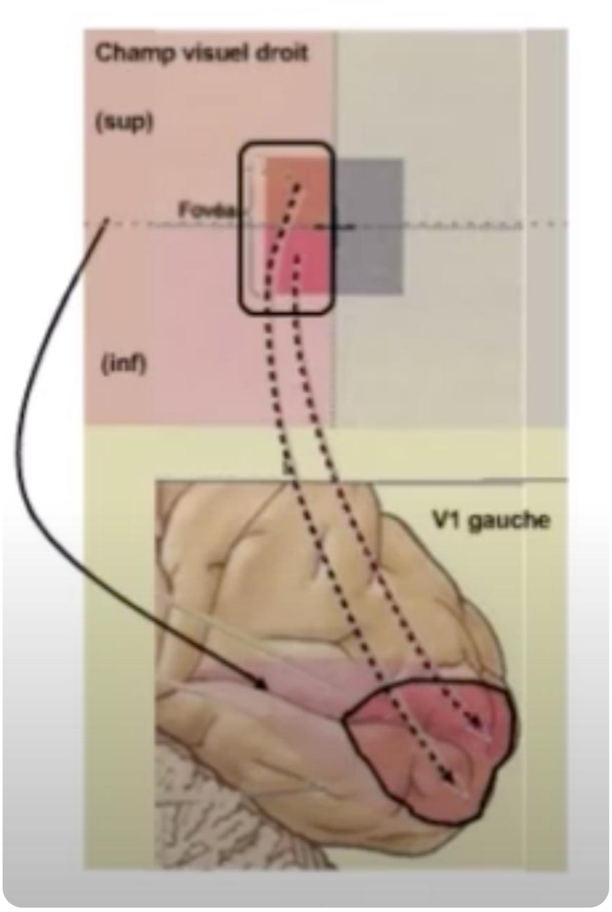


FIG. — Reinterpretation of perceived information

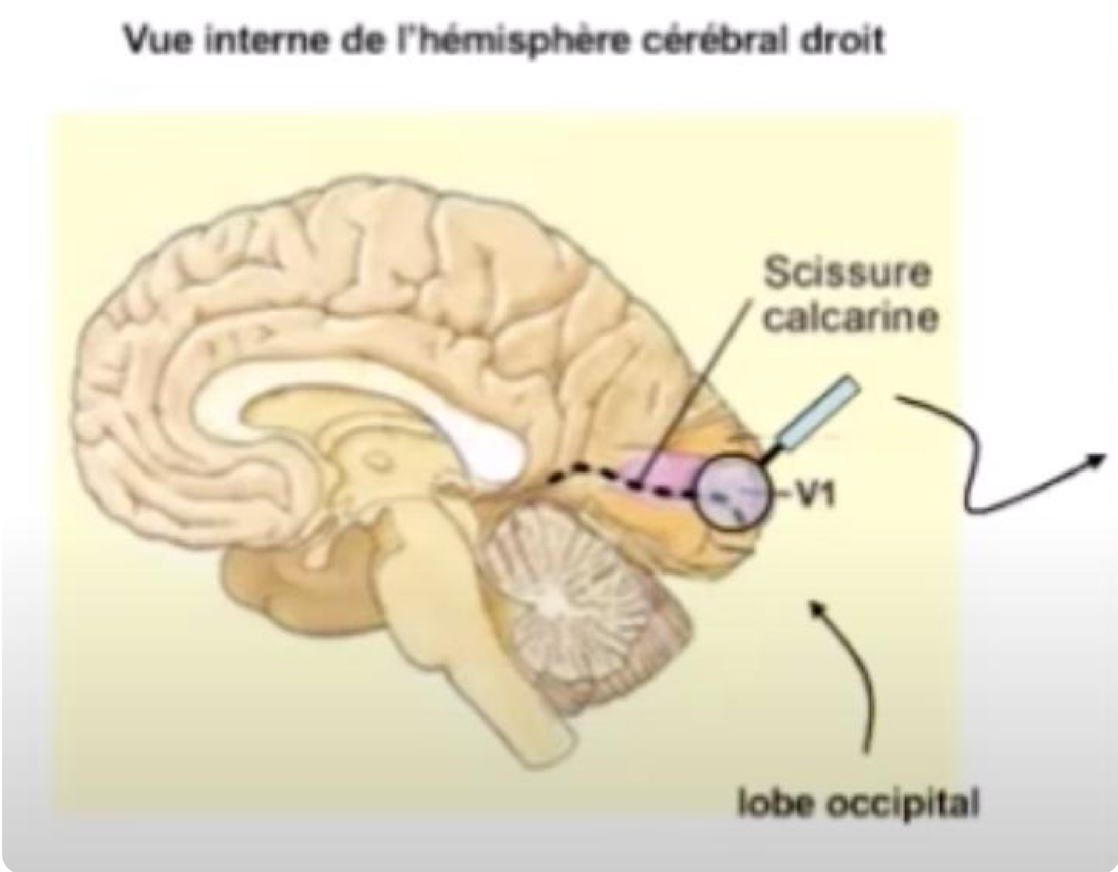


FIG. — *Interpretation of reality*

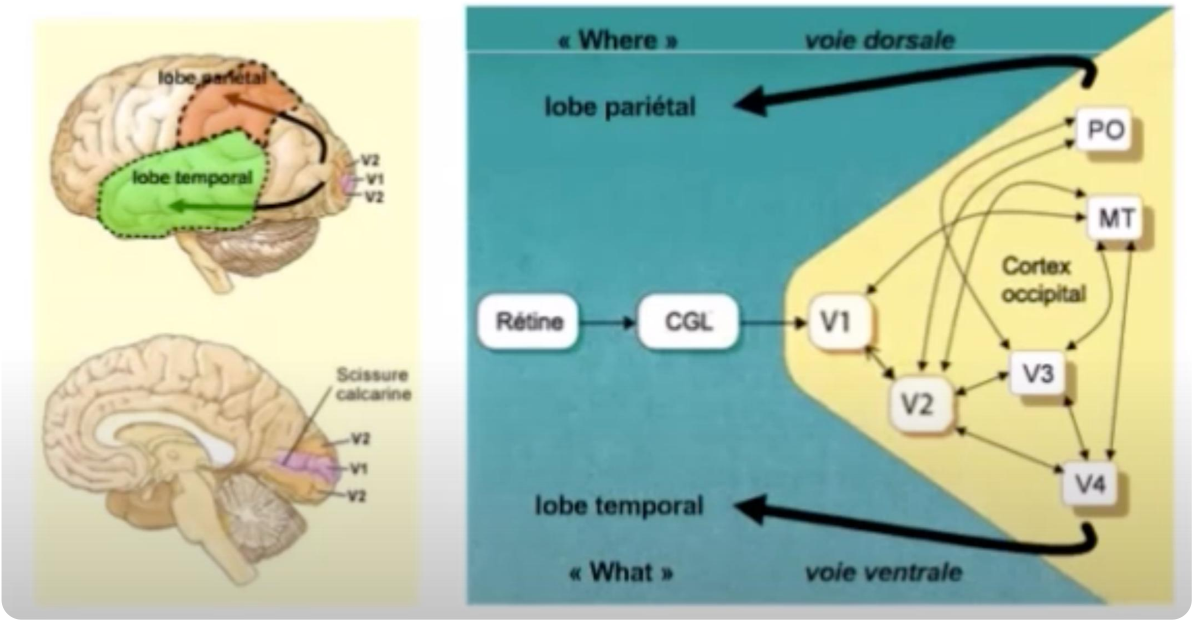


FIG. — *Importance of LGN in perception*

29. DISSOCIATION OF MEMORIES

DURATION : 03:20 STUDY SHEET

COURSE SUMMARY

This video explores the dissociation of emotional memories through the testimony of a significant personal experience. The speaker shares how stressful family-related events influenced his perceptions and physical health. Thanks to the intervention of a psychiatrist, he learned to dissociate emotions from events, offering a new perspective on his fears and bodily pains. The teaching focuses on the importance of releasing emotional associations and realigning one's experiences to foster self-understanding and act as a therapist. This re-evaluation process is fundamental to achieving true emotional liberation.

DISSOCIATION OF MEMORIES

During a significant experience, I found myself in a situation where my father was going to hospital. Everyone was worried, but as the youngest, I felt almost detached. My parents had just separated, and although I learned my mother was taking a plane to come, I felt a certain **satisfaction** regarding their arguments.

The next day, my sister was in a critical condition, and my brother and I were coming down the mountain. On the way to the hospital, I couldn't look out of the window, but I saw a plane pass by. At that moment, I had to take a plane, and a week before, I hadn't realised the emotional impact it had on me. I started to feel stomach pains, a **pressure** in my chest, and I looked for excuses to avoid taking the plane.

A prominent psychiatrist was present and asked me to stop and reflect on what I was experiencing. She questioned me about my fears, particularly that of planes, which I attributed to an incident I had seen. She then made me understand that these events were not my responsibility. She dissociated the events: my sister's **appendicitis**, the problems between my parents, and the plane crash. I realised I had nothing to do with these situations.

This realisation was liberating. I had accumulated all these emotions in different parts of my body: in my stomach, my testicles, my knees, my feet, my elbows, and my shoulders. Once all this was put into perspective, I felt a sense of relief.

This experience illustrates how our **fears** can be linked to repressed desires. For example, asking a parent if one can ride a bicycle can be tinged with the fear of falling. Every fear hides a desire. It is a complex memorisation process.

It is essential to free ourselves from these emotional associations and re-evaluate our experiences. It is a path towards **liberation** and self-understanding.

30. THE EYE PAIRS

DURATION : 04:11

STUDY SHEET

COURSE SUMMARY

This video addresses the fascinating theme of pairs of eyes as windows not only to the soul but also to various aspects of our being, ranging from physiology to spirituality. It explores how cultural elements, such as the activation of the pituitary gland in Asia, are linked to notions of clairvoyance and perception. The connections between the eyes, internal organs, and even joints such as the elbows and knees are deciphered, offering a new perspective on the human body and its memories. The concepts of harmony between the physical body and the energetic body are essential for understanding our personal journey and how our stories influence our perception of the world.

PAIRS OF EYES: AN EXPLORATION OF THE SENSES AND CONSCIOUSNESS

Polyvoyance is described as the ability to see **God** in everything, while the **pituitary gland** is the seat of **clairvoyance**. In Asia, the **third eye** is often mentioned, allowing access to the hidden side of things. This gland plays a crucial role in secreting the hormones necessary for the body.

In cultures such as **India** and **Nepal**, colours are applied to the forehead to activate this gland through the **frequency** and **vibration** of colours. The eyes are often considered the **windows to the soul**, but they also represent the eyes of the **mental body**.

The Hebrew term **ayin** translates to eye, but also to **source of light**. This highlights that the eye is not only a light sensor but also a light **emitter**. A deep gaze can cause **hormonal stimulation** in both women and men.

It is also interesting to note that the **angle of the jaw** is considered an eye, allowing exploration of the unconscious. Light sensors are also found in this region. The **parotid glands** are linked to this area, as are the **testicles**.

The **ring finger** is often called the finger of alliance and inner vision. The **heart** is perceived as the seat of **loving vision**, acting as the window to the spirit. The eyes are directly linked to the **ovaries** and the **knees**, forming a complex network.

The **D3 vertebra** is crucial, as it can alter vision. The **ovaries** and **testicles** are considered the eyes of the **energetic body**. The ovulation cycle is linked to a woman's evolution, while **menopause** marks the beginning of physical and spiritual maturity.

The **elbows** are perceived as the eyes that allow one to look behind the energetic body. The expression "playing elbows" evokes the idea of getting ahead in life. The **knees** represent the eyes of the physical body, where a woman evaluates her relationships with others and with herself.

The **heel** is also an eye, allowing one to look behind the physical body. It is considered the first eye of the earth, a symbol of peace. Each eye is connected to the others and carries **memories** in the physical, etheric, astral, and mental bodies.

These memories convey information that influences our interpretation and experiences. Thus, it is essential to reflect on what we do with these images that belong to us, influenced by our **genetic**, **karmic**, current history, and by **pure memory**.

31. PRACTICE: PINEAL AND PITUITARY GLANDS

DURATION : 10:39 STUDY SHEET

COURSE SUMMARY

This video offers an in-depth exploration of the fundamental roles of the pineal and pituitary glands in hormonal regulation and memory. It presents energy rebalancing techniques inspired by the embryonic journey, specifically concerning pituitary motility. Students will learn to identify the interactions between divine and current light, as well as to use clinical observations such as breathing and eye movements to restore balance between these two crucial glands. The course also highlights the importance of the pineal gland in regulating vital functions such as breathing, digestion, and sleep.

PRACTICE: EPIPHYSIS AND HYPOPHYSIS

The epiphysis and hypophysis play a crucial role in **hormonal regulation** and **memory**. The epiphysis, often called the pineal gland, is the **link between neurons** and the hormonal system. It is influenced by light and contains microorganisms that convert **serotonin** into **melatonin**. When light intensity decreases, melatonin production begins.

To rebalance the hypophysis and epiphysis, an energetic rebalancing technique is used, by revisiting the **embryonic journey**. This involves exploring the motility of the hypophysis through the entry points located at **nasion** and **bregma**.

At the time of the formation of the base of the skull, a space called **Rathke's pouch** forms. This invagination of the epiblast epithelium gives rise to the hypophysis, which receives both descending and ascending information. The hypophysis divides into anterior and posterior hypophysis, and its development is linked to the formation of the **prosencephalon**, which divides into **telencephalic** and **diencephalic** parts.

The epiphysis gland moves from front to back, influenced by divine light and the hypophysis. This interaction between divine light and current light is essential for the function of the hypophysis.

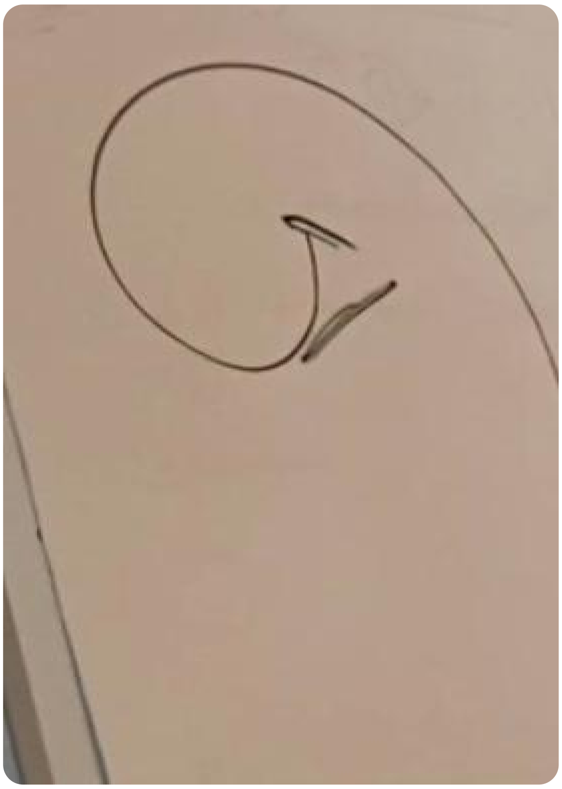


FIG. — Skull base formation

The hypophysis develops from the surface epithelium, and the flexion movement causes a suction towards the centre, thus forming the anterior hypophysis gland. In parallel, a suction movement in the brain gives rise to the infundibulum, which constitutes the posterior part of the hypophysis.

FIG. — Pituitary development

In practice, it is possible to use internal techniques to work on the energetic axis between the hypophysis and the epiphysis. This involves observing the patient's breathing and eye movements, as these elements are indicative of the energetic state.

FIG. — Infundibulum development

The **nervous system**, including the hypothalamus and thalamus, plays a key role in this dynamic. By restarting embryonic motility, it is possible to rebalance the anterior and posterior hypophysis. Eye movements can indicate a return to the hara, thus promoting stability.

The epiphysis is essential for the regulation of **respiration**, **digestion**, and **sleep**. It produces melatonin, a hormone that influences brain waves, thus facilitating falling asleep, deep sleep, and dreams. Although the hypophysis controls almost all glands, the epiphysis is considered the master gland, playing a central role in the body's hormonal and energetic balance.

31. BIS. INFLUENCES ON THE EYE

DURATION : 24:37 STUDY SHEET

COURSE SUMMARY

Examination of the eye's role within the neurosensory programme and its connection to Tenon's capsule / hypothalamo-ventricular system. Integration of postural chemistry, somatic feedback, and energetic mechanisms to demonstrate how visual afferents affect tone and psychic state. Analysis of fascial tissue as a vector of body memory and the cardiovascular system's influence on electromagnetic balance.

INFLUENCES ON THE EYE

The **Tenon's capsule** is connected to the entire body and plays a crucial role in a **neurosensorimotor programme**. This means it integrates information about **postural tone**, which can be considered as **postural chemistry**. At the level of the **hypothalamus**, a great deal of information is integrated, notably via the **thalamus** and the **pulvinar**, as well as other nuclei around the ventricular system.

This information can be classified into two types: **trophotropic** and **ergotropic**, corresponding respectively to the sympathetic and parasympathetic systems. Thus, the information received by the hypothalamus, including that from the eyes, influences postural chemistry at a cellular level, integrating elements such as **oxidation-reduction** and **nucleotides** like **guanosine** and **adenosine**. The hypothalamus uses this data to adjust bodily tonicity.

For example, a spiral movement can be oriented to the right or left, and can be associated with various energy states, such as **depression** or **overpressure**. Postural chemistry is thus integrated into a tone influenced by various afferents to the cortical and visual system, creating a continuous, positive or negative **feedback**.

The eye, as a source of information, plays an essential role in integrating these messages. It contributes to a specific postural outcome, representing the best possible adaptation at a given moment. There is no bad posture, only the most adequate functional response. It is crucial to trust what one sees, while being aware that the programme can sometimes loop. This requires a gaze, a liberating gesture, performed at the right time and in the right space.

Meditation can also help to break free from these habitual patterns, which can sometimes distort our perception of reality. Micro-lesions in the facial and ocular areas can alter the orientation of the eye and influence mental programmes. The eye is of paramount importance in the integration of neurosensory information, in connection with the **tectospinal system** and memories stored in the **facial tissue**.

The **fascia** is an intelligent tissue capable of storing information and memories. By being attentive, in the right place and at the right time, with an adequate mental map, it is possible to achieve liberation. This mental representation is influenceable and can even support the visual message.

The link between the eye and the meningeal system is fundamental. The eye acts as a beacon, integrating all information at this level. It is essential to understand that the eye cannot deceive, because it is pure. Light, as a photon, is a source of energy that influences our perception. The relationship between frequency and wavelength is crucial: a higher frequency corresponds to a shorter wavelength, leading to an increase in energy.

Light information, captured by the retina, can cause fluid mobilisation. The **cones** and **rods** of the retina are not only information sensors, but also energy sensors. Physical fatigue can often be linked to sympathetic lesions, leading to **mydriasis** and agitation of the rods, which can exhaust the sympathetic system.

Factors such as energetic stress can degrade our state. The right light information is essential to maintain our well-being. The aura, as a luminous field, is influenced by our position in life. Inner peace and wisdom must be cultivated to emanate positive energy.

The eye undergoes an energetic and interpretative programme, integrating information in the form of pre-programmes. This process leads us to an electrical and electromagnetic plane, where the heart plays a central role. The heart emits a powerful electromagnetic field, and the eye, as a mobile fulcrum, is also influenced by different currents.

There are significant vascular differences between the left and right eyes, influencing their functioning. For example, the right eye is linked to hepatic venous stasis, while the left eye is associated with arterial hypertension. These anatomical differences can lead to specific interferences on the visual cortex.

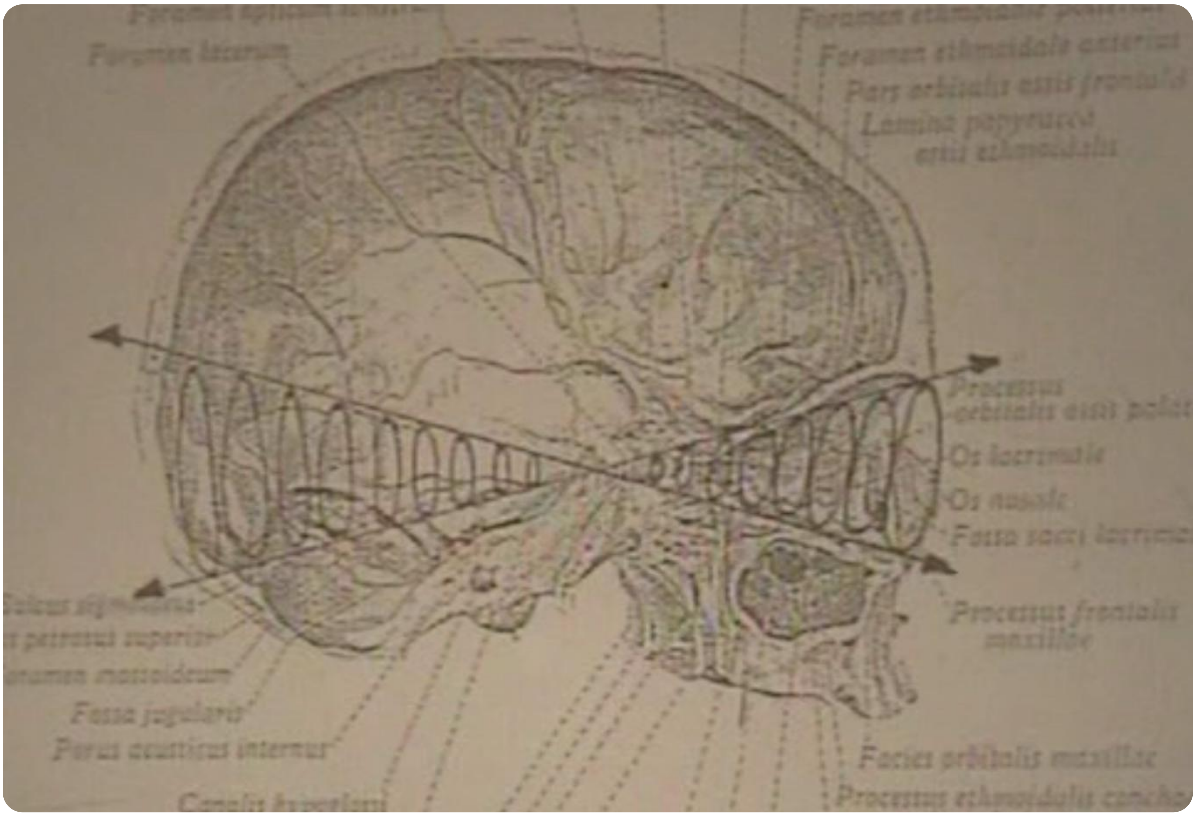


FIG. — *Spiral eyes*

Ocular reflexes, such as the fronto-orbital reflex and the auriculo-cardiac reflex, show how light pressure on the eyes can rebalance cardiac problems. Each eye has a postural role and a dominant eye, which are not always the same.

Eye movements are essential for rebalancing. Visual axes and angles of inclination play a role in neurosensory light integration. The eye has a freedom of light integration of approximately 23 degrees, a figure found in various contexts, such as the Earth's tilt.

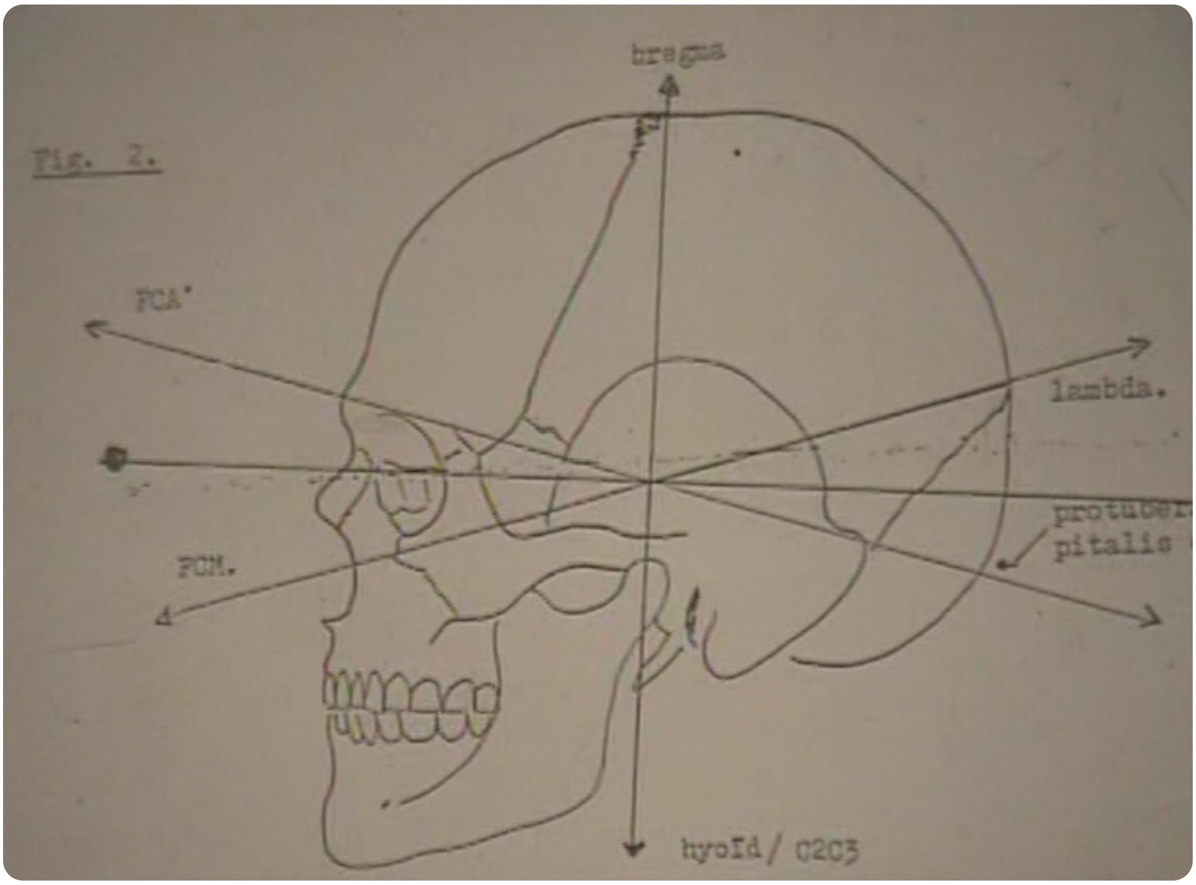


FIG. — Crossing the line

Finally, information axes, such as the petrous axis, show how ocular destabilisations can affect balance. Memories integrated into bones or electrical patterns can also influence our perception and balance.

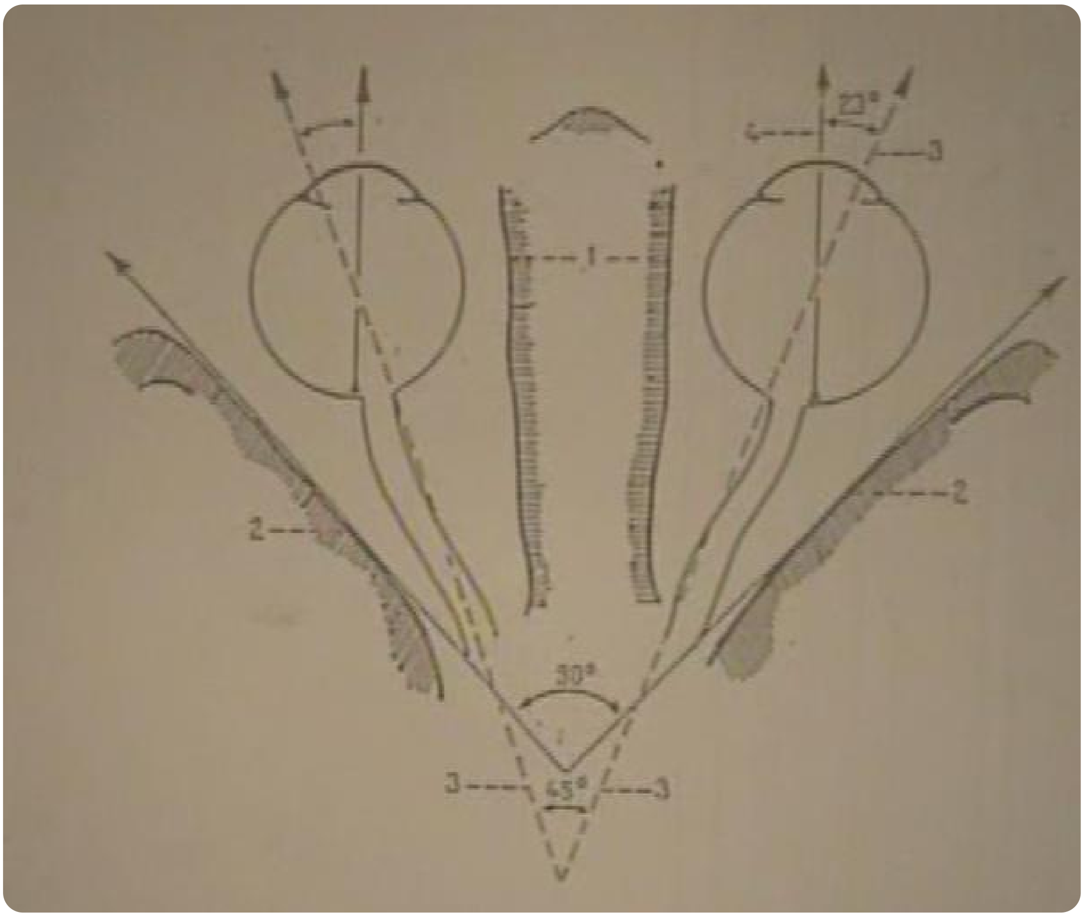


FIG. — *Axes and angles of visual integration*

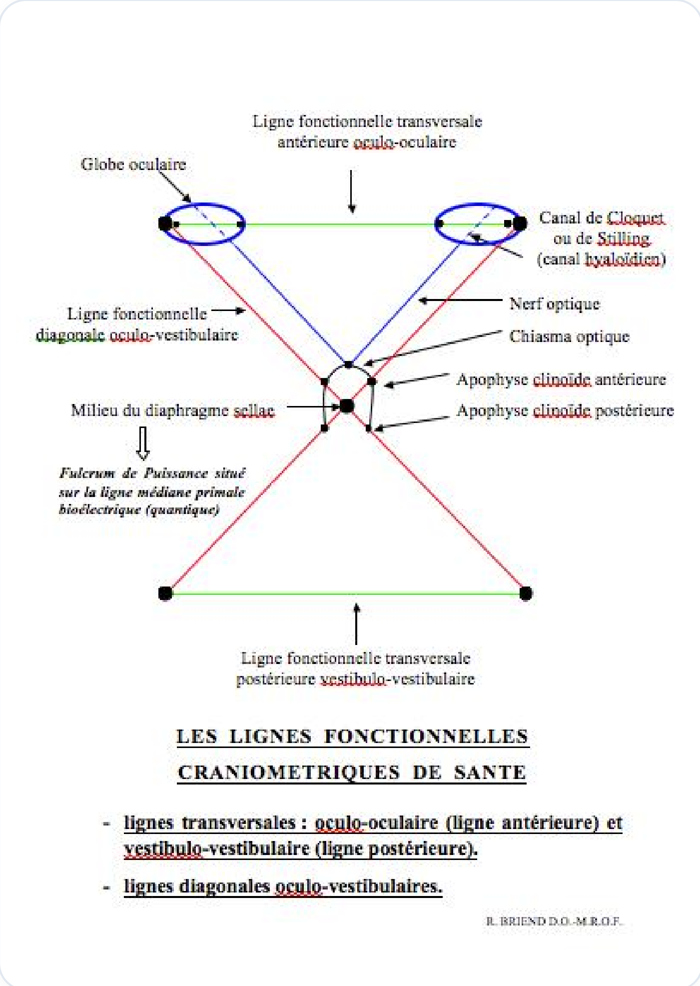


FIG. — Eye movements

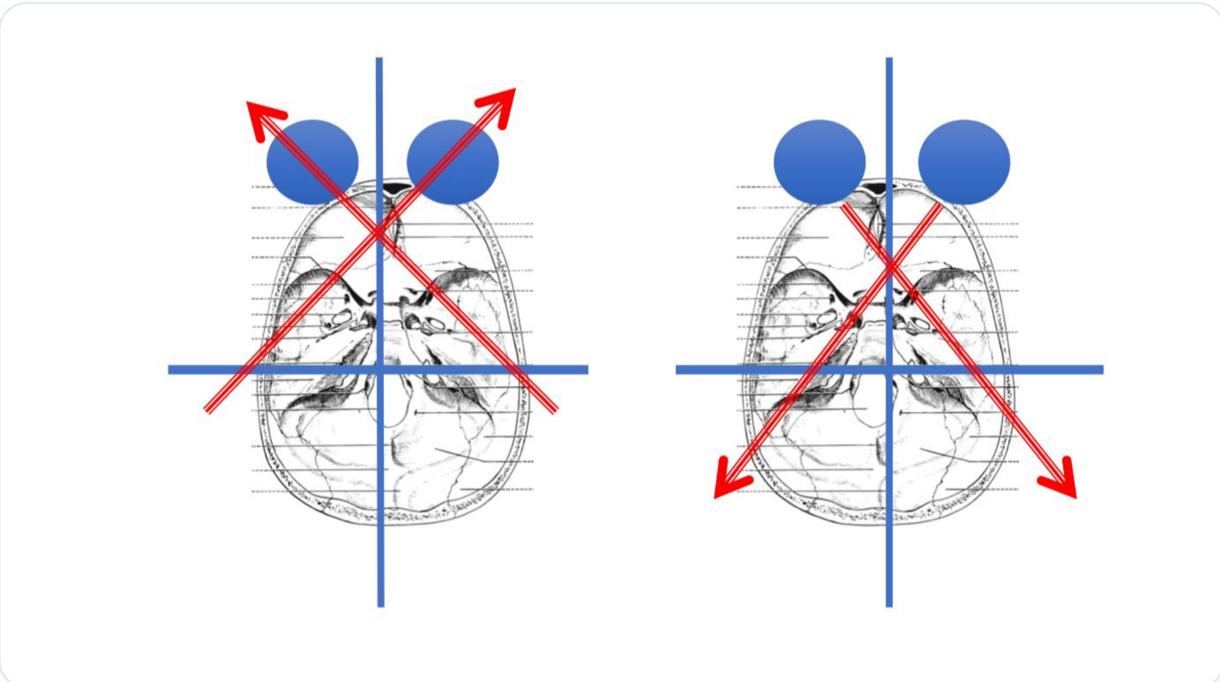


FIG. — 23 degrees of visual freedom

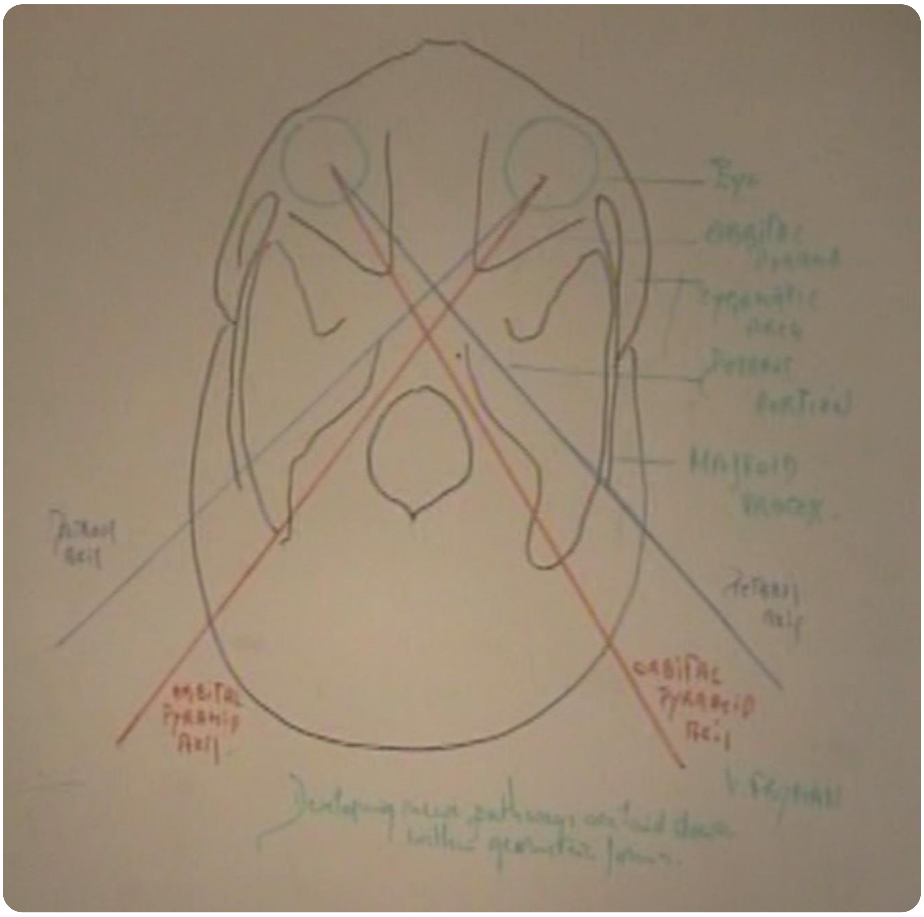


FIG. — *Orbital and pyramidal axes*

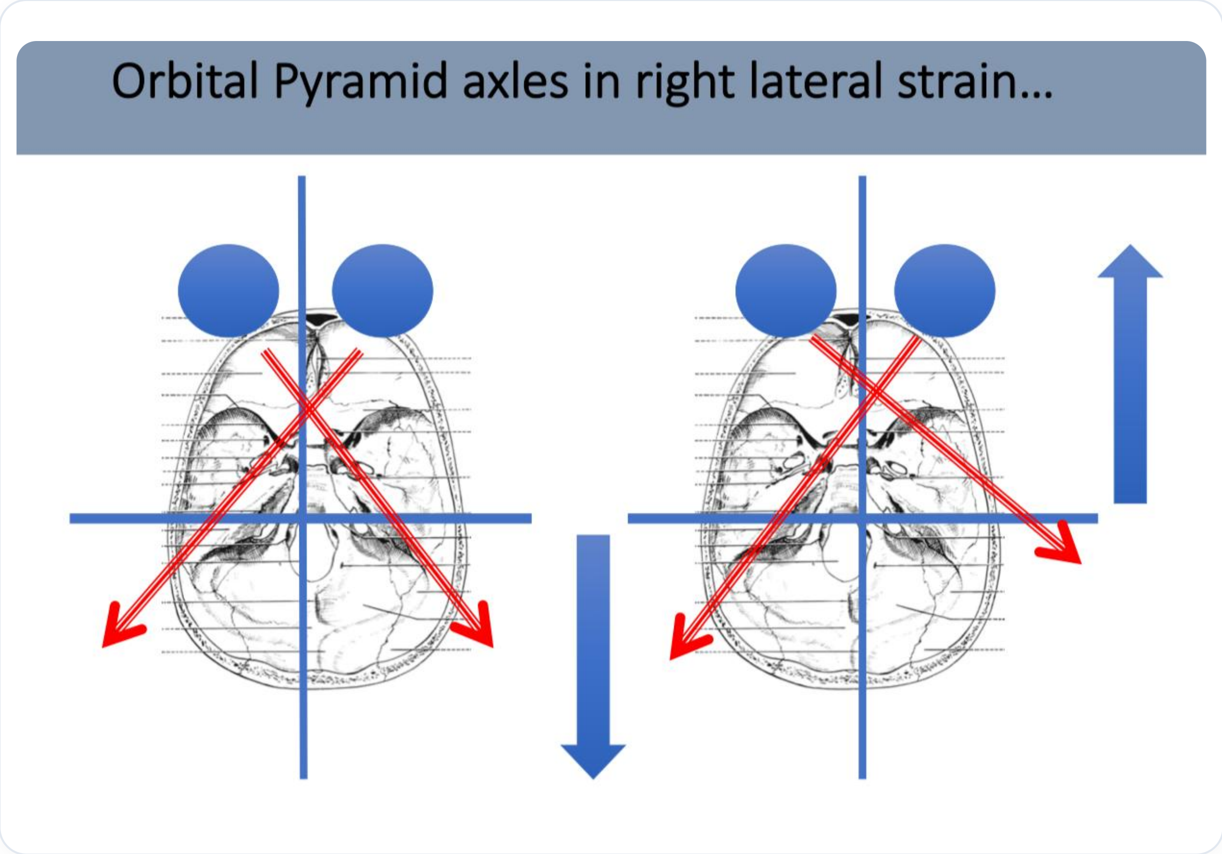


FIG. — *Memories in the bone*